

SEMESTER 3

**NAVAL ARCHITECTURE & SHIPBUILDING
ENGINEERING**

SEMESTER S3

MATHEMATICS FOR ELECTRICAL SCIENCE AND PHYSICAL SCIENCE – 3

(Common to B & C Groups)

Course Code	GYMAT301	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in complex numbers.	Course Type	Theory

Course Objectives:

1. To introduce the concept and applications of Fourier transforms in various engineering fields.
2. To introduce the basic theory of functions of a complex variable, including residue integration and conformal transformation, and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fourier Integral, From Fourier series to Fourier Integral, Fourier Cosine and Sine integrals, Fourier Cosine and Sine Transform, Linearity, Transforms of Derivatives, Fourier Transform and its inverse, Linearity, Transforms of Derivative. (Text 1: Relevant topics from sections 11.7, 11.8, 11.9)	9
2	Complex Function, Limit, Continuity, Derivative, Analytic functions, Cauchy-Riemann Equations (without proof), Laplace's Equations, Harmonic functions, Finding harmonic conjugate, Conformal mapping, Mappings of $w = z^2$, $w = e^z$, $w = \frac{1}{z}$, $w = \sin z$. (Text 1: Relevant topics from sections 13.3, 13.4, 17.1, 17.2, 17.4)	9

3	Complex Integration: Line integrals in the complex plane (Definition & Basic properties), First evaluation method, Second evaluation method, Cauchy's integral theorem (without proof) on simply connected domain, Independence of path, Cauchy integral theorem on multiply connected domain (without proof), Cauchy Integral formula (without proof). (Text 1: Relevant topics from sections 14.1, 14.2, 14.3)	9
4	Taylor series and Maclaurin series, Laurent series (without proof), Singularities and Zeros – Isolated Singularity, Poles, Essential Singularities, Removable singularities, Zeros of Analytic functions – Poles and Zeros, Formulas for Residues, Residue theorem (without proof), Residue Integration- Integral of Rational Functions of $\cos\theta$ and $\sin\theta$. (Text 1: Relevant topics from sections 15.4, 16.1, 16.2, 16.3, 16.4)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the Fourier transforms of functions and apply them to solve problems arising in engineering.	K3
CO2	Understand the analyticity of complex functions and apply it in conformal mapping.	K3
CO3	Compute complex integrals using Cauchy's integral theorem and Cauchy's integral formula.	K3
CO4	Understand the series expansion of complex function about a singularity and apply residue theorem to compute real integrals.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10 th edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Complex Analysis	Dennis G. Zill, Patrick D. Shanahan	Jones & Bartlett	3 rd edition, 2015
2	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39 th edition, 2023
3	Higher Engineering Mathematics	B.S. Grewal	Khanna Publishers	44 th edition, 2018
4	Fast Fourier Transform - Algorithms and Applications	K.R. Rao, Do Nyeon Kim, Jae Jeong Hwang	Springer	1 st edition, 2011

SEMESTER S3

FLUID MECHANICS FOR NAVAL ARCHITECTS

Course Code	PCNST302	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hr. 30 Mins
Prerequisites (if any)	Basic Calculus, Physics	Course Type	Theory

Course Objectives:

1. Understand basic fluid properties, their physical significance and units and to enable students appreciate that basic laws govern phenomena in the domain of the discipline.
2. Discern principles of fluid mechanics.
3. To help students know that an equation exists that can potentially solve all fluid problems.
4. To impart awareness that a ship's design calls for measures to countervail action of fluid forces.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fluids and liquids. Continuum. Liquids and their properties (density, specific weight, specific volume, specific gravity. viscosity), Newton's law of viscosity. Newtonian fluids, non-Newtonian fluids. Effect of temperature and pressure on viscosity, effect of viscosity on floating bodies, surface tension and its effect on ships, vapour pressure and cavitation, pressure at a point in a fluid, influence of depth of fluid on pressure, pressure measurement, hydrostatic forces on submerged bodies, metacentre, determination of metacentric height.	11

2	Fluid kinematics: Lagrangian method, Eulerian method, velocity and acceleration components of a fluid particle, types of flow and flow lines, conservation of mass, rate of flow, Continuity equation and its derivation, circulation and vorticity, velocity potential, stream function, relations between the functions. Fluid Dynamics: Euler's equation of motion, Bernoulli's equation for ideal and real fluid, application of BE to Venturimeter, momentum equation, moment of momentum equation.	11
3	Viscous flow of incompressible fluids, laminar flow, turbulent flow, Reynold's experiment, Reynolds number, Navier-Stokes equation, laminar flow through pipes, frictional loss in pipe for laminar and turbulent flows. Characteristics of turbulent flow, boundary layer approximation, boundary layer procedure applied to laminar boundary layer on a flat plate, displacement thickness, turbulent flat plate boundary layer, Momentum integral technique for boundary layers on flat plate.	11
4	External flow, drag and lift on foils, friction and pressure drag, flow separation, flow over flat plate, drag and friction coefficient, flow over cylinders, effect of surface roughness, application to drag on ship's hull, magnus effect. Six-dof motions of a floating body. CFD: Basics, introduction to solution of a simple problem	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
1. 2 Questions from each module. 2. Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	1. Each question carries 9 marks. 2. Two questions will be given from each module, out of which 1 question should be answered. 3. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To describe basic concepts in moderate width (what fluid/liquid is, fluid properties like density, specific gravity, specific weight with emphasis on viscosity, pressure)	K2
CO2	To identify hydrostatic forces on submerged surfaces and discuss buoyancy, stability and flotation of bodies in liquids	K2
CO3	To solve real engineering problems in fluid kinematics and kinetics and apply principles of mass, energy and momentum	K3
CO4	To interpret boundary layer, flow separation, and forces exerted by flowing fluid on a body and, 6-dof motions of floating vessel.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	1	1		1							
CO3	3	2	3		1							
CO4	3	2	1	1	1							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluid Mechanics: Fundamentals and Applications	Yunus A Cengel John M Cambala	Tata McGraw Hill Publishing Company Ltd: New Delhi	Edition 2, 2010
2	Fluid Mechanics	Frank M White	Tata McGraw Hill Publishing Company, Ltd: New Delhi	Edition 7, 2008
3	Fluid Mechanics, Hyrdraulic Machines	R. K Rajput	S Chand & Co Ltd	Edition 6, 2010
4	Fluid Mechanics and its Applications	Vijay Gupta	New Age International (P) Ltd Pub.	Edition 2, 2011
5	Ship Hydrostatics and Stability	Adrian Biran	Butterworth Heinemann	Edition 2, 2011
6	Mechanics of Fluids	Merie C Potter	Cengage Learning, N Delhi	Edition 5, 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanics of Fluids	Bernard Massey	Nelson Thomas	Edition 7, 1990
2	Engineering Fluid Mechanics	Graebel W. P.	Taylor and Francis	Edition 1 Indian Print, 2011
3	Fluid Mechanics	Piyush K Kundu, Ira M Cohen	Academia Press	Edition 4, 2008
4	Fluid Mechanics with Engineering Applications	J Frabzino	McGraw Hill	Edition 10, 2002

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/105103192 https://nptel.ac.in/courses/112105171
2	https://nptel.ac.in/courses/105103192 https://nptel.ac.in/courses/112105171
3	https://nptel.ac.in/courses/105103192 https://nptel.ac.in/courses/112105171
4	https://nptel.ac.in/courses/105103192 https://nptel.ac.in/courses/112105171

SEMESTER S3

SOLID MECHANICS FOR NAVAL ARCHITECTS

Course Code	PCNST303	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Engineering Mechanics	Course Type	Theory

Course Objectives:

1. Explain the behaviour of different types of solids made of different materials and having different cross-sections under various loading conditions
2. Explain the concepts of bending moment and shear stress in different types of beams
3. Explain the concepts of stresses in beams, beam deflection and buckling phenomenon in long columns
4. Explain the concept of torsion and buckling

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics in Mechanics of Solids: Types of Loads; Stress, Strain and their types. Stress-Strain Concepts: Hooke's Law; Stress-Strain relationship for mild steel- Elastic Limit; Yield point; Bars of varying sections; Bars of composite sections; Temperature Stresses; Poisson's Ratio; Working Stress; Factor of safety; Volumetric strain; Modulus of elasticity; Modulus of rigidity; Bulk Modulus; Relationship between the elastic constants.	11

2	Bending Moment and Shear Force: Definitions; Shear force and bending moment; Beams and Loads; Sign conventions; Shear force and Bending Moment Diagrams for Cantilever, simply supported and Overhanging beams with various types of loading (Point load, Distributed load, Couples).	11
3	Stresses in Beams: Simple bending; Theory of Simple Bending; Assumptions in Simple Bending; Neutral axis; Section Modulus; Flexural rigidity; Stresses in symmetrical sections; Bending Stress Distribution. Shear Stress: Shear stress distribution in beams; Assumptions; Shear stress distribution for rectangular, circular, triangular, I and T sections. Principal Stresses and Strains: Definition; Mohr's Circle.	11
4	Beam Deflection: Relation between curvature, slope and deflection. Double integration method. Theory of Columns: Introduction to columns; Buckling Theory; Euler's Formula for Long Columns; Assumptions and Limitations; Effect of End Conditions; Slenderness Ratio. Thin-Walled Pressure Vessels: Introduction; Biaxial Tension and Compression in Thin-Walled Pressure Vessels such as cylindrical and spherical. Torsion: Introduction to torsion; Theory of pure torsion; Torsion of Circular Shafts.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Assimilate the concepts on stress, strain, modulus of elasticity and the relationship between different elastic constants.	K1
CO2	Draw the bending moment diagram and shear force diagram for various types of beams under different boundary conditions and loading.	K2
CO3	Calculate the shear stress distribution under various cross –sections and determine principal stresses with the aid of equations and Mohr's circle.	K2
CO4	Conceive the concept on torsion and buckling and their applications.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1									
CO2	3	3										
CO3	3	2										
CO4	3	2										
CO5	2	2	2									

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Strength of Materials	Bansal R. K	Lakshmi Publications	Edition 4, 2007
2	Strength of Materials Part 1	Timoshenko S. P.	D. Van Nostrand Company Inc .	Edition 3, 2002
3	Strength of Materials	S. Ramamrutham	Dhanpat Rai Publishing Company	Edition 16, 2008
4	Engineering Mechanics of Solids	E P Popov	Prentice-Hall of India Pvt Ltd	Edition 2, 1998 (Reprint 2019)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Strength of Materials	S. S Bavikatti	Vikas Publishing House Pvt Ltd	Edition 4, 2014
2	Introduction to Solid Mechanics	Shames I. H., James M. Pitarresi	Pearson Education India	Edition 3, 2015
3	Analysis of Structures	V.N. Vazirani, M.M.Ratwan	Khanna Publishers	Edition 1, 2015
4	Mechanics of Materials	Punmia B. C. and A. K. Jain	Laxmi Publications (P) Ltd	Edition 1, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/105/104/105104160/
2	https://archive.nptel.ac.in/courses/105/104/105104160/
3	https://archive.nptel.ac.in/courses/105/104/105104160/
4	https://archive.nptel.ac.in/courses/105/104/105104160/

SEMESTER S3

SHIP STABILITY

Course Code	PBNST304	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hr. 30 Mins
Prerequisites (if any)	PCNST205	Course Type	Theory

Course Objectives:

1. Apply naval architectural principles to transverse stability of ships and estimate GZ values of ships
2. Apply naval architectural principles to longitudinal stability of ships and compute trim of ships for various loading conditions
3. Understand the concepts of floodable length and damage stability of ships
4. Check the compliance of ships with the requirements of regulatory bodies and Classification Societies

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Transverse stability: - Centre of gravity of ship, Estimation of BM_T and GM_T , Stability at small angles, Effect of addition, removal and transference (lateral and vertical) of weight, Effect of suspended weights, Free surface effect, Wall sided formula. Large angle stability - Statical stability curve, characteristics of GZ curve, effect of changes in ship's breadth, freeboard and superstructure on GZ curve. Cross curves of stability, Methods for calculating GZ (Krylov, Prohaska, etc.), Dynamical stability – diagram of dynamical stability	9

2	Longitudinal stability: – Trim, longitudinal metacentre, estimation of BM_L and GM_L , longitudinal centre of flotation, trimming moment; trim calculations – addition, removal and transference of weight, effect of change of density of water, fresh water allowance	9
3	Flooding and Damage stability: - Watertight integrity, Bilging, Permeability, Margin line, Flooding calculation, Floodable length, Factor of Subdivision, Permissible length, Curves of floodable length and permissible length, Compartment Standard, Damage Stability – Calculations by Lost Buoyancy and Added Weight Methods, Deterministic and Probabilistic Approach.	9
4	Stability of ships while docking and grounding, Inclining experiment, Heeling Moments due to wind, shift of cargo, passengers, turning and nonsymmetrical accumulation of ice. Stability in waves, Recommendations of IMO, Classification Societies and government authorities, Intact and damage stability rules. Stability criteria in weather condition. Stability booklet	9

Suggestion on Project Topic

Course Assessment Method (CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the transverse stability of ships and use standard methods to compute the GZ values of ships	K3
CO2	Explain the longitudinal stability of ships and compute trims of ships for various loading conditions	K3
CO3	Explain floodable length and damage stability calculations of ships.	K3
CO4	Analyse stability of ships and verify the compliance with rules/regulations /recommendations of regulatory bodies/Classification Societies.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2			1							
CO2	2	2			1							
CO3	2	2										
CO4	2	2	1		1				1	1		1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Naval Architecture	E. C. Tupper	Butterworth-Heinemann	Edition 5, 2013
2	Principles of Naval Architecture (Second Revision)	Edward V. Lewis (Editor)	The Society of Naval Architects and Marine Engineers	Edition 1, 1988
3	Basic Naval Architecture-Ship Stability	Philip A. Wilson	Springer	Edition 1, 2018
4	Ship Stability for Masters and Mates	C.B. Barrass & D.R. Derrett	Elsevier	Edition 7, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Basic Ship Theory	K. J. Rawson & E. C. Tupper	Butterworth-Heinemann	Edition 5, 2001
2	Ship Hydrostatics and Stability	Adrian B. Biran & Rubén López-Pulido	Butterworth-Heinemann	Edition 2, 2014
3	Contemporary Ideas on Ship Stability	D, Vassalos et. al	Elsevier	Edition 1, 2000
4	Statics and Dynamics of the Ship	V. Semyonov-Tyan-Shansky	Aalto University Publication Series, Science & Technology 6/2021	Edition 1, 2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105003
2	https://nptel.ac.in/courses/114105003
3	https://nptel.ac.in/courses/114105003
4	https://nptel.ac.in/courses/114105003

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. **Final Presentation (5 Marks)**

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. **Project Quality, Innovation, and Creativity (3 Marks)**

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	GNEST305	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Mins.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to AI and Machine Learning: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	11

2	Mathematical Foundations of AI and Data science: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	11
3	Applied Probability and Statistics for AI and Data Science: Basics of probability-random variables and statistical measures - rules in probability-Bayes theorem and its applications- statistical estimation-Maximum Likelihood Estimator (MLE) - statistical summaries- Correlation analysis-linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	11
4	Basics of Data Science: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, PowerBI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	11

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	K3
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems.	K3
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	K3
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								
CO2	3	3	3	3								
CO3	3	3	3	3								
CO4	3	3	3	3								
CO5	3	3	3	3								

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6 th edition, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Géron	O'Reilly Media, Inc.	2 nd edition, 2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1 st edition. 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9 th edition, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1 st edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2 nd edition, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1 st edition, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1 st edition, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1 st edition, 2019
5	Probability and Statistics -The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1 st edition, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome-extension://efaidnbmn nnibpcajpcglclefindm kaj/https://www.math. arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education,	6

	employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental	6

	Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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Course Assessment Method

(CIE: 50 marks, ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/ Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks			50	

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using

native plants and sustainable design.

- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S3

FLUID MECHANICS LAB

Course Code	PCNSL307	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST302	Course Type	Lab

Course Objectives:

1. Apply knowledge in the area of fluid mechanics and conduct experiments.
2. Understand with the arrangements and conduct of experiments in the fluid mechanics laboratory environment.
3. Comprehend the factors responsible for variation between theoretical and experimental results pertaining to the domain of fluid mechanics.
4. Record and analyse the results obtaining and interpret

Expt. No.	Experiments
1	Pressure measurement using U –Tube Manometer, Pitot Tube, Flow Meter, etc.,
2	Determination of Metacentric Height and Radius of Gyration of Floating bodies
3	Experimental Verification of Bernoulli's Theorem
4	Determination of Darcy's Constant and Chezy's constant for Pipe flow
5	Determination of Critical Velocity in Pipe flow
6	Determination of Minor losses in Pipe flow
7	Determination of type of flow using Reynolds Number
8	Determination of Hydraulic Coefficients of Orifices under Constant Head method
9	Determination of Hydraulic Coefficients of Orifices under Time of Emptying method
10	Calibration of Venturimeter
11	Calibration of Orificemeter
12	Calibration of Water meter
13	Study the use of different types of Valves and pipe fittings

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

5. Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.

6. Endorsement by External Examiner: The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify equipment and instruments used in determine fluid mechanics lab.	K1
CO2	Apply the theoretical Knowledge gained in the classroom and verify the same	K3
CO3	Compare different techniques and instruments for testing	K1
CO4	Record and analyse the results obtaining and interpret	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3			3			3			3		
CO2	3	1	1	3	1	1	3	1	1	3	1	1
CO3	2	2	2	2	2	2	2	2	2	2	2	2
CO4		3	3		3	3		3	3		3	3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluid Mechanics with Engineering Applications	J Frabzini	McGraw Hill	1997
2	Fluid mechanics and Machinery	Robert W Fox, Alan T McDonald, Philip J Pritchard.	John Wiley and sons	2015
3	Fluid Mechanics and Hydraulic Machines	Bansal R K	Lakshmi Publications	2011
4	Hydraulic and Fluid Mechanics including Hydraulic Machines	Modi P N and Seth S M	Standard book house New Delhi	Edition 20 2015
5	Fluid Mechanics	Frank M White	Tata McGraw Hill Publishing Company Ltd, New Delhi	Edition 7 2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluid Mechanics Fundamentals and Applications	Yunus A Cengner, John M. Cimbala.	McGraw Hill	2010
2	An introduction to Fluid dynamics	G K Batchelor	Cambridge University Press	1967
3	Fluid Flow hand book	Jamal M Saleh	McGraw Hill	2002
4	A working guide to process equipment	N P Lieberman, E T Lieberman	McGraw Hill	2022
5	Fluid Mechanics and its Applications	Vijay Gupta	New Age International (P) Ltd Pub.	Edition 2 2011

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/105103192 https://nptel.ac.in/courses/112105171
2	https://nptel.ac.in/courses/105103192 https://nptel.ac.in/courses/112105171
3	https://nptel.ac.in/courses/105103192 https://nptel.ac.in/courses/112105171
4	https://nptel.ac.in/courses/105103192 https://nptel.ac.in/courses/112105171

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S3

WELDING & MACHINE TOOLS LAB

Course Code	PCNSL308	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None / Course Code	Course Type	Lab

Course Objectives:

1. Apply knowledge in the area of ferrous and non ferrous welding
2. Prepare specified type of joint using various welding processes.
3. Understand with the arrangements and conduct of experiments in the welding and machine tool laboratory environment.
4. Prepare machined components for various engineering applications

Expt. No.	Experiments
1	Study of Precision tools and measuring instruments
2	Study of centre lathe
3	Performing various of operation in Lathe - Turning
4	Lathe - Grooving, Taper turning and Step cutting
5	Lathe - Thread cutting
6	Study of Machine tools - Shaper, Planner, Slotter, and Milling Machine
7	Study of different types of welding and ailed instruments.
8	Prepare a single V Butt joint using arc welding
9	Prepare a lap joint using arc welding
10	Prepare a butt joint using TIG welding
11	Prepare a butt joint using MIG welding
12	Study of Gas welding Equipment and operations

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

7. Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.

8. Endorsement by External Examiner: The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify different types of welding.	K2
CO2	Prepare specified type of joint using various welding processes.	K3
CO3	Understand with the arrangements and conduct of experiments in the welding and machine tool laboratory environment.	K1
CO4	Prepare machined components for various engineering applications	K1

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	2		1									
CO4	3											

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Text Book of Welding Technology	O P Khanna	Dhanpat Rai	Edition 2, 2015
2	The Welding of Aluminium and its Alloys	Gene Mathers	CRC Press/Woodhead Pub	Edition 1, 2015
3	Machine Tool Design	Acharkn N	McGraw Hill	Edition 4, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Workshop Technology	Chapman	CBS VOLI,II,III	2007
2	Workshop Technology	Hajra Choudary	Vol I , II	1988

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105004
2	https://nptel.ac.in/courses/114105004
3	https://nptel.ac.in/courses/114105004
4	https://nptel.ac.in/courses/112105233

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

1. Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
2. Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
3. Teamwork: Collaboration and participation in group experiments

3. Lab Reports and Record Keeping (6 Marks)

1. Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
2. Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

1. Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

7. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

1. Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
2. Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.

3. Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
4. Creativity and logic in algorithm or experimental design.

8. Conduct of Experiment/Execution of Work/Programming (15 Marks)

1. Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

9. Result with Valid Inference/Quality of Output (10 Marks)

1. Accuracy of Results: Precision and correctness of the obtained results.
2. Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

10. Viva Voce (10 Marks)

1. Ability to explain the experiment, procedure results and answer related questions
2. Proficiency in answering questions related to theoretical and practical aspects of the subject.

11. Record (5 Marks)

1. Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

**NAVAL ARCHITECTURE & SHIPBUILDING
ENGINEERING**

SEMESTER S4

MATHEMATICS FOR PHYSICAL SCIENCE– 4

Course Code	GCMAT401	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic calculus.	Course Type	Theory

Course Objectives:

1. To familiarize students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science.
2. To provide the students with the basics of various numerical methods to develop problem solving skills used in various engineering disciplines.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Binomial distribution, Poisson distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. [Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]	9
2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables. [Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1,5.2]	9

3	Confidence Intervals, Confidence Level, Confidence Intervals and One-side confidence intervals for a Population Mean for large and small samples (normal distribution and t -distribution), Hypotheses and Test Procedures, Type I and Type II error, z Tests for Hypotheses about a Population Mean (for large sample), t Test for Hypotheses about a Population Mean (for small sample), Tests concerning a population proportion for large and small samples. [Text 1: Relevant topics from 7.1, 7.2, 7.3, 8.1, 8.2, 8.3, 8.4]	9
4	Newton-Raphson Method, Gauss Elimination Method, Gauss - Jordan Method, Numerical solution of ordinary differential equations-Euler's method, Modified Euler's method, Runge - Kutta method of 2 nd Order, Numerical solution of Laplace equation –Jacobi's Method, Curve Fitting by Method of Least Squares - Straight lines, Parabola. (Text 2: Relevant topics from sections 2.5,4.2,7.5,8.4,8.5,9.4)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept, properties and important models of discrete random variables and to apply in suitable random phenomena.	K3
CO2	Understand the concept, properties and important models of continuous random variables and to apply in suitable random phenomena.	K3
CO3	Estimate population parameters, assess their certainty with confidence intervals, and test hypotheses about population means and proportions using z -tests and the one-sample t -test.	K3
CO4	Apply numerical methods to find solutions of linear system of equations, ordinary differential equations and Laplace equations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9 th edition, 2016
2	Introductory Methods of Numerical Analysis	S SSastry	PHI Learning Pvt Limited	5 th edition, 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability, Random Variables and Stochastic Processes,	Papoulis, A. & Pillai, S.U.,	McGraw Hill.	4 th edition, 2002
2	Introduction to Probability and Statistics for Engineers and Scientists	Ross, S. M.	Academic Press	6 th edition, 2020
3	Numerical methods for Engineers	Steven C. Chapra, Raymond P. Canale	McGraw Hill Education	8 th edition, 2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105085/
2	https://archive.nptel.ac.in/courses/117/105/117105085/
3	https://archive.nptel.ac.in/courses/117/105/117105085/
4	https://archive.nptel.ac.in/courses/111/107/111107105/

SEMESTER IV

SHIP RESISTANCE

Course Code	PCNST402	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand the flow around a ship and various components of resistance of ships.
2. Discern dimensional analysis in marine hydrodynamics and laws of comparison
3. Estimate effective power of ships using statistical / methodical series / model tests.
4. Understand the principles of operation of various high speed crafts and prediction of power requirement of such crafts.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Flow around a ship, ship wave system, Components of ship resistance, Viscous resistance, Viscous pressure resistance, Form factor, Hull roughness, Wave making resistance – pressure resistance, interference effects, wave breaking resistance, bulbous bows and their effects. Air and wind resistance, Resistance of appendages.	11
2	Dimensional analysis. Laws of comparison – geometrical, kinematical and dynamical similarity, Newton's, Froude's and Reynold's law, model-ship correlation.	11
3	Model testing – tank testing facilities, testing, prediction of resistance from model tests, extrapolation, Froude's concept, Determination of resistance from series test results - Guldhammer-Harvald's method, statistical prediction of resistance – Holtrop & Mennen method, effect of hull form on resistance	11

4	Added resistance in waves; Resistance in restricted waterways – resistance in shallow water, resistance in canals. Unconventional craft and power prediction - planning crafts, multi-hull vessels, hovercrafts/SES, hydrofoil craft	11
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Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

At the end of the course students should be able to:

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Resistance and Propulsion – Practical Estimation of Propulsive Power	Antony F. Moland, Stephen R. Turnock	Cambridge University Press	Edition 1, 2011
2	Principles of Naval Architecture series – Ship Resistance and flow	John Letcher, Randolph Paulling	SNAME, New Jersey	Edition 3, 2009
3	Resistance and Propulsion of Ships	Harvald S. A	John Wiley & Sons	Edition 1, 1983
4	Fundamentals of Ship Hydrodynamics – Fluid Mechanics, Ship Resistance and Propulsion	Lothar Birk	JohnWiley & Sons Ltd	Edition 1, 2019
5	Ship Resistance and Propulsion – Practical Estimation of Propulsive Power	Antony F. Moland, Stephen R. Turnock	Cambridge University Press	Edition 1, 2011
6	Principles of Naval Architecture series – Ship Resistance and flow	John Letcher, Randolph Paulling	SNAME, New Jersey	Edition 3, 2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mitigation of Hydrodynamic Resistance - Methods to Reduce Hydrodynamic Drag	Marc Perlin, Steven Ceccio	World Scientific Publishing Co. Pte. Ltd, Singapore	Edition 1, 2015
2	Basic Ship Theory Vol II	K J Rawson, E C Tupper	Butterworth-Heinemann Ltd	Edition 5, 2001
3	Practical Ship Hydrodynamics	Volker Bertram	Butterworth-Heinemann Ltd	Edition 1, 2000
4	Hydrodynamics of High-Speed Vehicles	O M Faltinsen	Cambridge University Press	Edition 1, 2005

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114106026
2	https://nptel.ac.in/courses/114106026
3	https://nptel.ac.in/courses/114106026
4	https://nptel.ac.in/courses/114106026

SEMESTER S4

ANALYSIS OF STRUCTURES

Course Code	PCNST403	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST303	Course Type	Theory

Course Objectives:

1. Enable to differentiate between a determinate and indeterminate structures and also to analyze continuous beams with and without settlement.
2. Perform step by step procedure involved in Moment Distribution Method, Accumulate knowledge on Principle of Virtual Work and basic theorems in structural analysis.
3. Understand the concepts of vibration of continuous system such as rod, string, beam, shaft, principles of plastic theory and its applications in structural analysis
4. Remember basic dynamics, understand the basic principles of structural dynamics and apply the same to simple structures.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to structural analysis, Determinate and Indeterminate structures, Static and Kinematic indeterminacy, Force and displacement method of analysis, Clapeyron's Theorem of Three Moments, Analysis of beams with different support condition and support settlement.	11
2	Moment Distribution method: Beams and Rigid Jointed Frames (with and without sway); Effect of support settlement. Strain Energy Methods: Principle of virtual work: Castigliano's Theorems, Clerk Maxwell's Reciprocal Theorem.	11
3	Vibrations Continuous Systems: Vibration of strings and rods; Vibration of beams: Vibration of shafts. Introduction to plastic analysis: Assumptions of plastic theory, Plastic	11

	Hinge, Shape Factor, Load Factor, Mechanism of Failure	
4	Structural dynamics: Introduction - degrees of freedom - equation of motion, D'Alembert's principle-dampingfree response of damped and undamped systems- logarithmic decrement-- single degree of freedom systems subjected to harmonic load - transient and steady state responses, simple portal frame problems	11

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Enable to differentiate between a determinate and indeterminate structures and also to analyze continuous beams with and without settlement.	K2
CO2	Perform step by step procedure involved in Moment Distribution Method, Accumulate knowledge on Principle of Virtual Work and basic theorems in structural analysis.	K1
CO3	Understand the concepts of vibration of continuous system such as rod, string, beam, shaft, principles of plastic theory and its applications in structural analysis	K2
CO4	Remember basic dynamics, understand the basic principles of structural dynamics and apply the same to simple structures.	K1

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	3	2			1							
CO3	2	2			1							
CO4	2	2										

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Structural Analysis –in theory and practice	Alan Williams	Butterworth-Heinemann	Edition 1, 2008
2	Basic Structural Analysis	C. S. Reddy	Tata McGraw-Hill	Edition 3, 2011
3	Structural Dynamics	Mario Paz	Springer	Edition 6, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Strength of Materials and Theory of Structures	Punmia B.C. and Jain A.K	Laxmi Publications (P) Ltd	Edition 2, 2005
2	Theory of Structures	Timoshenko & Young	McGraw Hill Publications	Edition 1, 1965
3	Theory of vibrations	V. P. Singh	Dhanpat Rai and Co (P) Ltd	2010
4	Dynamics of structures	Clough R.W. and Penzein	Tata McGraw Hill	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/105105166
2	https://nptel.ac.in/courses/105105166
3	https://nptel.ac.in/courses/101102090
4	https://nptel.ac.in/courses/105106151

SEMESTER S4

SHIP PROPULSION

Course Code	PBNST404	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST205	Course Type	Theory

Course Objectives:

1. Discern thrust development mechanism by screw propellers and its geometry.
2. Understand hull-propulsion interaction and resulting efficiencies and cavitation.
3. Be competent in the fundamentals of propulsion system design and propeller drawing as a project
4. Understand special types of propulsion systems and propeller strength.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Propeller as a thrust producing mechanism, Screw propeller, screw propeller geometry, sections, construction details. Propeller theories- Momentum theory, Blade element theory, Dimensional analysis	9
2	Interaction between Hull and propeller- Wake and wake fraction, Resistance augmentation and thrust deduction, propulsive efficiency in open water and behind conditions, hull efficiency, quasi propulsive coefficient, transmission efficiency; Powering Cavitation-Types, Cavitation Number, Effects of cavitation, Prevention of cavitation	9
3	Design of propellers-Propeller families and series; Open water tests, Presentation of data, Design charts- Kt-Kq, Bp- δ , T-J, P-J charts, Use of charts in propeller design including selection of prime movers and optimisation, propeller drawing. Design project report and drawing	9

	submission.	
4	Special types of propulsion systems - Shrouded propellers, Controllable Pitch propellers, Super cavitating propellers, Water jet propulsion, Vertical axis propellers, Sail, Paddle wheels, Electromagnetic propulsion etc. Ship standardisation trials. Propeller materials, propeller strength	9

Course Assessment Method
(CIE: 60 marks , ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks (8x2 =16marks)	• Each question carries 6 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 2 sub divisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Discern thrust development mechanism by screw propellers and its geometry.	K1
CO2	Understand hull-propulsion interaction and resulting efficiencies and cavitation.	K2
CO3	Be competent in the fundamentals of propulsion system design and propeller drawing as a project	K3
CO4	Understand special types of propulsion systems and propeller strength.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Basic Ship Propulsion	Ghose P. & Gokarn R.P	Knowledge World Publishers Pvt Ltd	Edition 1, 2015
2	Ship Resistance and Propulsion – Practical Estimation of Propulsive Power	Antony F. Moland, Stephen R. Turnock	Cambridge University Press	Edition 1, 2011
3	Marine Propellers and Propulsion	S Carlton	Elsevier Ltd.	Edition 4, 2019
4	Resistance and Propulsion of Ships	Harvald S. A	John Wiley & Sons	Edition 1, 1983
5	Fundamentals of Ship Hydrodynamics - Fluid Mechanics, Ship Resistance and Propulsion	Lothar Birk	JohnWiley & Sons Ltd	Edition 1, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Naval Architecture series – Propulsion	Justin E K, Jacques B H	SNAME, New Jersey	Edition 3, 2010
2	Basic Ship Theory Vol II	K J Rawson, E C Tupper	Butterworth-Heinemann Ltd	Edition 5, 2001
3	Practical Ship Hydrodynamics	Volker Bertram	Butterworth-Heinemann Ltd	Edition 1, 2000
4	Hydrodynamics of High-Speed Vehicles	O M Faltinsen	Cambridge University Press	Edition 1, 2005

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105047
2	https://nptel.ac.in/courses/114105047
3	https://nptel.ac.in/courses/114105047
4	https://nptel.ac.in/courses/114105047

PBL Course Elements			
L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity		
Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
	Total	30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S4

MECHANICAL ENGINEERING FOR SHIP DESIGN

Course Code	PENST411	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GCEST104 Part 1	Course Type	Theory

Course Objectives:

1. Understand the materials used in ship construction and for the main and auxiliary machinery systems onboard.
2. Study the material properties and heat treatment
3. Have an overview on thermodynamics and basic theories of heat transfer
4. Understand the various IC engines and their working.
5. Select a best suited engine for a particular application

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Materials for shipbuilding. Metallic materials and their properties. Production of shipbuilding steel. Various grades. Iron carbon diagram, TTT diagram, Heat treatment, Protection from corrosion. Ferrous and non ferrous alloys Non-metallic materials - Composites, Ceramics, Rubber and polymer.	9
2	Material testing. Destructive and non-destructive Cold and hot working of Metals and alloys. Failure of components and failure analysis. Stress strain curve. Elements of fracture mechanics. Griffiths crack theory. Fatigue Crack propagation. Selection of materials for various ship building applications.	9
3	Modes of heat transfer, Basics of Conduction, Convection and Radiation. Heat exchangers, different types, Performance of heat exchangers, Aspects of heat exchanger design.	9

4	Internal Combustion Engine. CI and SI engines, Performance evaluation of IC Engines, Testing of IC Engines and Heat balance. Multi cylinder engines Morse Test, Marine Engines.Engine Lubrication, Cooling System, Ignition system and Fuels	9
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Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Able to explain the various types of shipbuilding materials and their specifications	K1
CO2	Identify best suited material for the construction of structure and components of a ship. Also for the main and auxiliary machinery systems onboard	K2
CO3	Understand basic features of thermal systems of a ship using fundamental principles of thermodynamics.	K2
CO4	Recognize various types of IC engines used in ship	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	1	1		1							
CO3	3	2	3		1							
CO4	3	2	1	1	1							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Material Science and Engineering	P S Gill	S K Kataria & Sons	Edition 1, 2015
2	Strength of Materials	Sadhu Singh	Khanna Publishers	Edition 9, 2009
3	Heat and Mass Transfer	D S Kumar	S K Kataria & Sons	Edition 9, 2015
4	Fundamentals of Metal Fatigue Analysis	Julie A B, Jess J C, James L H	Prentice Hall	Edition 1, 1990

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Deformation and Fracture Mechanics of Engineering Materials	Richard W Hertzberg	John Wiley & Sons	Edition 4, 1995
2	Metal Fatigue in Engineering	H O Fuchs R I Stephens	John Wiley & Sons	Edition 1, 1980
3	Practical Fracture Mechanics in Design	Alexander Blake	Marcel Dekker, Inc.	Edition 1, 1996
4	Thermal Engineering	A S Sarao	Satya Prakashan	Edition 4, 1987

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://dituniversity.digimat.in/nptel/courses/video/114105029/ https://nptel.ac.in/courses/114105029
2	https://nptel.ac.in/courses/114105029
3	https://nptel.ac.in/courses/113107078
4	https://nptel.ac.in/courses/113107078

SEMESTER S4

MARINE POLLUTION, CONTROL AND RECOVERY SYSTEMS

Course Code	PENST412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand different sources of marine pollution including ships and floating structures, their impact on the marine environment.
2. Select suitable methods to control the marine pollution in different operating conditions
3. Apply various strategies and methods for marine pollution recovery to minimise environmental impact.
4. Understand role of various international and regional rules and regulations for marine environmental protection in design of ships.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definition of Marine Pollution, Types of Marine Pollution, Various Sources of Marine Pollution, Marine Pollution in the Coastal Zone, Pollution from ships and offshore assets Concerns and Consequences of Marine Pollution - Global Warming, Sea Level Changes, Carbon Emissivity and Green Shipping Biodiversity	9
2	Prevention of air and water pollution from Ships, Emissions from Ships Engines, Fuel Oil Quality, Requirements for Survey and Issuance of International Air Pollution Prevention Certificate (IAPP), Oil Pollution, Segregated Ballast Tanks, Oily Water Separator, Ballast Water Management (BWM) Convention	9
3	Introduction to Marine Oil Pollution Recovery Systems: Types of Recovery Systems, Skimming Systems, Oil storage system, Treatment of pollutant	9

	after recovery.	
4	IMO Marine Engine Regulations, Requirements for Survey and Issuance of International Air Pollution Prevention Certificate (IAPP), Energy Efficiency Design Index (EEDI), Ship Energy Efficiency Management Plan (SEEMP), Survey and Issuance of International Sewage Pollution Prevention Certificates (ISPP)	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand different sources of marine pollution including ships and floating structures, their impact on the marine environment.	K1
CO2	Select suitable methods to control the marine pollution in different operating conditions	K2
CO3	Apply various strategies and methods for marine pollution recovery to minimise environmental impact.	K2
CO4	Understand role of various international and regional rules and regulations for marine environmental protection in design of ships.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3						2					2
CO2	3	1	1		1							2
CO3	3	2	3		1							2
CO4	3	2	1	1	1							2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Marine Pollution	Judith S Weis	Oxford University Press	Edition 1, 2014
2	Marine Pollution	R.B. Clark, C. Frid, M. Attrill	Oxford University Press	Edition 4, 1997
3	Marine Pollution	Icardo Beiras	Elsevier	Edition 1, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The International Convention for the Prevention of Pollution from Ships (MARPOL) 1973		International Maritime Organisation	As Amended
2	The International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004		International Maritime Organisation	As Amended
3	Resolution MEPC.378(80) - Guidelines for the control and management of ships' biofouling to minimize the transfer of invasive aquatic species		International Maritime Organisation	As Amended
4	The Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships, 2009		International Maritime Organisation	As Amended

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/103107215
2	https://nptel.ac.in/courses/103107215
3	https://nptel.ac.in/courses/103107215
4	https://nptel.ac.in/courses/103107215

SEMESTER S4

INLAND WATER TRANSPORTATION

Course Code	PENST413	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST205	Course Type	Theory

Course Objectives:

1. Discern the national regulations, different components of inland water transport systems, different types of vessels and their characteristics for National waterways in India.
2. Identify the hull forms suitable in inland water transport and understand standard procedures and formulae to estimate the stability, resistance and propulsion characteristics of inland vessels.
3. Understand the general arrangement of inland vessels and various equipment, systems and outfit items on board
4. Identify the materials used for inland vessel construction and understand the methods and standards of construction of inland vessel.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	The Inland Vessels Act 2021, Inland Waterways, their importance and characteristics, Components of Inland Water Transport, Inland Water Authority of India, Classification of Inland Waterways, Kerala Inland Vessels Rules. Vessel types - self-propelled vessels and dumb vessels, Intermodal transportation with sea, road and rail	9
2	Dimensional Restrictions of waterways, bridges, bends, locks and gates, design using empirical relations, development of hull forms - chine and round bilge, multihull, weight estimation, stability, heel test. Resistance and propulsion of inland vessels, shallow water effect - determination of shallow water resistance. special features – tunnels, shrouded propeller.	9

3	General arrangement, Passenger accommodation, cargo handling & equipment, onboard systems – piping systems, fire fighting appliances, life saving appliances. Super structure arrangements, mooring and anchoring.	9
4	Structural design – Class rules for inland vessels, local rules and KIV Act 2021, structural design and analysis of bottom, sides, decks, bulkheads etc, materials of construction, methods of construction and production technologies.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Discern the national regulations, different components of inland water transport systems, different types of vessels and their characteristics for National waterways in India.	K1
CO2	Identify the hull forms suitable in inland water transport and understand standard procedures and formulae to estimate the stability, resistance and propulsion characteristics of inland vessels.	K2
CO3	Understand the general arrangement of inland vessels and various equipment, systems and outfit items on board	K2
CO4	Identify the materials used for inland vessel construction and understand the methods and standards of construction of inland vessel.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	1	1		1							
CO3	3	2	3		1							
CO4	3	2	1	1	1							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of Contemporary Inland Waterway Vessels	Dejan R, Aleksandar S, Nikola M, Milorad M, Benjamin F	Springer Nature, Switzerland	Edition 1, 2021
2	Rules & Regulations for the Construction and Classification of Inland Waterways Vessels – July 2022		Indian Register of Shipping	Edition 4, 2022
3	River – Sea Vessel notification, Directorate General Shipping, Govt of India		Govt of India	2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Kerala Inland Vessel Rules, 2021		Directorate of Ports, Govt. of Kerala.	Revision 1, 2021
2	Recommendations on Harmonized Europe- Wide Technical Requirements for Inland Navigation Vessels, Resolution No. 61		Economic Commission for Europe, Inland Transport Committee, United Nations	2011
3	Future Challenges for Inland Navigation	Christa sys, Therry Vanelslander	University press Antwerp	2011

SEMESTER S4

SHIPBUILDING MATERIALS, CORROSION PREVENTION AND PROTECTION

Course Code	PENST414	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Discern the Evolution of Shipbuilding materials
2. Understand types of corrosion of ship structures
3. Discern effective methods of material preservation and corrosion prevention
4. Understand the measures to be taken in design stage to minimise corrosion in ships.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to different types of materials used in shipbuilding, Material fabrication and service requirement, Classification society requirements, Types of shipbuilding quality steels, Mild steels, normal strength steels (A, B, D, E classes), High Tensile Steels (HTS) grades, High Strength Low Alloy (HSLA), Aluminium alloys, alloy designation, welding requirements, Strength of aluminium compared to steel, Composition of aluminium alloys used in ship building, Advantages of using aluminium over steel in ship building.	9
2	Types of Corrosion in Marine Environment, Corrosion Identification, Factors affecting Corrosion. Types of steel and tendency of corrosion in various types of steel, corrosion of Titanium and Nickel alloys, Copper and Copper based alloys, Zinc, Aluminium and its alloys. Corrosion on propeller, marine machinery and deck fittings.	9

3	Material storage and preservation in shipyard, treatment of steel in shipyards. degreasing, weathering, mechanical surface cleaning, pickling, blast cleaning, flame cleaning. rust converters, chemical pre-treatment and comparison of pre-treatment methods. Paint schemes in ships. classification of paints- common paint vehicles, drying oils, oleo-resins, alkyd resins, polymerizing chemicals and bitumen. Role of constituents of paints. suitability of each for various applications. typical paint schemes for underwater areas, boot top above water, weather decks, superstructures and tanks interiors	9
4	Corrosion control - importance of corrosion protection, measures to minimise corrosion, corrosion control by design, corrosion inhibitors. cathodic protection- mechanism of cathodic protection, sacrificial anode, design of sacrificial anode system for ships, advantages and disadvantages of sacrificial anode system. impressed current cathodic protection system in ships	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Discern the evolution of shipbuilding materials, and transition from wood to modern materials.	K1
CO2	Differentiate various types of corrosion of ship structures	K2
CO3	Implement effective methods of material preservation and corrosion prevention	K2
CO4	Use the measures to be taken in design stage to minimise corrosion in ships.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	1	1		1							
CO3	3	2	3		1							
CO4	3	2	1	1	1							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Material Science and Engineering	P.S.Gill	S K Kataria & Sons	Edition 1 2015
2	Marine Corrosion and Cathodic Protection	Chris Coogan	CRC Press	
3	Corrosion Control for Offshore Structures	Ramesh Singh	Gulf Publishing Company	Edition 1, 2014
4	Corrosion Engineering	Fontana M. G, Greene N. D	McGraw Hill,	Edition 2, 1978.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Corrosion and Protection of Steels in Marine Environments: State-of-the-Art and Emerging Research Trends	Philippe Refait Igor Chaves (Eds)	MDPI	Edition 1, 2022
2	Corrosion of Constructional Steels in Marine and Industrial Environment	Jayanta Kumar Saha	Springer	Edition 1, 2013
3	Designing Cathodic Protection Systems for Marine structures and vehicles	Harvey P Hack	SNAME	2000
4	Developments in Marine Corrosion	A Campbell, N. Campbell, F C Walsh	Royal Society of Chemistry	1998

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/113101098
2	https://nptel.ac.in/courses/113101098
3	https://nptel.ac.in/courses/113101098
4	https://nptel.ac.in/courses/113107078

SEMESTER S4

SHIP CONVERSION TECHNOLOGY

Course Code	PENST415	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST205 PBNST304	Course Type	Theory

Course Objectives:

1. Discern the basic concepts about conversion technology of ships
2. Illustrate the engine room layout, outfit, revised length original length etc
3. Design the modified propeller and outfit items
4. Select the main engine based on the revised powering calculations and revised shaft length

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic ship parameters including existing ship stability condition, principal particulars of the ship, general arrangement drawing of the ship. Fixation of the revised dimensions based on the cargo capacity and redesignated functions, layout general arrangement, including engine room lay out, deck plans, water tight bulk head arrangements, fore peak and aft peak arrangements, anchor and anchor handling arrangements, computation new displacement.	9
2	Form coefficients for new dimensions, mid ship section design, revised functions of stability criteria, cross curves stability, statical stability curves, damaged stability, sea keeping manoeuvring, rudder design, steering gear arrangement etc.	9

3	Shape of sectional area curve, bow and forward section forms, bulbous bow design, stern forms, conventional method of lines design, propeller clearances, propeller design, model testing, structural design, outfitting production (hull), outfitting production (machinery).	9
4	Selection of main engine based on the revised powering and shaft calculations, hull-propeller interaction, ship resistance under trial conditions, additional resistance under service condition, rudder design.	9

Course Assessment

Method (CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation(Evaluate and Analyse): 20 marks

Criteria for evaluation:

1. Problem Definition (K4 - 4 points)

- a. Clearly defines the conversion problem.
- b. Examine and identifies relevant contextual factors (constraints, resources, objectives).

2. Problem Analysis (K4 - 4 points)

- a. Break-down and presents a well-reasoned solution approach based on SWBS.
- b. Compare and justify the proposed solutions with technical evidence and financial reasoning.

3. Evaluate (K5 - 4 points)

- a. Thoroughly evaluate the proposed solutions for ship conversion.
- b. Compares trade-offs, advantages, and disadvantages.

c. *Considers feasibility and practical implications.*

4. Design Implementation (K5 - 4 points)

a. *Select the most feasible solution by implementing the proposed solutions.*

b. *Successfully translates the chosen solution into drawings and documents.*

5. Conclusion (K4- 2 points, K5 – 2 points)

a. *Summarizes findings and insights. State which solution is most appropriate for the problem. (K4)*

b. *Reflects critical thinking and informed decision-making. (K5)*

Scoring:

1. ***Accomplished (4 points):*** *Exceptional analysis, clear implementation, and depth of understanding.*

2. ***Competent (3 points):*** *Solid performance with minor areas for improvement.*

3. ***Developing (2 points):*** *Adequate effort but lacks depth or clarity.*

4. ***Minimal (1 point):*** *Incomplete or significantly flawed.*

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Discern the basic concepts about conversion technology of ships	K1
CO2	Illustrate the engine room layout, outfit, revised length original length etc	K2
CO3	Design the modified propeller and outfit items	K4
CO4	Select the main engine based on evaluation of the revised powering calculations and revised shaft length	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	3							
CO2	3	3		2	3							
CO3	3	3	3	2	2							
CO4	3	3		2	3							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Design and Construction	Thomas Lamb (Ed.)	The Society of Naval Architects and Marine Engineers	Edition 1, 2003
2	Elsevier Ocean Engineering Book Series- Volume 1 (Practical Ship Design)	D. G. M. Watson	Elsevier	Edition 1, 1998
3	Ship Design- Methodologies of Preliminary Design	Apostolos Papanikolaou	Springer	Edition 1, 2014
4	Ship Repairing - Analyses and Estimates	Arun Kr Dev, Makaraksha Saha, George Bruce	Springer Series on Naval Architecture, Marine Engineering, Shipbuilding and Shipping Vol 12	Edition 1, 2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Maritime Engineering Reference Book	Anthony F. Molland (Ed)	Elsevier	Edition1, 2008
2	Design Principles of Ships and Marine Structures	S. C. Misra	Taylor & Francis	Edition 1, 2016
3	Computational Ship Design	Myung-II Roh& Kyu-Yeul Lee	Springer	Edition 1, 2018
4	Ship Construction and Welding	Nisith R. Mandal	Springer Series on Naval Architecture, Marine Engineering, Shipbuilding and Shipping 2	Edition 1, 2017

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. <table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Cash flow</td><td>1000</td><td>900</td><td>800</td><td>700</td><td>600</td></tr></table> (In Crores)	Year	1	2	3	4	5	Cash flow	1000	900	800	700	600	CO4	(5)
	Year	1	2	3	4	5										
Cash flow	1000	900	800	700	600											
b)	Point out any three merits of NPV method.	CO4	(3)													

SEMESTER S3/S4
ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	6
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories	6

	(anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	6

Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/ Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S4

STRENGTH OF MATERIALS LAB

Course Code	PCNSL407	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST303	Course Type	Lab

Course Objectives:

1. Apply knowledge in the area of testing of materials and components of structural elements experimentally.
2. Understand with the arrangements and conduct of experiments in the Material Testing laboratory environment.
3. Able to comprehend the factors responsible for variation between theoretical and experimental results pertaining to the domain of Material Science.
4. Undertake the testing of materials when subjected to different types of loading.

Expt. No.	Experiments
1	Tests on Open Coiled Spring
2	Tests on Close Coiled Spring
3	Bending Test on Wooden Beams Using U. T. M.
4	Verification of Clerk Maxwell's Law of Reciprocal Deflection
5	Torsion Pendulum test for MS Wires
6	Torsion Pendulum test for Aluminium Wires
7	Torsion Pendulum test for Brass Wires
8	Tension Test Using on MS rod
9	Torsion Test on M S rod

10	Impact Test Using Izod Apparatus
11	Impact Test Using Charpy Apparatus
12	Double Shear Test on MS rod

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply knowledge in the area of testing of materials and components of structural elements experimentally.	K3
CO2	Understand with the arrangements and conduct of experiments in the Material Testing laboratory environment.	K1
CO3	Able to comprehend the factors responsible for variation between theoretical and experimental results pertaining to the domain of Material Science.	K1
CO4	Undertake the testing of materials when subjected to different types of loading.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1						1			
CO2	2	2	1						1			
CO3	2	1	1						1			
CO4	2	2	2						1			

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Strength of Materials	Bansal R. K	Lakshmi Publications	Edition 4, 2007
2	Strength of Materials Part 1	Timoshenko S. P.	D. Van Nostrand Company Inc .	Edition 3, 2002
3	Strength of Materials	S. Ramamrutham	Dhanpat Rai Publishing Company	Edition 16, 2008
4	Engineering Mechanics of Solids	E P Popov	Prentice-Hall of India Pvt Ltd	Edition 2, 1998 (Reprint 2019)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Strength of Materials Lab Manual	Anand Jayakumar A	Notion Press	2020
2	Applied Strength of Materials for Engineering Technology	Barry Dupen	Indiana University – Purdue University Fort Wayne	2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/105103192 https://nptel.ac.in/courses/112105171
2	https://nptel.ac.in/courses/105103192 https://nptel.ac.in/courses/112105171
3	https://nptel.ac.in/courses/105103192 https://nptel.ac.in/courses/112105171
4	https://nptel.ac.in/courses/105103192 https://nptel.ac.in/courses/112105171

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S4

MEASUREMENTS LAB

Course Code	PCNSL408	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Lab

Course Objectives:

1. Acquire knowledge in the field of measurement
2. Provide an exposure to the fundamentals of metrology
3. Understand the need of precision measurement and measuring instruments
4. Find out the possible errors in measurement

Expt. No.	Experiments
1	Measurement of tool angles of single point cutting tool using tool makers microscope
2	Measurement of thread parameters using profile projector
3	Evaluation of straightness error using autocollimator, spirit level, straight edge etc.,
4	Calibration and determination of uncertainties of Tachometers and stroboscopes
5	Calibration of temperature using thermo- couple equipment
6	Experiments on limit and fits.
7	Measurement of vibration using vibration analyzer
8	Study of different types of dial indicators - stands and holders for dial gauges
9	Study and determination of area using planometer
10	Polishing, etching and determination of grain size and micro structure studies using optical microscope
11	Study and use of different types of comparators.
12	Angular measurement using Sine bar

Course Assessment Method

(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify equipment and instruments used in measurements lab.	K1
CO2	Apply the theoretical Knowledge gained in the classroom and verify the same	K3
CO3	Compare different techniques and instruments for measurement	K1
CO4	Find out the possibility of error in measurement	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1						1			
CO2	2	2	1						1			
CO3	2	1	1						1			
CO4	2	2	2						1			

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Practical engineering metrology	SharpKWB	Sir Issac Pitman and Sons Ltd.,	First edition 1970
2	Fits Tolerance and engineering measurements volume III	Tarasevigh Y and Yavosih E	Forgn language publishing house	1963
3	Theory and design of mechanical measurements	Figiliola, Richard S and Beasley Donald E	John Wiley and sons	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Measurements	Collett C V and Hope A D	ELBS/Longman	2011
2	Metology for engineers	Shotbolt C R Gayler J F W	ELBS	1990

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

**NAVAL ARCHITECTURE & SHIPBUILDING
ENGINEERING**

SEMESTER S5
SHIP DYNAMICS

Course Code	PCNST501	CIE Marks	40
Teaching Hours/Week (L: T: P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST205, PCNST302	Course Type	Theory

Course Objectives:

1. To introduce regular wave theories and wave spectra to the students and to enable them to explain the ship motions in regular waves.
2. To explain ship motions in irregular waves and ship motion control using stabilisers.
3. To explain the concept of motion stability associated with ships and standard manoeuvres, and to illustrate the heel during turn.
4. To describe the methods for experimental determination of hydrodynamic derivatives and to enable the students to understand the design process for a ship's rudder

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to seakeeping, wind generated waves. Sinusoidal and trochoidal wave theories. Wave spectrum, types of spectra. Ship motions in regular waves, equations of motion, ship-wave encounter, strip theory.	11
2	Ship in a seaway, pitch and roll in irregular waves, Response Amplitude Operator (RAO). Ship motion control, control of roll and pitch, active and passive Stabilizers. Derived responses, local motions, relative motions, slamming, deck wetness, sea-sickness, added resistance, powering in waves, wave loads.	11
3	Introduction to manoeuvrability, the control loop, path keeping, various types of motion stability. Basic hydrodynamics and equations of motion of a manoeuvring body, control-fixed stability indices. Effect of hull configuration on controls-fixed stability. Turning circle manoeuvre, heel and speed loss during turn, zig-zag manoeuvre, spiral manoeuvre, pull-out	11

	manoeuvre. IMO Manoeuvring Standards.	
4	Experimental determination of hydrodynamic derivatives, straight line test, rotating arm technique, Planar Motion Mechanism (PMM) technique. Rudder geometry, rudder forces and moments, flow around rudder. Rudder design process, rudder types, location, size, section shape, and number of rudders, maximum rudder deflection angle. Other manoeuvring devices.	11

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the regular wave theories, wave spectra, and equations of motions of a ship in regular waves and determine wave characteristics, using the wave theories, and natural period of ship motions.	K2
CO2	Describe ship motions in irregular waves and the various methods of ship motion control using stabilisers.	K2
CO3	Describe the concept of motion stability associated with ships, explain the standard manoeuvres used to assess the manoeuvrability and compute angle of heel during turn.	K2
CO4	Describe the methods for experimental determination of hydrodynamic derivatives and predict the particulars of a rudder for a ship type.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											
CO2	3	2	1	1	1		1		1			1
CO3	3	2	1	1	1		1	1	1			
CO4	3	2	1		1	1	1		1	1		1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Naval Architecture	Tupper, E.C.	Butterworth-Heinemann	Edition 5, 2013
2	Principles of Naval Architecture (Second Revision)	Edward V. Lewis (Editor)	The Society of Naval Architects and Marine Engineers	Edition 1, 1988
3	Dynamics of Marine Vehicles	Rameswar Bhattacharya	Wiley-Interscience	Edition 1, 1978
4	Marine Rudders and Control Surfaces- Principles, Data, Design and Applications	Anthony F. Molland & Stephen R. Turnock	Elsevier	Edition 1, 2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Basic Ship Theory	K. J. Rawson & E. C. Tupper	Butterworth- Heinemann	Edition 5, 2001
2	Stability and Motion Control of Ocean Vehicles	Martin A. Abkowitz	The Massachusetts Institute of Technology	Edition 1, 1969
3	Practical Ship Hydrodynamics	Volker Bertram	Elsevier	Edition 2, 2012
4	Design Principles of Ships and Marine Structures	S. C. Misra	Taylor & Francis	Edition 1, 2016
5	Guide for Vessel Maneuverability	American Bureau of Shipping	American Bureau of Shipping	2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105006
2	https://nptel.ac.in/courses/114105006
3	https://nptel.ac.in/courses/114105006
4	https://nptel.ac.in/courses/114105006

SEMESTER S5

SHIP STRUCTURES I

Course Code	PCNST502	CIE Marks	40
Teaching Hours/Week (L: T: P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST303	Course Type	Theory

Course Objectives:

1. Understand the properties of various shipbuilding materials, the framing systems and the structural topology of ships.
2. Discern the major loads on the hull, shear lag and effects and understand classification society rules.
3. Apply classification society rules to finalise the scantlings of ship structure to meet longitudinal and transverse strength requirements and draw midship section and shell expansion.
4. Understand the theory of plates and shells applicable to marine structures.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Shipbuilding materials; wood; steel, properties of steel, stress-strain diagrams; higher tensile steel – properties and application; Aluminium alloys – properties and applications, proof stress; non metallic materials – GRP and FRP. Framing systems, longitudinal framing, transverse framing, combined framing, basic structural components, primary stiffeners, secondary stiffeners, bottom structure, side shell, decks and superstructure, bulkheads, fore end and aft end structures.	10
2	Loads acting on ship structures in still water, shear lag and effective breadth, classification society rules and their application to ship structural design.	10
3	Considerations in ship structural design, design of midship section of the assigned vessel, development of shell expansion drawing.	13

4	3-D Modelling of ship structures, different types of plates and shells, Kirchoff's theory, small deflection analysis of thin plates, transverse and combined loads, rectangular simply supported plates with lateral loading, Navier solution for simply supported rectangular plates, large deflection analysis buckling of thin plates- equilibrium method, analysis of stiffened plates, buckling of stiffened plates.	11
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Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the properties of various shipbuilding materials, the framing systems and the structural topology of ships.	K1
CO2	Discern the major loads on the hull, shear lag and effects and understand classification society rules.	K2
CO3	Apply classification society rules to finalise the scantlings of ship structure to meet longitudinal and transverse strength requirements and draw midship section and shell expansion.	K2
CO4	Understand the theory of plates and shells applicable to marine structures.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											
CO2	3	2	1	1								1
CO3	3	2	1	1	1	1	1	1	1	1		1
CO4	3	2	1		1							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Structural Analysis and Design	O F Hughes, Jeom Kee Paik	The Society of Naval Architects and Marine Engineers	Edition 1, 2010
2	Torsion and Shear Stresses in Ships	M Shama	Springer	Edition 1, 2010
3	Local Strength of Ship Structures	Piero Caridis	CRC Press	Edition 1, 2024
4	Ship Structural Design Concepts	J Harvey Evans	Cornel Maritime Press	Edition 2, 1983
5	Buckling of Ship Structures	M Shama	Springer	Edition 1, 2013
6	Ship Construction	D J Eyres	Butterworth Heinemann	Edition 7, 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Rules for classification of ships by various class societies, updated from time to time		LR, ABS, BV	Latest
2	Strength of Ship Structures	W Muckle	Edward Arnold Publishers Ltd	Edition 2, 1967
3	Strength of Ships and Ocean Structures, The Principles of Naval Architecture Series	Alaa Mansour	The Society of Naval Architects and Marine Engineers	Edition 3, 2008
4	Elementary Beam Theory and the Hull Girder	G W White	Standard Maritime London	Edition 1, 1974

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105031
2	https://nptel.ac.in/courses/114105031
3	https://nptel.ac.in/courses/114105031
4	https://nptel.ac.in/courses/114105031

SEMESTER S5
MARINE ENGINEERING

Course Code	PCNST503	CIE Marks	40
Teaching Hours/Week (L: T: P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GCEST104	Course Type	Theory

Course Objectives:

1. Impart basic knowledge on marine engineering useful for a naval architect and get a broad understanding of the layout of Engine Rooms.
2. Familiarize with various marine machinery used onboard ships.
3. Familiarize with various systems used onboard ships.
4. Get an understanding of the shafting system, steering gear and stabilisers.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Role of marine engineers; need of knowledge on marine engineering for a naval architect; Joint responsibilities of marine engineer and naval architect during design, survey and repair of vessels; IMO/MARPOL regulations. Engine room layout: Layout of main and auxiliary machinery in engine rooms of different ships.	9
2	Types and application of main propulsion system: Main engine- diesel engine, steam turbine, electric propulsion; diesel engine components; scavenging; supercharging; starting & reversing; hazards and maintenance of diesel engine; reduction gearing. Auxiliary machinery: Requirement of auxiliary machinery, list & functions; differentiate auxiliary engines from main engines; working principle and onboard applications of different types of marine air compressors, marine pumps, marine boilers, purifiers, marine valves, heat exchangers, oil water separators, incinerators, sewage treatment plant, fresh water generator.	9

3	Marine Piping and major marine Systems: Piping arrangement for steam; bilge; ballast and oil fuel systems; lube oil and cooling system with various fittings; domestic fresh water and sea water hydrophore system. Deck Machinery: Deck cranes; windlasses; mooring winches; anchors and anchor chains; davits and lifeboats lowering mechanism; cargo handling equipment.	9
4	Shafting and Propellers: stern tubes and glands, oil lubricated stern tubes; shaft seals; shaft alignment; thrust block; propeller types and construction details; maintenance and operation of marine propellers. Steering Gear: Operation and constructional details of various types of steering machinery. Ship Stabilizers: Types, operations of different types of passive and active ship stabilizing systems.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the relation between marine engineering and naval architecture, the IMO/MARPOL regulations and layouts of different engine rooms.	K1
CO2	Demonstrate knowledge of various marine machinery and systems onboard ships.	K2
CO3	Understand application of various marine auxiliary machinery onboard ships.	K2
CO4	Understand the working of shafting, propellers and steering systems.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											
CO2	3	2	1		1		1					1
CO3	3	2	1				1					
CO4	3	2	1			1	1					1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Marine Engineering	Edited by Roy L Harrington	SNAME, New York.	1992
2	Introduction to Marine Engineering	Taylor D.A	Butterworth-Heinemann, London.	Second Edition 1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles & Practices of Marine Diesel Engines	D.K. Sanyal	A Bhantarkar Pub.,	1981
2	Low Speed Marine Diesel	John B Woodward	Wiley	First Edition 1981
3	Marine Auxiliary Machinery	H.D. Mc George	Elsevier	1999
4	Reed's General Engineering Knowledge for Marine Engineers		Adlard Coles Nautical	1986
5	Modern Marine Engineer's Manual- Volumes 1 and 2	Edited by Everett C Hunt and others	Cornell Maritime Press	1999
6	Marine propellers and propulsion	Prof J S Carlton FREng	Elsevier	Third Edition 2010/11
7	Marine Propellers	Prof. A.H. Techet	MIT Handouts	2005

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114106534
2	https://nptel.ac.in/courses/114106534
3	https://nptel.ac.in/courses/114106534
4	https://nptel.ac.in/courses/114106534

SEMESTER S5

SHIP DESIGN I

Course Code	PBNST504	CIE Marks	60
Teaching Hours/Week (L: T: P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBNST304 PBNST404	Course Type	Theory

Course Objectives:

1. To introduce the structure of international marine industry and the economical aspects in ship design to the student
2. To provide the students an understanding of the methodologies for preliminary design of ships
3. To provide the students an understanding of the ship design methodologies to develop lines plans of typical merchant ships
4. To enable the student to understand the relevant regulations on freeboard, tonnage and stability

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction, categories of ships, various sectors of international marine industry. Types of trade and methods of employment of ships. Economics in ship design, economic criteria, shipbuilding cost, operating cost, required freight rate. Owner's requirements. Phases of ship design, design spiral. Ship parameters, deadweight, displacement, deadweight coefficient, hold capacity, stowage factor.	9
2	Weight equation and solution. Volume equation and solution. Basic design procedure for main categories of ships, deadweight carriers, capacity carriers, linear dimension ships, special purpose ships. Procedure for selection of main dimensions (length, breadth, depth, draught) and form coefficients. Estimation of weight, lightship weight, steel weight,	9

	machinery weight, outfit weight, deadweight. General Arrangement (GA); factors influencing GA; subdivision of ship's hull; superstructure and deckhouses; arrangement of cargo spaces; cargo handling arrangement; arrangement of machinery spaces, accommodation and associated spaces; tanks, and other spaces.	
3	Ship's lines plan, conventional method of lines design, lines design using existing lines plan, lines design using standard series, hull form section shapes, form of bow, bulbous bow, form of stern. Effect of hull form on ship's performance.	9
4	Freeboard and load line regulations. Stability and trim, watertight integrity. IMO regulations, intact stability criteria, damage stability criteria. Stability booklet. Tonnage regulations, gross tonnage, net tonnage.	9

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks (8x2 =16marks)	• Each question carries 6 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 2 sub divisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the structure of international marine industry and explain the economical aspects in ship design.	K2
CO2	Apply ship design methodologies for preliminary design of typical merchant ships.	K3
CO3	Apply ship design methodologies to develop lines plans of typical merchant ships.	K3
CO4	Understand the relevant regulations on freeboard, tonnage and stability applicable to ships and apply these to typical merchant ships	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											
CO2	3	2	1	1	1		1		1			1
CO3	3	2	1	1	1		1	1	1			
CO4	3	2	1		1	1	1		1	1		1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Design and Construction	Thomas Lamb (Ed.)	The Society of Naval Architects and Marine Engineers	Edition 1, 2003
2	Elsevier Ocean Engineering Book Series- Volume 1 (Practical Ship Design)	D. G. M. Watson	Elsevier	Edition 1, 1998
3	Ship Design- Methodologies of Preliminary Design	Apostolos Papanikolaou	Springer	Edition 1, 2014
4	Ship Design for Efficiency & Economy	H. Schneekluth & V. Betram	Butterworth-Heinemann	Edition 2, 1998
5	Ship Design and Construction	Thomas Lamb (Ed.)	The Society of Naval Architects and Marine Engineers	Edition 1, 2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Design and Construction	Robert Taggart (Ed.)	The Society of Naval Architects and Marine Engineers	Edition 1, 1980
2	The Maritime Engineering Reference Book	Anthony F. Molland (Ed)	Elsevier	Edition 1, 2008
3	Design Principles of Ships and Marine Structures	S. C. Misra	Taylor & Francis	Edition 1, 2016
4	Computational Ship Design	Myung-II Roh & Kyu-Yeul Lee	Springer	Edition 1, 2018
5	Principles of Naval Architecture (Second Revision)	Edward V. Lewis (Editor)	The Society of Naval Architects and Marine Engineers	Edition 1, 1988
6	Basic Ship Theory	K. J. Rawson & E. C. Tupper	Butterworth-Heinemann	Edition 5, 2001

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105004
2	https://nptel.ac.in/courses/114105004
3	https://nptel.ac.in/courses/114105004
4	https://nptel.ac.in/courses/114105003

PBL Course Elements			
L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity		
Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
	Total	30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S5

ELECTRICAL TECHNOLOGY AND INSTRUMENTATION

Course Code	PENST521	CIE Marks	40
Teaching Hours/Week (L: T: P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GCEST204	Course Type	Theory

Course Objectives:

1. Discern the basic working principle, construction, types, performance characteristics and applications of DC generators.
2. Understand the basic working principle, construction, types, performance characteristics and applications of DC motors.
3. Understand basic working principle, construction, types, performance characteristics and applications of AC machines and transformers
4. Identify the basic principles of instrumentation, measurement standards and types of errors in instruments and measurements.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	D.C. generator - working principle, construction, types, open circuit characteristics, condition for self excitation, field critical resistance, critical speed, load characteristics of generators, losses, efficiency, applications in ships and shipyards.	9
2	D.C. motors - working principle, construction, types, back EMF, necessity of starter, speed of DC motors, brake test, torque equation, efficiency, performance and operating characteristics of shunt, series and compound motors, applications in ships and shipyards.	9
3	AC machines- three phase induction motor: working principle, construction, types, slip, performance characteristics and efficiency, single phase induction motor -working, types-split phase and capacitor start, alternators-working, construction, types, working principles and types of transformers, applications in ships and shipyards	9

4	Introduction to instrumentation - classification of instruments, standards and calibration, types of errors, strain gauge, L.V.D.T. (Linear Variable Differential Transformer), application of various instruments in shipbuilding.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electrical Machinery	P. S. Bimbra	Khanna Publishers	Edition 7, 2021
2	Theory and principles of Electrical Machines	J. B. Gupta	S. K.Kataria and Sons	Edition 9, 2023
3	Instrumentation and Control	D Patranabis	PHI Learning Pvt Ltd	Edition 3, 2011
4	Performance and Design of AC Machines	Say M. G	ELBS	2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory of AC Machinery	Alexander Langsdorf A. S	Mc-Graw Hill	Edition 2, 1985
2	Basic Electrical Engineering	U.A.Bakshi, V.U.Bakshi	Technical Publications	Edition 2, 2009
3	Instrumentation for Engineering Measurements.	James.W.Dally, William.F. Riley, Kenneth G. McConnell	Wiley India Private Limited	Edition 2, 2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108102185
2	https://nptel.ac.in/courses/108102185
3	https://nptel.ac.in/courses/108102185
4	https://nptel.ac.in/courses/114106049

SEMESTER S5

SHIP RECYCLING

Course Code	PENST523	CIE Marks	40
Teaching Hours/Week (L: T: P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs 30 Min.
Prerequisites (if any)	PCNST205	Course Type	Theory

Course Objectives:

1. To introduce the concept of ship recycling and to explain its relevance
2. To describe the ship recycling stages and the activities prior to commencement of actual dismantling of ships in recycling yards.
3. To describe ship recycling plan and various processes involved in ship recycling
4. To introduce various rules and regulations applicable to recycling of ships

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to ship recycling, role of maritime industrial sector in sustainable development, relevance of ship recycling. Statistics of global shipping and shipbuilding. Ship life cycle stages, various stages of life cycle of ships. Operations in life cycle stages and effective management of the stages. Importance of ship recycling in life cycle stage management.	9
2	Recycling stages. Decision on decommissioning of ships. Preparations for transferring obsolete vessels to recycling yards, planning, commercial matters. Transportation methods. Survey before positioning, legal matters. Positioning of ships for recycling- beaching method, dry dock method, slipway method, alongside method.	9
3	Ship recycling process, ship dismantling, access, cleaning, marking, cutting, handling, lifting, sorting, stacking, storing, marshalling. Concept of recycling, reuse and land-filling in ship recycling. Health, safety, environmental issues in ship recycling. Ship recycling plan. Risk assessment. Hazardous waste management. Ship recycling yard layout.	9

	Design for ship recycling.	
4	Rules and regulations in ship recycling, rules and guidelines of various international and national agencies, IMO, UNEP (Basel Convention), EPA (USA), ILO, European Union, Government of India. Statutory certificates for ship recycling, inventory of hazardous materials, green passport. Role of NGOs.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the concept of ship recycling and its relevance.	K1
CO2	Explain the stages of ship recycling and the preparations necessary prior to dismantling of ships.	K2
CO3	Describe ship recycling plan and the various processes in ship recycling and explain the concept of design for recycling.	K3
CO4	Describe the rules and regulations applicable in ship recycling	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	1	1		1							
CO3	3	2	3		1							
CO4	3	2	1	1	1							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Recycling - A Hand Book for Mariners	Purnendu Mishra & Anjan Mukherjee	Narosa Publication	Edition 1, 2009
2	Ship Recycling- Practice and Regulation Today	Lloyd's Register	Lloyd's Register	2011
3	Disposal Options for Ships	Ronald W. Hess et al.	RAND	Edition 1, 2001
4	A Guide for Ship Scrappers: Tips for Regulatory Compliance	U. S. Environmental Protection Agency	U. S. Environmental Protection Agency	2000

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Safety and Health in Shipbreaking - Guidelines for Asian Countries and Turkey	International Labour Organization (ILO)	International Labour Organization (ILO)	Edition 1, 2004
2	Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal.	Secretariat of the Basel Convention	Secretariat of the Basel Convention	2014
3	Technical Guidelines for the Environmentally Sound Management of the Full and Partial Dismantling of Ships	Secretariat of the Basel Convention	Secretariat of the Basel Convention	2003
4	Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships, 2009	International Maritime Organisation (IMO)	International Maritime Organization (IMO)	2009
5	Safe Work Practices for Shipbreaking	Occupational Safety and Health Administration (OSHA)	Occupational Safety and Health Administration (OSHA)	2010
6	Overview of Ship Recycling in the UK	Department` of Environment, Food and Rural Affairs (DEFRA)	Department` of Environment, Food and Rural Affairs (DEFRA)	2007
7	Ship Dismantling and Pre-cleaning of Ships: Final Report	European Commission Directorate General, Environment	European Commission Directorate General, Environment	2007
8	Ship Design and Construction	Robert Taggart (Ed.)	The Society of Naval Architects and Marine Engineers	Edition 1, 1980

SEMESTER S5

NON FERROUS MATERIALS IN SHIPBUILDING

Course Code	PENST523	CIE Marks	40
Teaching Hours/Week (L: T: P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand the historic importance of wood as a shipbuilding material and its relevance today
2. Understand the use of non ferrous metallic materials in marine applications
3. Discern the properties of composites used in shipbuilding
4. Understand the use of advanced and unconventional materials for shipbuilding

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Properties of wood, strength of wood, Dugouts and Early Planked Boats, Lashed/Sewn Boats in the Mediterranean Basin, Khufu's Barge and Lisht Timbers, planking, erections and joiner work, preservation.	9
2	Aluminium and its alloys, material characteristics of marine grade aluminium, structural design considerations, welding and fabrication, TIG, MIG, riveting, joining aluminium and steel, residual stresses and distortion, fatigue and fracture design and analysis procedures, fire protection, vibration, maintenance and repair.	9
3	Classification of polymers, types of polymerization, preparation, properties and uses of some important polymers, fabrication of plastics, rubber, synthetic rubbers, composites, difference between thermoplastics and thermosets, types of resins, glass and carbon fibres, different types of fabrics and mats such as chopped stranded mats, woven roving and their properties, FRP, GRP materials, different types of moulding techniques, lay-up techniques, manufacturing requirements, advantages and disadvantages of composites over steel and aluminium in shipbuilding.	9

4	Marine concrete, history of concrete ships, prestressed concrete vessels, advantages and disadvantages, future trends in concrete ships, advanced materials, use of adhesive bonding on superstructures, laser-welded steel sandwich panels.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the historic importance of wood as a shipbuilding material and its relevance today	K1
CO2	Understand the use of non ferrous metallic materials in marine applications	K2
CO3	Discern the properties of composites used in shipbuilding	K3
CO4	Understand the use of advanced and unconventional materials for shipbuilding	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											
CO2	2	1	1									
CO3	2	1	1									
CO4	2	1	1									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Wooden Shipbuilding Techniques at Vila Do Conde's Shipyards	António José Carmo	Shipbuilding: Knowledge And Heritage	1988
2	The Elements of Wood Ship Construction	W. H. Curtis	Mcgraw-Hill Book Company, Inc.	1919
3	Ship Design & Construction	Thomas Lamb, Editor	The Society of Naval Architects and Marine engineers	Edition 3, 2004
4	Marine Composites	Erric Greene	Erric Greene Associates	Edition 2, 1999
5	Concrete Ships	Mark L. Lavache	University of Rhode Island	1978

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Aluminum Structure Design and Fabrication Guid	R A Sielski	Ship Structure Committee	Edition 1, 2007
2	Cambridge Ocean Technology Series 4 & 5 - Composite Materials in Maritime Structures Vol 1 & 2	R A Shenoy (ed)	Cambridge University Press	Edition 1, 2008
3	Concrete Ships and Vessels Past, Present and Future	Tony. Liu, James E. McDonald	Concrete Laboratory, U. S. Army Engineer Waterways Experiment Station	1977
4	The Maritime Engineering Reference Book	Anthony F. Molland (Ed)	Elsevier	Edition1, 2008

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/113107078
2	https://nptel.ac.in/courses/113106105
3	https://nptel.ac.in/courses/112104168
4	https://nptel.ac.in/courses/113105081

SEMESTER S5

ELECTRICAL SYSTEMS IN SHIPS AND SHIPYARDS

Course Code	PENST524	CIE Marks	40
Teaching Hours/Week (L: T: P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GCEST204	Course Type	Theory

Course Objectives:

1. Understand the ship's electrical system.
2. Understand AC distribution systems, emergency power and shore supply.
3. Understand direct current machines and starters for alternating current motors
4. Identify automatic voltage regulators and electrical system in shipyards.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Components of electrical system on board ships, merits and demerits of AC and DC on board. Standard voltages, difference between marine and industrial circumstances. Safety and quality of supply. Standard IEC symbols. Electrical power generation on board ships- comparison of diesel, thermal and nuclear power plants as prime movers, shaft driven generators, brushless generators, specification of generators. Specification of motors- speed based and torque based motors.	9
2	Identify the general concept of alternators, Specific systems for distribution of AC power in ships, general distribution scheme onboard a ship, relevant regulations. Method of supplying emergency power, actions by engineers during a blackout situation, shore supply, relevant rules.	9
3	Introduction to DC generators, types of direct current generators, series wound generator, shunt wound generator, compound wound generators. introduction to dc motors, principle of dc motor operation, shunt wound motor, series wound motor, compound motors. Basics of starters, Direct-	9

	On-Line or D.O.L. Starter, star-delta starter, autotransformer starter.	
4	Understand the performance requirements of alternators, operating principle of an AVR, thyristor-based static automatic voltage regulator, effect of kW loading, effect of kVA loading. Basics, manual synchronising, auto-synchronising, parallel operation, load sharing, speed droop and power generation. Power factor improvement, power tariff calculation, essential regulations, and main loads. Potential hazards, Maritime Labour/industry recommendations.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Discern the basic working principle, construction, types, performance characteristics and applications of DC generators.	K2
CO2	Understand the basic working principle, construction, types, performance characteristics and applications of DC motors.	K2
CO3	Understand basic working principle, construction, types, performance characteristics and applications of AC machines and transformers	K2
CO4	Identify the basic principles of instrumentation, measurement standards and types of errors in instruments and measurements.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	1	1									
CO3	3	2	1									
CO4	3	2	1									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electrical Machinery	P. S. Bimbra	Khanna Publishers	Edition 7, 2021
2	Theory and principles of Electrical Machines	J. B. Gupta	S. K.Kataria and Sons	Edition 9, 2023
3	Instrumentation and Control	D Patranabis	PHI Learning Pvt Ltd	Edition 3, 2011
4	Performance and Design of AC Machines	Say M.G	ELBS	2007
5	Shipboard electrical power systems	Mukund R Patel	CRC Press	Edition 1, 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Marine Electrical Practice	G. O. Watson	Butterworths	Edition 6, 1990
2	Marine Electrical Technology	Elstan A Fernandez	Shroff Publishers & Distributors Pvt. Ltd	Edition 12, 2014
3	Ships Electrical System	Rene Borstlap, Hans ten Katen	Dokmar Maritime Publishers	Edition 1, 2011

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108102185
2	https://nptel.ac.in/courses/108102185
3	https://nptel.ac.in/courses/114106534
4	https://nptel.ac.in/courses/114106049

SEMESTER S5

MARITIME LAW AND SHIPPING MANAGEMENT

Course Code	PENST525	CIE Marks	40
Teaching Hours/Week (L: T: P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST205	Course Type	Theory

Course Objectives:

1. Understand the role of international conventions in formulating Maritime Law, the roles of Maritime Labour Convention and international conventions on the Law of the Sea.
2. Understand the roles of international conventions for the Safety of Life at Sea, International Ship and Port Facility Security Code (ISPS Code), and other conventions
3. To enable the student to understand the basic structure of shipping industry and the role of various stakeholders in the industry
4. To provide the student an understanding of shipping economics and the shipping market

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to maritime law; need and source of maritime law; implementation of maritime law through national legislation; role of United Nations (UN), International Maritime Organization (IMO) and International Labour Organization (ILO); role of Flag state and Port state in implementation of law; introduction to SOLAS, MARPOL, STCW and Maritime Labour Convention (MLC); need of survey and certification. Classification Societies, need and role of Classification Societies. United Nations Convention on the Law of the Sea (UNCLOS), various zones and limits. Maritime Labour Convention, minimum requirements for seafarers to work on a ship, conditions of employment, accommodation, recreational facilities, food and catering, health protection, medical care, welfare and social security.	11

2	SOLAS- Introduction to SOLAS; Subdivision and stability; Fire protection, detection and extinction; Lifesaving appliances and arrangements; Carriage of grain; Carriage of dangerous goods. Load Line Convention, important definitions and markings, survey and certification. ISPS Code, ship security plan, security levels, declaration of security, ship security alert system, piracy, best management practices to deter piracy. Introduction to other Conventions; International Convention on Salvage, Convention on Limitation of Liability for Maritime Claims, Oil Pollution Compensation Conventions.	11
3	Shipping Management- Types of cargoes, types of ships which carry them; Ship's tonnages (GT, NT); Cargo carrying capacity. Ship registration, ship insurance. Ship owners, operators and managers. Types of shipping services- liner and tramp. Role of intermediaries in shipping business- freight brokers, clearing and forwarding agents, stevedores, shipbrokers, bunker and stores suppliers, shipping agencies.	11
4	Shipping Economics-Sea transport and the global economy, role of shipping in economic development. Shipping market cycles, shipping risk, shipping market supply and demand model. The four shipping markets- freight market, sale and purchase market, newbuilding market and demolition market. Classification of costs, Financing ships and shipping companies, Risk and returns.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Criteria for evaluation:

1. Problem Definition (K4 - 4 points)

- a. Clearly defines the legal or shipping management problem.
- b. Examine and identifies relevant contextual factors (constraints, resources, objectives).

2. Problem Analysis (K4 - 4 points)

- a. Break-down and present a well-reasoned solution approach based on legal or management perspectives.
- b. Compare and justify the proposed solutions with technical evidence.

3. Evaluate (K5 - 4 points)

- a. Thoroughly evaluate the proposed solutions.
- b. Compare trade-offs, advantages, and disadvantages.
- c. Consider feasibility and practical implications.

4. Design Implementation (K5 - 4 points)

- a. Select the most feasible solution by implementing the proposed solutions.
- b. Successfully translate the chosen solution into tangible outputs.

5. Conclusion (K4- 2 points, K5 – 2 points)

- a. Summarizes findings and insights. State which solution is most appropriate for the problem. **(K4)**
- b. Reflects critical thinking and informed decision-making. **(K5)**

Scoring:

- 1. **Accomplished (4 points):** Exceptional analysis, clear implementation, and depth of understanding.
- 2. **Competent (3 points):** Solid performance with minor areas for improvement.
- 3. **Developing (2 points):** Adequate effort but lacks depth or clarity.

4. **Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basics of international maritime law and describe the roles of international conventions in formulating Maritime Law, the roles of Maritime Labour Convention and international conventions on the Law of the Sea.	K2
CO2	Describe the roles of international conventions for the Safety of Life at Sea, International Ship and Port Facility Security Code (ISPS Code) and other conventions	K2
CO3	Understand various elements of shipping management and the role of various stakeholders in shipping business	K1
CO4	Understand the basics of shipping economics and the shipping market	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1				1					1
CO2	1	1	1				1					1
CO3	1	1	1									
CO4	1	1	1									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Maritime Economics	Martin Stopford	Taylor & Francis	Edition 3, 2008
2	Economics of Shipping Practice and Management	Alan E. Branch	Chapman and Hall	Edition 2, 1988
3	Elements of Shipping	Alan E. Branch	Routledge	Edition 8, 2007
4	Reeds 21 st Century Ship Management	Capt. J W Dickie	Adlard Coles Nautical	Edition 1, 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Shipping and Logistics Management	Y. H. V Lun, K. H. Lai, & T. C. E. Cheng	Springer	Edition1, 2010
2	The Maritime Engineering Reference Book	Anthony F. Molland (Ed)	Elsevier	Edition1, 2008
3	Ship Design and Construction	Thomas Lamb (Ed.)	The Society of Naval Architects and Marine Engineers	Edition 1, 2003
4	Elsevier Ocean Engineering Book Series- Volume 1 (Practical Ship Design)	D. G. M. Watson	Elsevier	Edition 1, 1998

SEMESTER S5

MARINE HYDRODYNAMICS AND HYDRAULICS LAB

Course Code	PCNSL507	CIE Marks	50
Teaching Hours/Week (L: T: P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST205 PBNST304 PCNST402	Course Type	Lab

Course Objectives:

1. Apply knowledge in marine hydrodynamics and hydraulic machinery experimentally.
2. Understand the arrangement and conduct of experiments in the Marine Hydrodynamics & Hydraulic Machinery laboratory.
3. Comprehend the factors responsible for variation between theoretical and experimental results pertaining to the domain of Marine Hydrodynamics & Hydraulic Machinery.
4. Able to note down relevant readings and perform calculations while an experiment is in progress thereby correlating theoretical concepts of marine hydrodynamics and their practical implications on ship.

Expt. No.	Experiments (Minimum 12 are mandatory)
1	Performance Characteristic Tests on Pelton Wheel (Load test & best speed). - Equipment: Pelton Wheel Turbine Test Rig.
2	Performance Characteristic Tests on Francis Turbine (Load test & best gate opening). - Equipment: Francis Turbine Test Rig.
3	Performance Characteristic Tests on Kaplan Turbine (Load test & best gate, vane angle opening) - Equipment: Kaplan Turbine Test Rig.
4	Performance characteristic Tests on Single Stage, Multi Stage Centrifugal Pumps at Constant Speed & at Variable Speed. (Actual & predicted curves) - Equipment:

	Centrifugal Pump Test Rig.
5	Performance Characteristic Tests on Self-priming, Jet, Airlift and Deep Well Pumps - Equipment: Self Priming Pump Test Rig.
6	Performance Characteristic Tests on Hydraulic Ram - Equipment: Hydraulic Ram Test Rig.
7	Performance Characteristic Tests on Reciprocating Pump at Constant Speed - Equipment: Reciprocating Pump Test Rig.
8	Performance Characteristic Tests on Gear Pump - Equipment: Gear Oil Pump Test Rig.
9	Performance Characteristic Tests on Screw Pump - Equipment: Screw Pump Test Rig.
10	Impact of Water jet on Flat Plate - Equipment: Impact of Jet on Vane Apparatus.
11	Impact of Water jet on Curved Plate - Equipment: Impact of Jet on Vane Apparatus.
12	Prediction of Ship Hull Resistance - Equipment: Ship Design Software / test facilities.
13	Prediction of Propeller Performance - Equipment: Ship Design Software / test facilities.
14	Prediction of Ship Resistance Using Data Obtained from Model Test - Equipment: Ship Design Software / test facilities.
15	Study of Roll Decay Tests and Calculation of Roll Period - Equipment: Ship Design Software / test facilities.
16	Prediction of Seakeeping Characteristics - Equipment: Ship Design Software / test facilities.
17	Study on Open Water Tests - Equipment: Ship Design Software / test facilities.
18	Study on Manoeuvring Performance of Ships. Equipment: Ship Design Software / test facilities

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply knowledge in marine hydrodynamics and hydraulic machinery experimentally.	K2
CO2	Understand the arrangement and conduct of experiments in the Marine Hydrodynamics & Hydraulic Machinery laboratory.	K2
CO3	Comprehend the factors responsible for variation between theoretical and experimental results pertaining to the domain of Marine Hydrodynamics & Hydraulic Machinery.	K2
CO4	Able to note down relevant readings and perform calculations while an experiment is in progress thereby correlating theoretical concepts of marine hydrodynamics and their practical implications on ship.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2		1							
CO2	3	3	2		1							
CO3	3	3	3		1							
CO4	3	3	3	2	1							

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Laboratory Manual Hydraulics and Hydraulic Machines	Raikar, R. V.	PHI Learning	Edition 1, 2003
2	A Textbook of Fluid Mechanics and Hydraulic Machines	RK Rajput	S Chand	Edition 1, 2015
3	Ship Resistance and Propulsion – Practical Estimation of Propulsive Power	Antony F. Moland, Stephen R. Turnock	Cambridge University Press	Edition 1, 2011
4	Principles of Naval Architecture series – ShipResistance and flow	John Letcher, Randolph Paulling	SNAME, New Jersey	Edition 3, 2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mitigation of Hydrodynamic Resistance Methods to Reduce Hydrodynamic Drag	Marc Perlin, Steven Ceccio	World Scientific Publishing Co. Pte. Ltd, Singapore	Edition 1, 2015
2	Basic Ship Theory Vol II	K J Rawson, E C Tupper	Butterworth-Heinemann Ltd	Edition 5, 2001
3	Practical Ship Hydrodynamics	Volker Bertram	Butterworth-Heinemann Ltd	Edition 1, 2000
4	Hydrodynamics of High-Speed Vehicles	OM Faltinsen	Cambridge University Press	Edition 1, 2005
5	Principles of Naval Architecture (Second Revision)	Edward V. Lewis (Editor)	The Society of Naval Architects and Marine Engineers	Edition 1, 1988

Video Links (NPTEL, SWAYAM...)	
Sl No.	Link ID
1	https://nptel.ac.in/courses/114105003
2	https://nptel.ac.in/courses/114105006
3	https://nptel.ac.in/courses/114106026

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S5
SHIP DESIGN LAB

Course Code	PCNSL508	CIE Marks	50
Teaching Hours/Week (L: T: P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST205 PBNST304 PCNST402	Course Type	Lab

Course Objectives:

1. Understand the methodologies for modelling 3D bodies like ships and submarines
2. Generate faired lines plan for given offset table.
3. Generate hydrostatics particulars and Bonjean data and get the plots of the curves.
4. Analyse stability (intact and damaged) parameters of the ship using the generated model

Expt. No.	Experiments (Minimum 12 are mandatory)
1	Study various approaches in generating lines plan of ships.
2	Modelling of hull surface from offset data.
3	Modelling of hull surface by modifying general hull form from software database.
4	Solid modelling from boundary surfaces (e.g. superstructure).
5	Solid modelling by revolving closed areas about an axis (e.g. Submarine hull).
6	Bow thruster tunnel modelling.
7	Modelling of tanks and cargo spaces.
8	Plotting lines plan of a vessel from given offset table.
9	Computation and plotting of Bonjean and sectional area curve.
10	Computation of ship hydrostatic particulars.
11	Calculate equilibrium condition of a given ship at given loading conditions.
12	Computation of transverse metacentric height.
13	Computation of stability at small and large angles of heel.

14	Computation of static stability and cross curves of stability.
15	Dynamic stability computations.
16	Generate stability booklet for given ship particulars and conditions.
17	Carry out floodable length calculations for a ship at given loading condition.
18	Damage stability computations at given loading condition

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the methodologies for modelling 3D bodies like ships and submarines	K2
CO2	Generate faired lines plan for given offset table.	K3
CO3	Generate hydrostatics particulars and Bonjean data and get the plots of the curves.	K3
CO4	Analyse stability (intact and damaged) parameters of the ship using the generated model	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2		1							
CO2	3	3	2		1							
CO3	3	3	3		1							
CO4	3	3	3	1	1							

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Resistance and Propulsion – Practical Estimation of Propulsive Power	Antony F. Moland, Stephen R. Turnock	Cambridge University Press	Edition 1, 2011
2	Principles of Naval Architecture series – Ship Resistance and flow	John Letcher, Randolph Paulling	SNAME, New Jersey	Edition 3, 2009
3	Resistance and Propulsion of Ships	Harvald S. A	John Wiley & Sons	Edition 1, 1983
4	Fundamentals of Ship Hydrodynamics-Fluid Mechanics, Ship Resistance and Propulsion	Lothar Birk	John Wiley & Sons Ltd	Edition 1, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mitigation of Hydrodynamic Resistance Methods to Reduce Hydrodynamic Drag	Marc Perlin, Steven Ceccio	World Scientific Publishing Co. Pte. Ltd, Singapore	Edition 1, 2015
2	Basic Ship Theory Vol II	K J Rawson, E C Tupper	Butterworth-Heinemann Ltd	Edition 5, 2001
3	Practical Ship Hydrodynamics	Volker Bertram	Butterworth-Heinemann Ltd	Edition 1, 2000
4	Hydrodynamics of High-Speed Vehicles	O M Faltinsen	Cambridge University Press	Edition 1, 2005
5	Hydrodynamics In Ship Design Vol 1-3	H E Saunders	SNAME, New Jersey	Edition 1, 1957

Video Links (NPTEL, SWAYAM...)	
SL No.	Link ID
1	https://nptel.ac.in/courses/114105003

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S6

**NAVAL ARCHITECTURE & SHIPBUILDING
ENGINEERING**

SEMESTER S6
SHIP STRUCTURES II

Course Code	PCNST601	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-1-0-0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST502	Course Type	Theory

Course Objectives:

1. Understand the loads acting on the hull girder, modes of failure and hull girder global response
2. Understand the response of hull girder to wave loads,
3. Understand the effect of torsion and buckling on hull girder strength and the concept of limit state analysis.
4. Understand the basics of vibration and fatigue as applicable to marine structures.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Loads and moments acting on ship structures in still water and in waves, modes of failure in sagging, hogging and torsion, weight and weight distribution, buoyancy and buoyancy distribution, load curve, shear force curve, bending moment curve, deflection curve, effect of thermal loads.	11
2	Loads imposed by regular waves and oblique waves, representation of irregular seaway, short term and long term distribution of loads, spectral approach to response of ship structures, effect of slamming and shipping of green seas.	11
3	Torsional loads due to oblique waves and cargo distribution, torsion warping deformations and stresses, torsion of container ships, shear stress distribution procedure, limit states - ultimate, serviceability, fatigue,	11

	accidental limit states, ultimate strength assessment of hull girder using progressive collapse analysis.	
4	Classification of vibration, introduction to hull structure vibration, modes of hull structure vibration, sources of vibration, measures for control of vibration, fatigue failures of marine structures, Main Contributing Factors to Fatigue, Fatigue Design Strategies	11

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the loads acting on the hull girder, modes of failure and hull girder global response	K1
CO2	Understand the response of hull girder to wave loads,	K2

CO3	Understand the effect of torsion and buckling on hull girder strength and the concept of limit state analysis.	K2
CO4	Understand the basics of vibration and fatigue as applicable to marine structures.	K2

Note: *K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	2											
CO3	2	2	2									
CO4	2	2	2		1							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Structural Analysis and Design	O F Hughes, Jeom Kee Paik	The Society of Naval Architects and Marine Engineers	Edition 1, 2010
2	Torsion and Shear Stresses in Ships	M Shama	Springer	Edition 1, 2010
3	Local Strength of Ship Structures	Piero Caridis	CRC Press	Edition 1, 2024
4	Ship Structural Design Concepts	J Harvey Evans	Cornell Maritime Press	Edition 2, 1983
5	Buckling of Ship Structures	M Shama	Springer	Edition 1, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Aluminium Structural Design and Fabrication Guide	Edward F Noonan	Ship Structure Committee	Edition 1, 1990
2	Strength of Ship Structures	W Muckle	Edward Arnold Publishers Ltd	Edition 2, 1967
3	Strength of Ships and Ocean Structures, The Principles of Naval Architecture Series	Alaa Mansour	The Society of Naval Architects and Marine Engineers	Edition 3, 2008
4	Elementary Beam Theory and the Hull Girder	G W White	Standard Maritime London	Edition 1, 1974
5	Fatigue Design of Marine Structures	Inge Lotsberg	Cambridge University Press	Edition 1, 2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105031
2	https://nptel.ac.in/courses/114105031
3	https://nptel.ac.in/courses/114105031
4	https://nptel.ac.in/courses/114105031 https://nptel.ac.in/courses/113105106

SEMESTER S6

OFFSHORE STRUCTURES

Course Code	PCNST602	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand the basics features of offshore structures
2. Discern the basic principles of fixed offshore platform design
3. Discern the basic principles of Compliant offshore structures design
4. Understand the basics of offshore installation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definition and functions of Offshore structures, Brief Historical Development, Different types of Offshore Structure Configurations: Bottom Supported Fixed Structures, Compliant Structures, Floating Structures and Novel & Small Field Offshore Structures, Offshore field development concepts and design phases	9
2	Waves and wave theories, Wave force on offshore structures: Small diameter and large diameter structures, vertical and arbitrarily oriented cylindrical members, Ocean currents and current forces exerted on offshore structures, Wind forces, Earthquake forces, Ice forces and Tides, Marine growth, Response of offshore structures to ocean environment, Introduction to corrosion and protection of offshore structures	9
3	Design spiral, Various types of loads acting on offshore structures, Introduction to design tools for offshore structure design, Jacket structure: Basic configuration, sizing considerations, design strategy, bracing system, jacket design, pile design, deck design, top sides, appurtenances, Floating Structures: Basic configurations, sizing considerations, design strategy, design consideration for components. Compliant Structures: Basic configurations, sizing considerations, design strategy, design consideration	9

	for components. Gravity structures: Basic configuration, sizing considerations, design concepts for foundation, Material considerations for offshore structures	
4	Introduction to construction of various types of offshore structures, Load out methods, Transportation facilities and methods for different types of offshore structures, Installation methods for different offshore structures	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basics features of offshore structures	K2
CO2	Discern the basic principles of fixed offshore platform design	K2
CO3	Discern the basic principles of Compliant offshore structures design	K2
CO4	Understand the basics of offshore installation	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1									
CO2	2	1	1									
CO3	2	1	1									
CO4	2	1	1									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Offshore Structures Design, Construction and Maintenance	Mohamed A El-Reedy	Elsevier	Edition 1, 2012
2	Introduction to Offshore Structures- Design, Fabrication, Installation	W J Graff	Gulf Publishing Co	Edition 1, 1981
3	Offshore Engineering – An Introduction	Mather A	Witherby Publishers	Edition 2, 2000
4	Essentials of Offshore Structures - Framed & Gravity Platforms	DV Reddy, A S J Swamidas	CRC Press	Edition 2, 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Dynamics of Offshore Structures	Minoo H Patel	Butterworths	Edition 1, 1989
2	Compliant Offshore Structures	Minoo H Patel Joel Witz	Butterworths	Edition 1, 1991
3	Handbook of Offshore Engineering (Vol I & II), 2005	S.K.Chakrabarti	Elsevier	2005

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114106015
2	https://nptel.ac.in/courses/114106015
3	https://nptel.ac.in/courses/114106015
4	https://nptel.ac.in/courses/114106015

SEMESTER S6
SHIP PRODUCTION

Course Code	PENST631	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST502 PBNST504	Course Type	Theory

Course Objectives:

1. Understand various types of shipyard layouts, material handling systems.
2. Discern various production management methodologies in modern shipbuilding.
3. Discern the various assembly techniques, welding symbols and standards used in shipbuilding.
4. Explicate the various types of outfitting and launching methodologies adopted in modern shipbuilding.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	General principles for location and layout of shipyards. Functional and geographic aspects, transport, supply industry and human resource. Evolution of first to fourth generation shipyards and their typical layouts. Product and Process Oriented Work Breakdown Structure. Integrated hull construction, outfitting and painting (IHOP), grouping and work breakdown structure. Zone construction method for hull blocks, outfitting and painting, pipe piece family manufacturing. Material handling, storage and transport systems. Material preparation. Straightening of plates, sand blasting, painting, lofting, nesting.	9
2	Production management. Pre-production, production and post production stages. Planning, design, production, material management. Quality control/ quality assurance. Accuracy control. Risk management. Project monitoring – build strategy, progress monitoring, PERT charts. Cutting processes and tools mechanical cutting, thermal cutting, general description of the various machines, photoelectric and NC-control devices. Bending of plates and sections. Heat-line bending	9

3	Assembly of Ship's Structure. Assembly jigs. Structural arrangement of the fore and aft end structure, superstructures, machinery foundation, cranes and davits. Welding in shipbuilding, welding methods, standards, symbols. Erection of ship's hull. Handling of pre-assembled units in the erection area, Different methods of hull assembly. Block erection methods – floor, ring and pyramid. Single/ multi point erection methods, Technological processes in the installation of main machinery, shafting and propeller.	9
4	Weld defects, distortions and their preventive measures. Testing – destructive/ non destructive. Launching- Mechanical Launching Methods (Slip, Lift), Construction of slipways for end launch and side launch. Launching curves. Arresting arrangements. Tipping, Pivoting/ plunging, drop off. Other Methods of launching - Launching by Floating Off (Building Dock, Floating Dock), Launching using air bags.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Production	Richard Lee	Society of Naval Architects and Marine Engineers	Edition 2, 1995
2	Engineering for Ship Production	Thomas Lamb	Society of Naval Architects and Marine Engineers	Edition 1, 1986
3	Ship Design and Construction Vol I & II	Thomas Lamb (Ed)	Society of Naval Architects and Marine Engineers	Edition 3, 2004
4	Reed's ship construction For marine students	E A Stokoe	Thomas Reed Publications	Edition 5, 1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Construction and Welding	Nisith R. Mandal	Springer Series on Naval Architecture, Marine Engineering, Shipbuilding and Shipping 2	Edition 1, 2017
2	Merchant Ship Construction	H J Pursey	Nautical Press	1978
3	Shipbuilding Technology	Dormidontov VK	Mir Publishers, Moscow	1966

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105004
2	https://nptel.ac.in/courses/114105004
3	https://nptel.ac.in/courses/114105004
4	https://nptel.ac.in/courses/114105004

SEMESTER S6

SHIPBUILDING TECHNOLOGY

Course Code	PENST632	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST502 PBNST504	Course Type	Theory

Course Objectives:

1. Identify the operations in different outfitting departments.
2. Explain various processes involved in the hull and machinery installation works.
3. Understand the concept of design for production and need of quality control and assurance.
4. Understand typical shipyard layout and productivity.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Workshops, piping shop, fitters shop, Carpenters shop (wood, plastics), Mechanical workshop, Machine shop (preassembly of blocks), other workshops (electrical installation, painting, insulation, etc.).	9
2	Technological process in the hull and machinery installation, technological process in installing the main machinery, installation of shafting and propeller, installation of the main machinery, installing of auxiliary machinery and boilers, installation of piping systems, electrical installation, hull installation work, pre and advanced outfitting.	9
3	Production planning and control in shipbuilding, production design, application of the principles of design for production in shipbuilding, joining of parts, relations between structural design and prefabrication, simplifications in structural design (design for welding), quality control.	9

4	Capacity planning and cost estimation, productivity in shipyard, measurement and monitoring, shipyard capacity estimation, role of process and procedure in shipyard capacity evaluation, Shipyard layout, factors affecting design of shipyard layout, production facility layout, some aspects of shipyard capacity augmentation, developing shipbuilding strategy, Cost Estimation methods in ship production.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the operations in different outfitting departments.	K2
CO2	Explain various processes involved in the hull and machinery installation works.	K2
CO3	Understand the concept of design for production and need of quality control and assurance.	K2
CO4	Understand typical shipyard layout and productivity.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1									
CO2	2	1	1									
CO3	2	1	1									
CO4	2	1	1									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Production	Richard Lee	Society of Naval Architects and Marine Engineers	Edition 2, 1995
2	Engineering for Ship Production	Thomas Lamb	Society of Naval Architects and Marine Engineers	Edition 1, 1986
3	Ship Design and Construction Vol I & II	Thomas Lamb (Ed)	Society of Naval Architects and Marine Engineers	Edition 3, 2004
4	Reed's ship construction For marine students	E A Stokoe	Thomas Reed Publications	Edition 5, 1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Construction and Welding	Nisith R. Mandal	Springer Series on Naval Architecture, Marine Engineering, Shipbuilding and Shipping 2	Edition 1, 2017
2	Merchant Ship Construction	H J Pursey	Nautical Press	1978
3	Shipbuilding Technology	Dormidontov VK	Mir Publishers, Moscow	1966

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105004
2	https://nptel.ac.in/courses/114105004
3	https://nptel.ac.in/courses/114105004
4	https://nptel.ac.in/courses/114105004

SEMESTER S6

SHIP PRODUCTION MANAGEMENT

Course Code	PENST633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand a standard ship production system.
2. Discern production planning and control for ship production activities
3. Apply operation research, operation planning in shipbuilding process.
4. Understand the information exchange during ship production .

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Ship production systems, systems approach, subsystems, comprehensive system model, ship as a system, product standardization and work simplification, product mix, quality assurance and quality control activities in shipyards, problems of accuracy, tolerances, standards, measuring techniques	9
2	Production planning and control, planning for operations, interconnection between production design and process planning. production and process analysis, assembly charts, operation process charts, flow process charts, process selection, application of models for process planning, scheduling and control, Gantt charts, CPM & PERT, transportation models.	9
3	Introduction to operation research, operation planning and control production planning scheduling network models, quality control maintenance analysis, production standards in several parts of the ship production process, work measurement systems, methods of man- hour determination.	9

4	Information exchange during ship production - communication between departments and other stakeholders - classification society, regulatory body, ship owner, design office, Database Management Systems (DBMS) in ship production, the inventory problem, functions of inventory, inventory costs.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand a standard ship production system.	K2
CO2	Discern production planning and control for ship production activities	K2
CO3	Apply operation research, operation planning in shipbuilding process.	K2
CO4	Understand the information exchange during ship production.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	1	1							1
CO2	2	2	2			1						
CO3	2	2	2	1	2	1					1	1
CO4	2	2	2			1						1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Production	Richard Lee	Society of Naval Architects and Marine Engineers	Edition 2, 1995
2	Engineering for Ship Production	Thomas Lamb	Society of Naval Architects and Marine Engineers	Edition 1, 1986
3	Ship Design and Construction Vol I & II	Thomas Lamb (Ed)	Society of Naval Architects and Marine Engineers	Edition 3, 2004
4	Reed's ship construction For marine students	E A Stokoe	Thomas Reed Publications	Edition 5, 1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Construction and Welding	Nisith R. Mandal	Springer Series on Naval Architecture, Marine Engineering, Shipbuilding and Shipping 2	Edition 1, 2017
2	Merchant Ship Construction	H J Pursey	Nautical Press	1978
3	Shipbuilding Technology	Dormidontov VK	Mir Publishers, Moscow	1966

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105004
2	https://nptel.ac.in/courses/114105004
3	https://nptel.ac.in/courses/114105004
4	https://nptel.ac.in/courses/114105004

SEMESTER S6

SUBMARINES AND SUBMERSIBLES

Course Code	PENST634	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST205	Course Type	Theory

Course Objectives:

1. Understand the submersible System.
2. Discern submarine hydrostatics.
3. Understand submarine hydrodynamics.
4. Understand the basic structural aspects and engineering systems of a submarine.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	The submersible system, basic design of manned submersibles, design objectives, characteristics and development of submersibles, environment – physical properties of sea water, dynamical processes	9
2	Submarine hydrostatics, principles of flotation, submarines on the surface, arrangement of the main ballast tanks, submarine submerged, buoyancy elements, weight elements, trim and compensating tanks, special tanks, weight / space relationship, general form and arrangement	9
3	Submarine hydrodynamics, Resistance, appendage resistance, sway and heave resistance, resistance estimation, propulsion, dynamics and control, equation of motion of a submarine, hydrodynamic derivatives, stability and control in the horizontal plane, stability and control in the vertical plane, steering and depth control system.	9

4	Structure and systems, submarine structure, operational requirement for depth, pressure hull design, shape of the pressure vessel, buckling deformation of the shell, failure modes, pressure hull penetration, diving and surfacing system, Hydraulic systems - high pressure air systems - water systems -systems for hydrostatic control - environmental control system - provision for escape, - electrical systems.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the submersible System.	K2
CO2	Discern submarine hydrostatics.	K2
CO3	Understand submarine hydrodynamics.	K2
CO4	Understand the basic structural aspects and engineering systems of a submarine.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1									
CO2	2	1	1									
CO3	2	1	1									
CO4	2	1	1									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Submarine Design and Development	Norman Friedman	Conway Maritime Press Ltd	Edition , 1984
2	Concepts In Submarine Design - Cambridge Ocean Technology Series 02	Roy Burcher, Louis J. Rydill	Cambridge University Press	Edition 1, 2014
3	Theory of submarine Design	V.N. Kormilitsin, O.A. Kha Lizev	Reivera Maritime Medias	Edition 1, 2001
4	Submersible Vehicle Systems Designs	E Eugene Allmendinger	Society of Naval Architecture and Marine Engineering	1990

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Weapons and Warfare: Submarines - An Illustrated History of Their Impact	Paul E. Fontenoy	ABC-CLIO	Edition 1, 2007
2	Submarines in Color	Bill Gunston	Blandford color series	

SEMESTER S6

COMPUTATIONAL FLUID DYNAMICS IN MARINE TECHNOLOGY

Course Code	PENST635	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST205 PCNST302	Course Type	Theory

Course Objectives:

1. Understand the governing flow equations for common flow problems and to understand discretisation of a domain into a mesh
2. Understand the boundary conditions to be applied.
3. Understand numerical solution methods and their limitations and to understand the role of turbulence and its influence on fluid flows
4. Provide the student with a significant level of experience in the use of modern CFD software for the analysis of complex fluid-flow systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Computational Fluid Dynamics , CFD applications, numerical vs analytical vs experimental, modelling vs experimentation, fundamental principles of conservation, Reynolds transport theorem, conservation equations for mass, momentum and energy, convective forms of the equations and general description	9
2	Numerical methods for solving CFD problems, classification and overview of numerical methods, classification into various types of equations such as parabolic, elliptic and hyperbolic, boundary and initial conditions, overview of numerical methods, Finite Difference Method, Finite Element Method and Finite Volume Method	9
3	The basic structure of a CFD code, pre-processor, solver and postprocessor, user-defined subroutines, numerical grid generation, transformation and mapping, solution to some basic problems in fluid flow – free surface flows and N-S equations around a body in water, numerical flow simulation.	9
4	Features of turbulent flow, vorticity transport equation, general properties of	9

	turbulent quantities, Reynolds Average Navier Stokes (RANS) equation, closure problem in turbulence, types of turbulence models like eddy viscosity models, mixing length model, turbulent kinetic energy and dissipation, κ-ϵ model, RNG κ-ϵ model and κ-ω model, Reynolds Stress Model (RSM), Large Eddy Simulation (LES), Direct Numerical Simulation (DNS). A minor CFD project in ship hydrodynamics using any commercial or open-source CFD package.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Criteria for evaluation:

1. Problem Definition (K4 - 4 points)

- a. Clearly defines the CFD problem.
- b. Examine and identify relevant contextual factors (constraints, resources, objectives).

2. Problem Analysis (K4 - 4 points)

- a. Break-down and present a well-reasoned solution approach based on CFD.
- b. Compare and justify the proposed solutions with technical evidence.

3. Evaluate (K5 - 4 points)

- a. Thoroughly evaluate the proposed solutions for flow problems.
- b. Compare trade-offs, advantages, and disadvantages.
- c. Consider feasibility and practical implications.

4. Design Implementation (K5 - 4 points)

- a. Select the most feasible solution by implementing the proposed solutions.
- b. Successfully translate the chosen solution into tangible outputs.

5. Conclusion (K4- 2 points, K5 – 2 points)

- a. Summarizes findings and insights. State which solution is most appropriate for the problem. **(K4)**
- b. Reflects critical thinking and informed decision-making. **(K5)**

Scoring:

1. **Accomplished (4 points):** Exceptional analysis, clear implementation, and depth of understanding.
2. **Competent (3 points):** Solid performance with minor areas for improvement.
3. **Developing (2 points):** Adequate effort but lacks depth or clarity.
4. **Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the governing flow equations for common flow problems and to understand discretisation of a domain into a mesh	K1
CO2	Understand the boundary conditions to be applied.	K2
CO3	Analyse numerical solution methods and their limitations and to understand the role of turbulence and its influence on fluid flows	K4
CO4	Evaluate the output of modern CFD software for the analysis of complex fluid-flow systems	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3	3	3									1
CO4	3	3	3	2	3							1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computational Fluid Dynamics	Chung TJ	Cambridge University Press	Edition 2, 2016
2	Computational Methods for Fluid Dynamics	Joel H Ferziger, Milovan Peric	Springer	Edition 1, 2014
3	An Introduction to Computational Fluid Dynamics: The Finite Volume Method.	Versteeg, H.K, Malasekera, W	Pearson education.	Edition 3, 2002
4	Computational Fluid Dynamics for Engineers	T Cebeci, J R Shao, F Kafyeke, E Laurendeau	Horizons Publishing Inc	Edition 1, 2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Marine CFD 2011, Proceedings of Developments in Marine CFD, 22-23 March 2011, London, UK	RINA	Royal Institution of Naval Architects	2011
2	CFD 2003: Computational Fluid Dynamics Technology in Ship Hydrodynamics	RINA	Royal Institution of Naval Architects	2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_ae23/preview
2	https://onlinecourses.nptel.ac.in/noc24_ae23/preview
3	https://onlinecourses.nptel.ac.in/noc24_ae23/preview
4	https://onlinecourses.nptel.ac.in/noc24_ae23/preview

SEMESTER S6

SHIP DESIGN II

Course Code	PBNST604	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBNST504	Course Type	Theory

Course Objectives:

1. To enable the student to understand the standardization concept, cargo handling arrangements in ships and the design process of ship's rudder
2. To enable the student to understand the anchoring and mooring systems in ships and to determine the numbers and sizes of anchors, chains and mooring ropes
3. To provide the student an understanding of ship design methodologies to design general arrangement of typical merchant ships
4. To enable the student to understand the relevant regulations on life saving appliances and fire fighting appliances and to apply these regulations to typical merchant ships

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Standardisation, process and product standard. Cargo handling equipment, lifting devices. Cargo access, cargo hatches, design of hatch coamings. Steering gear, types, design aspects, design of rudder.	9
2	Anchoring and mooring arrangement. Types of anchors, anchor chain, anchor windlass, capstan, chain stopper, hawse pipe, spurling pipe, design of chain locker, bitter end connection. Equipment number calculation. Mooring systems, mooring machinery, deck fittings and structural arrangement.	9
3	Design of General Arrangement (GA) of a ship, factors influencing GA, subdivision of ship's hull, superstructure and deckhouses, arrangement of cargo spaces, cargo handling arrangement, arrangement of machinery spaces, arrangement of accommodation and associated spaces, arrangement of tanks and other spaces. Accommodation standards, space allocation, modular arrangement. Access equipment, hatches, manholes, doors, other closing and	9

	opening devices, load line rules on closing and opening devices, gangways and ladders. Guard rails and stanchions.	
4	Lights and sound signals, navigational mast. Life Saving Appliances (LSA), international regulations on LSA. Fire Fighting Appliances (FFA), international regulations on FFA. Structural fire protection. Ventilation, insulation, lining, partitions, floor covering and painting in accommodation spaces. Air pipes and sounding pipes in tanks. Ventilators in holds. Sea inlets.	9

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16marks)	- Each question carries 6 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 subdivisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the concept of standardisation in shipbuilding and the cargo handling arrangement in ships and explain the design process of rudder	K2
CO2	Carry out the equipment number calculations for ships and determine the numbers and sizes of anchors, chains and mooring ropes	K3
CO3	Apply ship design methodologies to develop general arrangement of typical merchant ships.	K3
CO4	Understand the relevant regulations on life saving appliances and fire fighting appliances applicable to ships and apply these regulations to typical merchant ships	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											1
CO2	3	2	2									
CO3	3	2	2									1
CO4	3					1	1					1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Design and Construction	Thomas Lamb (Ed.)	The Society of Naval Architects and Marine Engineers	Edition 1, 2003
2	Elsevier Ocean Engineering Book Series- Volume 1 (Practical Ship Design)	D. G. M. Watson	Elsevier	Edition 1, 1998
3	Ship Design and Construction	Robert Taggart (Ed.)	The Society of Naval Architects and Marine Engineers	Edition 1, 1980
4	Computational Ship Design	Myung-II Roh & Kyu-Yeul Lee	Springer	Edition 1, 2018
5	Ship construction	D. J. Eyres	Butterworth-Heinemann	Edition 6, 2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Merchant Ship Construction	D A Taylor	The Institute of Marine Engineers	Edition 4, 1998
2	The Maritime Engineering Reference Book	Anthony F. Molland (Ed)	Elsevier	Edition 1, 2008
3	Design Principles of Ships and Marine Structures	S. C. Misra	Taylor & Francis	Edition 1, 2016
4	Reeds Marine Engineering and Technology- Ship Construction for Marine Engineers	Paul A. Russel & E. A. Stokoe	Adlard Coles Nautical	Edition 6, 2016
5	Basic Ship Theory	K. J. Rawson & E. C. Tupper	Butterworth-Heinemann	Edition 5, 2001
6	Ship Design- Methodologies of Preliminary Design	Apostolos Papanikolaou	Springer	Edition 1, 2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105004
2	https://nptel.ac.in/courses/114105004
3	https://nptel.ac.in/courses/114105004
4	https://nptel.ac.in/courses/114105004

PBL Course Elements			
L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity		
Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
	Total	30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S6

FUNDAMENTALS OF FLOATING BODIES

Course Code	OENST611	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs 30 Min
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Acquire knowledge on various types of floating bodies such as ships.
2. Develop fundamental concepts of Naval Architecture.
3. Identify various types of materials used for construction of floating bodies and to identify various major and minor structural components of a ship.
4. Identify various main and auxiliary machineries onboard a vessel.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Types of ships, classification, cargo ships, general cargo ships, bulk carriers, container ships, ro-ro ships, barge carriers, tankers, fishing vessels, factory ships, supply ships, cable ships, ice breakers, research vessels, and warships, high speed crafts such as hydrofoils, air cushion vehicles etc, small pleasure crafts like yachts, ketches, etc, the ship's form, main dimensions and hydrostatic particulars, lines plan, form coefficients.	6
2	Physical fundamentals and forces acting on a ship - Archimedes principle, laws of floatation, stability of floating bodies, six degrees of freedom, forces acting on a ship in static condition and in waves, heel, trim, hydrostatic parameters, basic stability parameters.	9
3	Shipbuilding Materials and their properties, compositions, introduction to ship structural members, structural components such as bottom structure, shell plating and framing, decks, hatches and hatch covers, superstructures, bulkheads, tanks, holds, fore and aft structure, stern and rudder.	9

4	Propulsion machinery - development of ship propulsion, main engines - diesel engines, steam engines & turbines, gas turbines, diesel-electric drive, nuclear power plants, auxiliary machinery- power supply, auxiliary engines for ship, auxiliary engines for engine plant operation, steering gear	12
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify various types of ships.	K1
CO2	Understand fundamental concepts of Naval Architecture	K2
CO3	Identify and explain the various types of shipbuilding materials and various major and minor structural components of a ship	K2
CO4	Identify the main and auxiliary machinery systems onboard.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1									
CO2	2	1	1									
CO3	2	1	1									
CO4	2	1	1									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Naval Architecture	Tupper, E.C.	Butterworth-Heinemann	Edition 5, 2013
2	Ship Stability for Masters & Mates	C.B. Barrass & D.R. Derrett	Elsevier	Edition 7, 2019
3	Ship Construction	Eyres D J	Elsevier	Edition 7, 2012
4	Introduction to Marine Engineering	Taylor D A	Elsevier	Edition 2, 1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Maritime Engineering Reference Book	Molland A F (Ed)	Elsevier	Edition 1, 2008
2	Encyclopaedia of Ship Technology	Jan Babicz	Wärtsilä Corporation	Edition 2, 2015
3	Ship Knowledge – A Modern Encyclopaedia	Klaas van Dokkum	DOKMAR	2003
4	Lecture Notes in Basic Naval Architecture	Spyros Hirdaris	Aalto University Publication Series, Science & Technology 6/2021	Edition 1, 2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105003
2	https://nptel.ac.in/courses/114105003
3	https://nptel.ac.in/courses/114105003
4	https://nptel.ac.in/courses/114105031

SEMESTER S6

MARINE MATERIALS, CORROSION PREVENTION AND PROTECTION

Course Code	OENST612	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs 30 Min
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Discern the Evolution of marine materials
2. Understand types of corrosion of marine structures
3. Discern effective methods of material preservation and corrosion prevention
4. Understand the measures to be taken in design stage to minimise corrosion in marine structures.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to different types of materials used in marine construction/shipbuilding, Material fabrication and service requirement, Classification society requirements, Types of shipbuilding quality steels, Mild steels, normal strength steels (A, B, D, E classes), High Tensile Steels (HTS) grades, High Strength Low Alloy (HSLA), Aluminium alloys, alloy designation, welding requirements, Strength of aluminium compared to steel, Composition of aluminium alloys used in ship building, Advantages of using aluminium over steel in ship building.	9
2	Types of Corrosion in Marine Environment, Corrosion Identification, Factors affecting Corrosion. Types of steel and tendency of corrosion in various types of steel, corrosion of Titanium and Nickel alloys, Copper and Copper based alloys, Zinc, Aluminium and its alloys. Corrosion on propeller, marine machinery and deck fittings.	8
3	Material storage and preservation in shipyard, treatment of steel in shipyards. degreasing, weathering, mechanical surface cleaning, pickling, blast cleaning, flame cleaning. rust converters, chemical pre-treatment and	11

	comparison of pre-treatment methods. Paint schemes in ships and marine structures, classification of paints- common paint vehicles, drying oils, oleo-resins, alkyd resins, polymerizing chemicals and bitumen. Role of constituents of paints. suitability of each for various applications. typical paint schemes for underwater areas, boot top above water, weather decks, superstructures and tanks interiors	
4	Corrosion control - importance of corrosion protection, measures to minimise corrosion, corrosion control by design, corrosion inhibitors. cathodic protection- mechanism of cathodic protection, sacrificial anode, design of sacrificial anode system for ships and marine structures, advantages and disadvantages of sacrificial anode system. impressed current cathodic protection system in ships	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Discern the evolution of shipbuilding materials, and transition from wood to modern materials.	K1
CO2	Differentiate various types of corrosion of ship structures	K2
CO3	Implement effective methods of material preservation and corrosion prevention	K2
CO4	Use the measures to be taken in design stage to minimise corrosion in ships.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1									
CO2	2	1	1									
CO3	2	1	1									
CO4	2	1	1									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Material Science and Engineering	P.S.Gill	S K Kataria & Sons	Edition 1 2015
2	Marine Corrosion and Cathodic Protection	Chris Coogan	CRC Press	
3	Corrosion Control for Offshore Structures	Ramesh Singh	Gulf Publishing Company	Edition 1, 2014
4	Corrosion Engineering	Fontana M. G, Greene N. D	McGraw Hill,	Edition 2, 1978.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Corrosion and Protection of Steels in Marine Environments: State-of-the-Art and Emerging Research Trends	Philippe Refait Igor Chaves (Eds)	MDPI	Edition 1, 2022
2	Corrosion of Constructional Steels in Marine and Industrial Environment	Jayanta Kumar Saha	Springer	Edition 1, 2013
3	Designing Cathodic Protection Systems for Marine structures and vehicles	Harvey P Hack	SNAME	2000
4	Developments in Marine Corrosion	A Campbell, N. Campbell, F C Walsh	Royal Society of Chemistry	1998

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/113101098
2	https://nptel.ac.in/courses/113101098
3	https://nptel.ac.in/courses/113101098
4	https://nptel.ac.in/courses/113101098

SEMESTER S6

MARINE POLLUTION

Course Code	OENST613	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs 30 Min
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand different sources of marine pollution including ships and floating structures, their impact on the marine environment.
2. Select suitable methods to control the marine pollution in different operating conditions
3. Apply various strategies and methods for marine pollution recovery to minimise environmental impact.
4. Understand role of various international and regional rules and regulations for marine environmental protection in design of ships.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definition of Marine Pollution, Types of Marine Pollution, Various Sources of Marine Pollution, Marine Pollution in the Coastal Zone, Pollution from ships and offshore assets Concerns and Consequences of Marine Pollution - Global Warming, Sea Level Changes, Carbon Emissivity and Green Shipping Biodiversity	9
2	Prevention of air and water pollution from Ships, Emissions from Ships Engines, Fuel Oil Quality, Requirements for Survey and Issuance of International Air Pollution Prevention Certificate (IAPP), Oil Pollution, Segregated Ballast Tanks, Oily Water Separator, Ballast Water Management (BWM) Convention	9
3	Introduction to Marine Oil Pollution Recovery Systems: Types of Recovery Systems, Skimming Systems, Oil storage system, Treatment of pollutant after recovery.	9

4	IMO Marine Engine Regulations, Requirements for Survey and Issuance of International Air Pollution Prevention Certificate (IAPP), Energy Efficiency Design Index (EEDI), Ship Energy Efficiency Management Plan (SEEMP), Survey and Issuance of International Sewage Pollution Prevention Certificates (ISPP)	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand different sources of marine pollution including ships and floating structures, their impact on the marine environment.	K1
CO2	Select suitable methods to control the marine pollution in different operating conditions	K2
CO3	Apply various strategies and methods for marine pollution recovery to minimise environmental impact.	K2
CO4	Understand the role of various international and regional rules and regulations for marine environmental protection in design of ships.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1									
CO2	2	1	1									
CO3	2	1	1									
CO4	2	1	1									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Marine Pollution	Judith S Weis	Oxford University Press	Edition 1, 2014
2	Marine Pollution	R.B. Clark, C. Frid, M. Attrill	Oxford University Press	Edition 4, 1997
3	Marine Pollution	Icardo Beiras	Elsevier	Edition 1, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The International Convention for the Prevention of Pollution from Ships (MARPOL) 1973		International Maritime Organisation	As Amended
2	The International Convention for the Control and Management of Ships' Ballast Water and Sediments, 2004		International Maritime Organisation	As Amended
3	Resolution MEPC.378(80) - Guidelines for the control and management of ships' biofouling to minimize the transfer of invasive aquatic species		International Maritime Organisation	As Amended
4	The Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships, 2009		International Maritime Organisation	As Amended

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/103107215
2	https://nptel.ac.in/courses/103107215
3	https://nptel.ac.in/courses/103107215
4	https://nptel.ac.in/courses/103107215

SEMESTER S6

BASIC SHIP STRUCTURES

Course Code	OENST614	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Discern shipbuilding materials, role of classification societies and ship framing systems
2. Understand the shell and bottom structures of a ship
3. Discern basic structural drawings and understand the structure of bulkheads and tanks
4. Understand the structure of decks, superstructures and the end structures

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Shipbuilding materials, wood, steel, stress-strain diagrams, higher tensile steel, aluminium alloys, proof stress, GRP and FRP Classification societies, longitudinal and transverse strength. Framing system, longitudinal, transverse and combined framing, basic structural components - primary and, secondary stiffeners	9
2	Bottom construction, single and double bottom, functions, inner bottom plating, floors, transversely and longitudinally framed double bottom, additional stiffening in critical areas, testing of double bottom compartments, foundations Shell plating- bottom shell plating, side shell plating, additional stiffening for panting, navigation in ice, bilge keel	9
3	Typical ship structural drawings such as midship section, shell expansion, structural profile. Bulkheads, spacing of watertight bulkheads, construction of watertight bulkheads, testing of watertight bulkheads, watertight doors. Tanks, deep tanks, testing of deep tanks, topside tanks, shaft tunnel	9

4	Decks and superstructure, deck plating, deck stiffening, pillars, hatch coamings, hatch covers, bulwarks, forecastle, poop structure, superstructures and deck houses, weather tight doors. Fore end structure, stem, bulbous bows, chain locker, hawse pipes, bow thruster units, aft end structure, stern construction, stern frame, rudders, steering gear; stern tube, shaft bossing, P and A brackets	9
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Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Discern shipbuilding materials, role of classification societies and ship framing systems	K1
CO2	Understand the shell and bottom structures of a ship	K2
CO3	Discern basic structural drawings and understand the structure of bulkheads and tanks	K2
CO4	Understand the structure of decks, superstructures and the end structures	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1									
CO2	2	1	1									
CO3	2	1	1									
CO4	2	1	1									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Structural Analysis and Design	O F Hughes, Jeom Kee Paik	Society of Naval Architects and Marine Engineers	Edition 1, 2010
2	Torsion and Shear Stresses in Ships	M Shama	Spinger	Edition 1, 2010
3	Local Strength of Ship Structures	Piero Caridis	CRC Press	Edition 1, 2024
4	Ship Structural Design Concepts	J Harvey Evans	Cornel Maritime Press	Edition 2, 1983
5	Buckling of Ship Structures	M Shama	Spinger	Edition 1, 2013
6	Ship Construction	D J Eyres	Butterwoth Heinemann	Edition 7, 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Rules for classification of ships by various class societies, updated from time to time		LR, ABS, BV	Latest
2	Strength of Ship Structures	W Muckle	Edward Arnold Publishers Ltd	Edition 2, 1967
3	Strength of Ships and Ocean Structures, The Principles of Naval Architecture Series	Alaa Mansour	The Society of Naval Architects and Marine Engineers	Edition 3, 2008
4	Elementary Beam Theory and the Hull Girder	G W White	Standard Maritime London	Edition 1, 1974

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105031
2	https://nptel.ac.in/courses/114105031
3	https://nptel.ac.in/courses/114105031
4	https://nptel.ac.in/courses/114105031

SEMESTER S6

MARINE ENGINEERING LAB

Course Code	PCNSL607	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST503	Course Type	Lab

Course Objectives:

1. Familiarise with various machineries used in the ship.
2. Dismantle & Assemble Various Marine Auxiliary Machineries.
3. Analyse the results to understand the performance characteristics of engines.
4. Learn about the advanced technologies and research areas in engines.

Expt. No.	Experiments (Minimum 12 are mandatory)
1	Study of Internal Combustion Engine and its Components. Equipment: Internal Combustion Engine.
2	Study of Boilers, Preparation for Firing Up of Boiler. Equipment: Water Tube Boiler
3	Valve Timing Diagram of Four Stroke Vertical Diesel Engine & Post Timing Diagram of Two Stroke Petrol Engine. Equipment: Internal Combustion Engine.
4	Load Test on Four Stroke Single Cylinder Vertical Diesel Engine. Equipment: Four Stroke Single Cylinder Vertical Diesel Engine.
5	Morse Test on Four Stroke Four Cylinder Petrol Engine. Equipment: Four Stroke Four Cylinder Petrol Engine.
6	Load Test on Four Stroke Single Cylinder Horizontal Diesel Engine. Equipment: Four Stroke Single Cylinder Horizontal Diesel Engine.
7	Speed Variation Test on Four Stroke Horizontal Diesel Engine. Equipment: Four Stroke Horizontal Diesel Engine.
8	Retardation Test on Four Stroke Single Cylinder Four Stroke Single Cylinder Vertical Diesel Engine. Equipment: Four Stroke Single Cylinder Vertical Diesel Engine.
9	Load Test on Slow Speed Diesel Engine. Equipment: Four Stroke Single Cylinder Vertical Diesel Engine.
10	Testing of Fuels and Lubricants - Determination of flash and fire points of petroleum

	products - Determination of kinematic and absolute viscosity of lubricating oils - Determination of calorific values. Equipment: Calorimeter, Redwood Viscometer Etc.
11	Energy Balance of a Diesel Engine. Equipment: Four Stroke Horizontal Diesel Engine.
12	Determination of the Characteristic Curves of Centrifugal Pumps. Equipment: Centrifugal Pump Test Rig.
13	Determination of the Characteristic Curves of Screw Pump. Equipment: Screw Pump Test Rig.
14	Overhauling of Gate Valve, Globe Valve Equipment: Gate Valve, Globe Valve.
15	Overhauling of Reciprocating Pump/ Centrifugal Pump/ Screw Pump Equipment: Reciprocating Pump/ Centrifugal Pump/ Screw Pump.
16	Overhauling of Diesel Engines - Equipment: Four Stroke Horizontal Diesel Engine.
17	Study of Fuel Injection Valve and Pump. Equipment: Fuel Injection Valve and Pump
18	Study of Various Types of Filters and Incinerator. Equipment: Fuel Oil Filters and Incinerator
19	Study of Cold starting of diesel engines techniques Equipment: Diesel Engine.
20	Study of Internal Combustion Engine and its Components. Equipment: Internal Combustion Engine.

Course Assessment Method

(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- **Submission of Record:** Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- **Endorsement by External Examiner:** The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Familiarise with various machineries used in the ship.	K2
CO2	Dismantle & Assemble Various Marine Auxiliary Machineries.	K2
CO3	Analyse the results to understand the performance characteristics of engines and Various Marine Auxiliary Machineries.	K4
CO4	Learn about the advanced technologies and research areas in engines	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3	3										
CO4	3											2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Marine Engineering	Edited by Roy L Harrington	SNAME, New York.	1992
2	Introduction to Marine Engineering	Taylor D.A	Butterworth-Heinemann, London.	Second Edition 1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles & Practices of Marine Diesel Engines	D.K. Sanyal	A Bhandarkar Pub.,	1981
2	Low Speed Marine Diesel	John B Woodward	Wiley	First Edition 1981
3	Marine Auxiliary Machinery	H.D. Mc George	Elsevier	1999
4	Reed's General Engineering Knowledge for Marine Engineers	Leslie Jackson, Paul Anthony Russell, and Thomas D Morton	Adlard Coles Nautical	1986
5	Modern Marine Engineer's Manual	Everett C. Hunt	Schiffer Publishing Ltd	Edition 3, 2009

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

**NAVAL ARCHITECTURE & SHIPBUILDING
ENGINEERING**

SEMESTER S7

JOINING TECHNIQUES IN SHIPBUILDING TECHNOLOGY

Course Code	PENST741	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PENST641	Course Type	Theory

Course Objectives:

1. Explain types of welding, welding process and welding parameters used in shipbuilding.
2. Discern various special welding arrangements in modern shipbuilding.
3. Discern the radiant welding, underwater welding and robotic welding used in shipbuilding.
4. Explicate the various types of weld defects and testing in modern shipbuilding.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to various joining techniques in ship building industry. Understand welding and classification of welding Fuel gases, GMAW process, different metal transfers and shielding gas. Different types of oxy-acetylene flames, Fore hand and back hand welding techniques. Function of fillers and fluxes. Carbon arc welding, Shielded metal arc, submerged arc, TIG, MIG, Electro slag, Electro gas, Plasma arc, arc spot welding, stud welding	9
2	Principle, welding arrangements and advantages/ disadvantages of resistance welding, spot welding, seam welding, projection welding, resistance upset welding, flash butt, percussion welding, tube welding, high frequency welding, solid state welding, cold welding, diffusion welding, ultrasonic welding, explosive, friction and inertia welding, thermo-chemical welding (thermit and atomic hydrogen welding. Forging,	11
3	Radiant energy welding (electron beam and laser beam). Under water welding. Wet, dry, hyper baric and cavity welding. Welding arrangements and special precautions to be taken. Robotic Welding. Application of Robotic Welding in Ship Production, Types of Welding Robots and their control system. Brazing and	10

	soldering. Difference between brazing, soldering and welding. Adhesives and bonding methods.	
4	Weld Defects and distortion control. Tests destructive/ non-destructive. Welding performance and procedure qualification, test reports, acceptance Standards. Variables essential/ non-essential. Consumable Classification & Coding. Crystallization of a Pure Metal, Phase Transformation in Iron-Carbon Diagram	7

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain types of welding, welding process and welding parameters used in shipbuilding.	K1
CO2	Discern various special welding arrangements in modern shipbuilding.	K2
CO3	Discern the radiant welding, underwater welding and robotic welding used in shipbuilding.	K2
CO4	Explicate the various types of weld defects and testing in modern shipbuilding.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyze, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											
CO2	3	2	1	1	1		1		1			1
CO3	3	2	1	1	1		1	1	1			
CO4	3	2	1		1	1	1		1	1		1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Construction and Welding	Nisith R. Mandal	Springer Series on Naval Architecture, Marine Engineering, Shipbuilding and Shipping 2	Edition 1, 2017
2	Welding Engineering and Technology	R.S. Parmar	Khanna Publishers	Edition 3, 2022
3	A Textbook of Welding Technology	O P Khanna	Dhanpat Rai & Sons	Edition 11, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Standard symbols for welding, brazing, and nondestructive examination		American Welding Society	Edition 7, 2012
2	Merchant Ship Construction	H J Pursey	Nautical Press	1978
3	Shipbuilding Technology	Dormidontov VK	Mir Publishers, Moscow	1966

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105004
2	https://nptel.ac.in/courses/114105004
3	https://nptel.ac.in/courses/114105004
4	https://nptel.ac.in/courses/114105004

SEMESTER S7
COMPOSITE BOAT DESIGN

Course Code	PENST742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand composition and processing of composite materials.
2. Explain various processes of composite boat construction.
3. Understand mechanical and structural properties of composites.
4. Undertake structural design and analysis of composite ships.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to composite materials, history of fibre reinforced composites, constituent materials, fibres, matrix, fillers, additives, properties of typical composite materials, application of composites, processing FRP composites, molding, spray up, hand lay-up, compression moulding, matched dies moulding, forming methods employing gas pressure, low pressure closed moulding, pultrusion, filament winding.	9
2	FRP boat construction, single skin, sandwich construction, types of frames (top hat, with or without core), bulkhead, typical midship section of FRP boat, connections, hull to deck, bulkhead to hull, hull moulding, deck moulding, environmental control, temperature and humidity control, FRP boat construction quality control, destructive and non destructive testing	9
3	Micro mechanical analysis, volume and weight fractions, longitudinal strength and stiffness, transverse section modulus, shear modulus, Poisson's ratio, structural analysis, elastic properties of unidirectional lamina, stress strain relationships, analysis of laminated composites, basic assumptions, stress- displacement / stress-strain relationship, coupling effect, types of laminate configuration (symmetric, anti-symmetric)	9

4	Structural design of FRP boats, basic structural arrangement using classification society rules, Determination of boat shell lamination using classification society rules, Determination of stiffener lamination and size using classification society rules.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand composition and processing of composite materials.	K2
CO2	Explain various processes of composite boat construction.	K2
CO3	Understand mechanical and structural properties of composites.	K2
CO4	Carry out structural design and analysis of composite ships.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2						1					
CO2	2	1	1		1							1
CO3	2	2	1				1					
CO4	2	2	1	1	1							1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Marine Composites	Erric Greene	Erric Greene Associates	Edition 2, 1999
2	Cambridge Ocean Technology Series 4 & 5 - Composite Materials in Maritime Structures Vol 1 & 2	R A Shenoy (ed)	Cambridge University Press	Edition 1, 2008
3	Mechanics of composite materials and structures	Mukhopadhyay M	Universities Press	2005
4	Mechanics of composite materials.	Jones, Robert M	CRC press	1998

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Rules and regulations for the construction and classification of high speed crafts and light crafts, 2016.	-	Indian Register of Shipping	2016
2	Durability of Composites in a Marine Environment	Peter Davies, Y D S Rajapakse	Springer	Edition 1, 2014
3	Marine Composites Design and Performance	R Pemberton (Ed)	Elsevier	Edition 1, 2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/101104010
2	https://nptel.ac.in/courses/101104010
3	https://nptel.ac.in/courses/101104010
4	https://nptel.ac.in/courses/101104010

SEMESTER S7
REFRIGERATION AND AIR CONDITIONING OF SHIPS

Course Code	PENST743	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand basic principles of refrigeration on board ships.
2. Explain details of refrigerating machinery.
3. Understand basic principles of air-conditioning on-board ships.
4. Understand standards for marine air conditioning systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction, marine applications of mechanical refrigeration, refrigerated ship's stores refrigeration at sea, cargoes and carrying conditions, thermodynamical principles vapour, compression cycles (CARNOT, superheating, subcooling, multistage operation), absorption cycles.	9
2	Refrigerating machinery, principal components, compressors, condensers, evaporators, regulators. refrigerant properties- demands, choice, safety, lubricants, refrigerant numbering system, alternative refrigerants, secondary refrigerants, methods of cooling- cargo chamber cooling in conventional reefers, methods of cooling containers, insulations, heat leakage estimation, total load on the refrigeration plant	10
3	Air conditioning fundamentals, definitions, psychrometric chart, air conditioning (heating), heat sources, heat exchangers, heat pump, air conditioning (cooling), principal arrangement and equipment.	7
4	Standards for marine air conditioning systems, air flow & air conditioning capacity calculations, air conditioning systems, HVAC systems and components- single zone system, multiple zone systems, terminal reheat system, dual duct system, variable air volume systems, water systems, unitary systems, cargo hold dehumidification systems.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions.

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand basic principles of refrigeration on board ships.	K2
CO2	Explain details of refrigerating machinery.	K2
CO3	Understand basic principles of air-conditioning on-board ships.	K2
CO4	Understand standards for marine air conditioning systems.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1					1			1		
CO2	2	1	1		1		1			2		1
CO3	2	2	1		1		1			2		1
CO4	2	2	1	1	1		1			2		1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Marine Refrigeration and Air-Conditioning	James A. Harbach	Cornell Maritime Press	2005
2	Textbook of Refrigeration and Air Conditioning	R.S. Khurmi, J. K. Gupta	S. Chand	2016
3	Refrigeration and Air Conditioning	P.I. Ballaney	Khanna Publishers	1985

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Refrigeration and Air Conditioning	Hundy G.H., Trott A.R., Welch T.C.	Butterworth-Heinemann	Edition 4, 2008
2	Modern Refrigeration and Air Conditioning	Andrew D. Althouse, Carl H. Turnquist, A.F. Bracciano	Goodheart-Willcox Co	Edition 1, 2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/112105128
2	https://nptel.ac.in/courses/112105128
3	https://nptel.ac.in/courses/112105128
4	https://nptel.ac.in/courses/112105128

SEMESTER S7
DESIGN OF MACHINE ELEMENTS

Course Code	PENST744	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs 30 Mins
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand basic principles of design of machine elements.
2. Explain details of joints.
3. Understand basic principles of drive elements and shaft couplings.
4. Understand standards bearings and gears

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of machine design, definitions, design process, design principles, design criteria, stresses in machine parts-working stress, safe stress, factor of safety, endurance limits, fatigue factors elastic springs-classification and uses of springs-allowable stresses and deflections-design for fluctuating loads	9
2	Joints, principles of force transmission, detachable joints (pins, keys, spline and bolted joints), non detachable joints, welded, soldered and glued joints, riveted joints, strength of welded and riveted joints.	8
3	Drive elements, shafts, torsion and bending of shafts, design of shafts for strength and deflection, effect of key ways, crank shafts, shaft couplings, rigid coupling (flange and compression couplings), couplings with kinematics flexibility slip couplings, fluid couplings.	9
4	Bearings, slide bearings-introduction to lubrication, hydrodynamic bearings, bearing materials, design of slide bearings. roller bearings, types, static & dynamic load, capacity, bearing life and selection of bearings. gears: - types (spur and parallel helical gears) and function of gears, strength of gear teeth, stresses and stress concentration in gears-design of gears.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand basic principles of design of machine elements.	K2
CO2	Explain details of joints.	K2
CO3	Understand basic principles of drive elements and shaft couplings.	K2
CO4	Understand standards bearings and gears	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	2	2	1	2			1		1
CO2	2	1	1	1	2	1	2			1		1
CO3	2	2	2	2	2	1	2			1		1
CO4	2	2	2	2	2	1	2			1		1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanical Engineering Design	J.E.Shigley	McGraw-Hill	2003
2	Machine Design.	R.K.Jain	Khanna Publications, New Delhi	2017
3	Design of machine elements	Bhandari.V	McGraw Hill education	2017
4	A text book of machine design	P.C.Sharma, D.K.Aggarwal	S.K.Kataria& sons	2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design data hand book	K. Mahadevan, Dr. K. Balaveera Reddy	CBS Publishers	2014
2	Machine design	U.C. Jindal	Pearson education	2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/112105125
2	https://nptel.ac.in/courses/112105125
3	https://nptel.ac.in/courses/112105125
4	https://nptel.ac.in/courses/112105125

SEMESTER S7
DESIGN OF HIGH SPEED CRAFTS

Course Code	PENST745	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBNST504 PBNST604	Course Type	Theory

Course Objectives:

1. Understand the requirements for high speed craft market.
2. Develop the dimensions and hull form for high speed craft.
3. Select the propulsion package for the selected ship
4. Optimise the structure of high speed craft

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Basic differences between conventional and high-performance crafts , special features, types, monohull , planning craft, hydrofoil craft , air cushion vehicles, multihulls – catamaran – trimaran, SWATH vessels, Unconventional hulls	7
2	Hull forms, aero and hydrodynamic forces, stability criteria, resistance of very slender hulls, in particular importance of form factors, transom stern effects prediction of form factors for transom stern vessels, effect of water depth on HSE resistance.	8
3	Resistance prediction methods for HSC, regression methods, , thin strip theory, assumptions, approximation, CFD, hybrid methods, powering and propulsion, seaworthiness, maneuvering and control, comparative performance and applications of propulsion devices of water jets and other devices	12
4	Structural design of high-speed craft, slamming loads and structure responses, materials and structures, structural design philosophy, modeling and analysis, class society rules	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Criteria for evaluation:

1. Problem Definition (K4 - 4 points)

- a. *Clearly define owner's requirements of a high speed craft*
- b. *Examine and identifies relevant contextual factors (constraints, resources, objectives).*

2. Problem Analysis (K4 - 4 points)

- a. *Analyse the owners requirements and bring out possible dimensions as solution*
- b. *Compare and justify the proposed solutions with technical evidence.*

3. Evaluate (K5 - 4 points)

- a. *Thoroughly evaluate the proposed solutions.*
- b. *Compare trade-offs, advantages, and disadvantages.*
- c. *Consider feasibility and practical implications.*

4. Design Implementation (K5 - 4 points)

- a. *Select the most feasible design by implementing the proposed solutions.*
- b. *Successfully translate the chosen solution into tangible outputs.*

5. Conclusion (K4- 2 points, K5 – 2 points)

- a. *Summarizes findings and insights. State which design is most appropriate for the problem. (K4)*
- b. *Reflects critical thinking and informed decision-making. (K5)*

Scoring:

1. **Accomplished (4 points):** *Exceptional analysis, clear implementation, and depth of*

understanding.

2. **Competent (3 points):** Solid performance with minor areas for improvement.
3. **Developing (2 points):** Adequate effort but lacks depth or clarity.
4. **Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. - Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the requirements for high speed craft market.	K1
CO2	Develop the dimensions and hull form for high speed craft.	K3
CO3	Select the propulsion package for the selected ship	K4
CO4	Optimise the structure of high speed craft	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	2	1		1	1			1
CO2	3	2	2	2	2	1			1			1
CO3	3	2	2	2	2	1	2		1			1
CO4	3	2	2	2	2	1			1			1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hydrodynamics of High-Speed Vehicles	Odd M. Faltinsen	Cambridge University Press	2005
2	High Performance Marine Vessels	Liang Yun , Alan Bliault	Springer	2012
3	Rules for Building and Classing High-Speed Craft		American Bureau of Shipping	2013
4	Ship Structure Committee Report No 439 - Comparative Structural Requirements for High Speed Craft	Kevin F. Stone	Ship Structure Committee	2005
5	Ship Structure Committee Report No 464 - Design and Detailing for High Speed Aluminum Vessels Design	Mish et al	Ship Structure Committee	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Innovations in High Speed Marine Vessels	RINA Conference	The Royal Institution of Naval Architects	2009
2	Recent Developments in High-Speed Transport	Dmitry Yu. Strelets, Oleg N. Korsun (Ed)	Springer	2023
3	Considerations on the Structural Design of High Performance Marine Vehicles	Allen R.G., Jones R.R	Society of Naval Architects and Marine Engineers	1977

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105030
2	https://nptel.ac.in/courses/114105030
3	https://nptel.ac.in/courses/114105030
4	https://nptel.ac.in/courses/114105030

SEMESTER S7
OCEAN WAVE HYDRODYNAMICS

Course Code	PENST751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST302	Course Type	Theory

Course Objectives:

1. Understand the Ocean environment, graphical and mathematical representation of waves
2. Discern various types of wave loads on structures
3. Understand wave theories
4. Discern wave deformations and understand random wave theory

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Wave Motion : Classification of Waves, Derivation of the Velocity Potential, Dispersion Relationship, Celerity in Different Water Depth Conditions, Local Fluid Particle Velocities and Acceleration Under Progressive Waves, Water Particle Displacement Under Progressive Waves, Pressure Distribution Under Progressive Waves, Group Celerity, Wave Energy, Wave Power, Simple Problems on Wave Motion.	9
2	Wave Loads: Force Regimes, Design Wave Approach, Morison Equation- Fixed Cylinder in Waves, Fixed Cylinder in Waves and Current, Flexible Cylinder in Waves, Wave Forces on an Inclined Cylinder, Wave Force on a Vertical Cylinder in Deep water, Wave Forces on Piles in Shallow Water, Submarine Pipelines, Froude-Krylov Forces, Diffraction Regime, Simple Problems on Wave Loads.	9
3	Finite Amplitude Waves: Stoke's Wave Theory- Comparison Between Wave Theories, Solitary Wave Theory, Cnoidal Wave Theory, Stream Function Theory, Simple Problems on Finite Amplitude Waves.	9

4	Wave Deformation: Wave Refraction, Wave Diffraction, Wave Breaking- Types of Wave Breaking, Waves on Currents, Simple Problems on Wave Deformation, Random Waves: Generation of Ocean Waves, Importance of Study of Wave Data, Collection of Wave Data, Analysis of Ocean Waves- Statistical Method, Spectral Method, Fast Fourier Transform Method, Irregular Sea Way- Wave-Wave Spectrum Relationship (Theory Only).	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the Ocean environment, graphical and mathematical representation of waves	K2
CO2	Discern various types of wave loads on structures	K3
CO3	Understand wave theories	K2
CO4	Discern wave deformations and understand random wave theory	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	2	2	1	2	1					1
CO2	2	1	2		3	2	2	2		2		
CO3	2	1	2	2	2	1	2	1	1	3	1	1
CO4	2	4	2	2	3	1	3				4	1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ocean Surface Waves: Their Physics and Prediction	Stanisław Ryszard Massel	World Scientific	Edition 1, 2017
2	Ocean Waves: The Stochastic Approach	Michel K. Ochi	Cambridge University Press	Edition 1, 2005
3	Water Wave Mechanics of Engineers and Scientists	Robert G. Dean, Robert A. Dalrymple	Allied Publishers Limited	Edition 1, 1991
4	Sea Loads on Ships and Offshore Structures	O. Faltinsen	Cambridge University Press	Edition 1, 1993

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Wave Mechanics for Ocean Engineering	P. Boccotti (Eds.)	Elsevier oceanography series	Edition 1, 2000
2	Handbook of Ocean Wave Energy	Arthur Pecher, Jens Peter Kofoed (eds.)	Springer International Publishing	Edition 1, 2017
3	Water wave mechanics for engineers and scientists	Robert G. Dean, Robert A. Dalrymple	World Scientific	Edition 1, 1991

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114106033
2	https://nptel.ac.in/courses/114106033
3	https://nptel.ac.in/courses/114106033
4	https://nptel.ac.in/courses/114106033

SEMESTER S7
COMPUTER AIDED SHIP DESIGN

Course Code	PENST 752	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBNST 504 PBNST 604	Course Type	Theory

Course Objectives:

1. Understand Computer aided Ship Design Process.
2. Discern Numerical representation method of Hull form and Ship Geometry.
3. Demonstrate CAD Tool in developing preliminary faired lines design for hull form generation and presenting in 3D form.
4. Apply optimization techniques in generating basic data in development of detailed design from preliminary design.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Numerical Techniques in Computer Aided Ship Design - Numerical Interpolation: Differences, Newton's Forward Difference interpolation formula, Lagrangian Interpolation formula. Numerical Integration: Integration formulas. Curve Fitting: Fitting of Polynomials, Least Square curve fitting technique, choosing the degree of the polynomial, Ill-conditioning difficulties, Orthogonal Polynomial fitting	9
2	Preliminary Ship Design, Preliminary Parameter Estimation: Displacement, Length, Breadth, Depth, Draught, Block Coefficient, Check on Transverse & longitudinal Stability, Freeboard, etc., Estimation of Power, Capacity, Basic Ship Method, Computer Aided Preliminary Ship Design: Preliminary Parameter Selection,	11
3	Lines Design and Fairing: Manual Graphical method, Computer Aided Curve fitting Techniques Fairing Principles, Spline Fitting: Cubic Spline curve, Bezier Curve, B-Spline Curve.	10

4	Optimisation Methods in Ship Design Introduction-Modelling of Design as Optimisation, Different Optimisation Methods, Linear and Non-linear optimization, Application to Ship Design	7
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand Computer aided Ship Design Process.	K2
CO2	Discern Numerical representation method of Hull form and Ship Geometry.	K1
CO3	Demonstrate CAD Tool in developing preliminary faired lines design for hull form generation and presenting in 3D form.	K3
CO4	Apply optimization techniques in generating basic data and development of detailed design from preliminary design.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	1							1
CO2	3	2	2			1				2		
CO3	3	2	2	2	2	1		1	1	1	2	
CO4	3	2	2			1	1					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computational Geometry for ships	H. Nowacki, et.al.	World Scientific Publishing Co. Pvt. Ltd	Edition 1, 1995
2	Mathematical Elements for Computer Graphics, 1990	David F. Rogers & J Alan Adams	McGraw-Hill International Editions,	Edition 1, 1990
3	Engineering Mathematics in Ship Design	Cristiano Fragassa et.al	MDPI	Edition 1, 2019
2	Computational Ship Design	Myung-II Roh, Kyu-Yeul Lee	Springer	Edition 1, 2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Methods for Ship Surface Design	Chengi Kuo	Longman,	1971
2	Computer Applications in Ship Technology	Chengi Kuo	Heyden& Son Ltd.	1977
3	Computer Methods in Preliminary Ship Design	L.K Kupras	Delft University Press	1983

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/112104031
2	https://nptel.ac.in/courses/112104031
3	https://nptel.ac.in/courses/112104031
4	https://nptel.ac.in/courses/112104031

SEMESTER S7

FINITE ELEMENT ANALYSIS IN NAVAL ARCHITECTURE

Course Code	PENST753	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST403	Course Type	Theory

Course Objectives:

1. Understand basics of finite element methods
2. Discern finite element fundamentals like shape functions, convergence, stiffness matrix etc
3. Understand FEA applications in ship structural analysis
4. Discern pre processing, processing and post processing considerations in ship structural analysis.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Scope of finite element method in solving engineering problems, General steps in finite element analysis, variational formulations and weighted residual methods, Analysis of trusses, beams, and frames, Analysis of plates, Analysis of three-dimensional problems	9
2	Shape functions, Convergence criteria, General equations for calculation of stiffness matrix, Derivation of stiffness matrix for truss beam, Plane stress, plane strain, axisymmetric elements.	9
3	Ship Structural Analysis using FEM- formulation of plate finite elements, issues associated with plate formulation of finite elements. Numerical examples on simple plate analysis. One dimensional and two dimensional finite element modelling of ship structure	9
4	Preprocessing - preliminary checks, engineering model checks - analysis type and assumptions, geometry assumptions, material properties, stiffness and mass properties, loads and boundary conditions, finite element model checks, element types, mesh design, substructures and sub-modelling, FE model loads and boundary conditions, processing, post processing – methods, displacement results, stress results, fatigue and fracture results	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand basics of finite element methods	K1
CO2	Discern finite element fundamentals like shape functions, convergence, stiffness matrix etc	K1
CO3	Understand FEA applications in ship structural analysis	K2
CO4	Discern preprocessing, processing and post processing considerations in ship structural analysis.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	1	1							
CO2	2	1	2	2		1	2			2		
CO3	2	2	1	1	2	1		1	1	1		
CO4	2	2	2			1	1					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Finite Element Method	O.C.Zienkiewicz	Mc Graw Hill	Edition 7, 2013
2	Fundamentals of Finite Element Analysis	David V. Hutton	Mcgraw Hill Education	Edition 3, 2017
3	Finite Element Modeling for Stress Analysis	R D Cook	John Wiley & Sons	1995

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Structure Committee Report 387 - Guideline for Evaluation of Finite Elements and Results	R.1. Basu, K.J. Kirkhope, J. Srinivasan	Ship Structure Committee	1997
2	Ship Structure Committee Report 475 - Guideline for Evaluation of Marine Finite Element Analysis	E.D. Wang, J.S. Bone, M. Ma and A. Dinovitzer	Ship Structure Committee	2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/112106135
2	https://nptel.ac.in/courses/112106135
3	https://nptel.ac.in/courses/112106135
4	https://nptel.ac.in/courses/112106135

SEMESTER S7
DESIGN OF FISHING VESSELS

Course Code	PENST754	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Acquire basic knowledge on fishing vessels and systems onboard.
2. Understand basic design procedure of fishing vessels
3. Understand resistance and propulsion performance characteristics of fishing vessels.
4. Understand basic structural design concepts of fishing vessels.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: definitions of fishing vessel, special features of fishing vessels, regulations for the safety of fishing vessels, classification of fishing vessels, Characteristics of fish ground; Fishing gear and methods, drift net, long line, drag net, seine net, dredging, with electric light, harpoon/whale catching, trawling (side and stern trawlers, single and pair trawling, pelagic & bottom trawling).	9
2	Design Procedure: Owner's specifications, Economy, fuel efficiency, hull form, investment cost, operating revenues and costs. Design of Main Dimensions and form: parent vessel data analysis, space requirement (capacity) of the whole ship, estimation of main dimensions, estimation of form coefficients, estimation of light ship weight, estimation of dead weight, design of lines. General arrangement: Engine room, fish holds, erections, deck machinery arrangement, crew accommodation, fuel, fresh-water, ballast tanks, bulkhead positions.	9
3	Resistance, powering and propeller selection: friction resistance, wave making resistance, eddy resistance, net resistance, powering calculations, propeller selection, Propulsion Systems and other machinery/equipment, different	9

	propulsion systems and selection of main engine, selection of equipment's/instruments for fish finding, navigation, communication, firefighting, lifesaving and net monitoring Seakeeping and manoeuvring considerations	
4	Material and construction methods: mechanical properties of materials, comparison of hulls of different material, type of construction, details of steel construction, structural design methods, construction methods in FRP/GRP, Aluminium and Ferro-cement.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Acquire basic knowledge on fishing vessels and systems onboard.	K1
CO2	Understand basic design procedure of fishing vessels	K1
CO3	Understand resistance and propulsion performance characteristics of fishing vessels.	K2
CO4	Understand basic structural design concepts of fishing vessels	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	1	1							
CO2	2	2	2			1						
CO3	2	2	2	1	2	1		1	1			
CO4	2	2	2			1	1					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of small fishing vessels	John F. Fyson	Food and Agriculture Organization	Edition 1, 1985
2	Modern Fishing Gear Technology	M Shahul Hameed		Edition 3, 2022
3	Fishing boat designs 1. Flat bottom boats	A F Haug	Food and Agriculture Organization	Edition 1, 1972
4	Fishing boat designs 2. V-bottom boats of planked and plywood construction (Rev.2)	O Gulbrandsen	Food and Agriculture Organization	Edition 2, 2004
5	Fishing boat designs 3. Small Trawlers	John F. Fyson	Food and Agriculture Organization	Edition 2, 1991
6	Fishing boat designs 3. Small Steel Fishing Boats	D J Eyres	Food and Agriculture Organization	Edition 1, 1984

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fishing boats of the world	Latest edition		
2	Guidelines for the Design, Construction and Equipment of Small Fishing Vessels		FAO/ILO/IMO	2005

SEMESTER S7
RENEWABLE ENERGY IN MARITIME SECTOR

Course Code	PENST755	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand ocean renewable energy sources.
2. Discern the tidal energy sources and power generation
3. Explore the offshore wind energy sources and power generation
4. Explore the wave energy sources, OTEC and power generation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	The global energy mix, climate change & sustainability, Introduction to ocean energy sources and types, Methods for ocean observations: Water level Measurements - Tidal poles, tidal gauges, pressure sensors, Radar sensors. Current measurements, Wave measurements	9
2	Tidal Energy, Tide generating forces, progressive waves, Cotidal charts, standing waves, resonance, Coriolis forces, Kelvin Waves, tidal analysis & prediction, compound tides, over tides and tidal asymmetry, characterizing tides at site and power density, tidal stream devices, basic hydrodynamics of horizontal axis turbines and power coefficients & Betz limit, tidalrange: lagoons and barrages.	9
3	Offshore Wind Energy, introduction and history of offshore wind energy, offshore wind turbines, aerodynamics of wind turbines, power curves, assessment of wind energy at site, case study with calculation of power output & capacity factor, marine spatial planning.	9

4	Wave Energy and Other forms of Ocean Energy, Wave processes, linear wave theory, dispersion equation, wave energy & wave power, irregular & nonlinear waves, wave transformation due to shoaling water, wave energy convertors, wave resource assessment, ocean currents, Ocean Thermal Energy Conversion (OTEC): Closed & open cycle, Environmental impacts of OTEC, Salinity gradients	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Criteria for evaluation:

1. Problem Definition (K4 - 4 points)

- a. Clearly define a renewable energy problem in marine environment
- b. Examine and identifies relevant contextual factors (constraints, resources, objectives).

2. Problem Analysis (K4 - 4 points)

- a. Break-down and present a well-reasoned solution approach for the renewable energy problem..
- b. Compare and justify the proposed solutions with technical evidence.

3. Evaluate (K5 - 4 points)

- a. Thoroughly evaluate the proposed solutions.
- b. Compare trade-offs, advantages, and disadvantages.
- c. Consider feasibility and practical implications.

4. Design Implementation (K5 - 4 points)

- a. Select the most feasible solution by implementing the proposed solutions.
- b. Successfully translate the chosen solution into tangible outputs.

5. Conclusion (K4- 2 points, K5 – 2 points)

- a. Summarizes findings and insights. State which solution is most appropriate for the problem. **(K4)**
- b. Reflects critical thinking and informed decision-making. **(K5)**

Scoring:

1. **Accomplished (4 points):** Exceptional analysis, clear implementation, and depth of understanding.
2. **Competent (3 points):** Solid performance with minor areas for improvement.
3. **Developing (2 points):** Adequate effort but lacks depth or clarity.
4. **Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. - Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand ocean renewable energy sources.	K1
CO2	Discern the tidal energy sources and power generation	K3
CO3	Analyse the offshore wind energy sources and power generation	K4
CO4	Evaluate the wave energy sources, OTEC and power generation	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	1							
CO2	3	2	2			1						
CO3	4	2	2	1	2	1		1	1			
CO4	5	5	2			1	1					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Ocean Renewable Energy,	Simon P. Neill and M. Reza Hashemi,	Academic Press (Elsevier).	Edition 1, 2018
2	Wave and Tidal Energy,	Deborah Greaves and Gregorio Iglesias	John Wiley & Sons Ltd	Edition 1, 2018
3	Ocean Energy: Tide & Tidal Power	Roger H. Charlier and Charles W. Finkl	Springer.	Edition 1, 2009
4	Wave Energy Conversion	R. Bhattacharya and M.E. McCormik	Elsevier Ocean Engineering Book Series, Elsevier	Vol 6, 2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ocean Energies : Environmental, Economic and Technological Aspects of Alternative Power Sources	Roger H. Charlier and John R. Justus	Elsevier Oceanography series	1993
2	Renewable Energy from the Ocean: A Guide to OTEC	William H. Avery and Chih Wu	Oxford University Press.	1994
3	Handbook of Ocean Wave Energy	Arthur Pecher, Jens Peter Kofoed	Springer.	2016

SEMESTER S7

BASIC PRINCIPLES OF MARINE VESSEL DESIGN

Course Code	OENST721	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs 30 Mins
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. To introduce the structure of international marine industry and the economical aspects in ship design to the student
2. To provide the students an understanding of the methodologies for preliminary design of ships
3. To provide the students an understanding of the ship design methodologies to develop lines plans of typical merchant ships
4. To enable the student to understand the relevant regulations on freeboard, tonnage and stability applicable to ships and to apply these regulations to typical merchant ships

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Types of ships: - terms and definitions, cargo ships (general cargo ships, bulk carriers, container ships, Ro-Ro ships, barge carriers, tankers), fishing vessels, factory ships, supply ships, Cable ships, ice breakers, research vessels, warships, hydrofoils, air cushion vehicles, small pleasure crafts (yachts, ketches, etc). Phases of ship design, design spiral. Ship parameters, deadweight, displacement, deadweight coefficient, hold capacity, stowage factor.	9
2	Weight equation and volume equation and solution. Basic design procedure for main categories of ships, deadweight carriers, capacity carriers, linear dimension ships, special purpose ships. Procedure for selection of main dimensions and form coefficients. Estimation of weight, lightship weight,	11

	steel weight, machinery weight, outfit weight, deadweight. General Arrangement (GA), factors influencing GA, subdivision of ship's hull, superstructure and deckhouses, arrangement of cargo spaces, cargo handling arrangement, arrangement of machinery spaces, accommodation and associated spaces, tanks, and other spaces.	
3	Ship's lines plan, conventional method of lines design, lines design using existing lines plan, lines design using standard series, hull form section shapes, form of bow, bulbous bow, form of stern. Effect of hull form on ship's performance.	10
4	Freeboard and load line regulations. Stability and trim, watertight integrity. IMO regulations, intact stability criteria, damage stability criteria. Stability booklet. Tonnage regulations, gross tonnage, net tonnage.	7

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Introduce the structure of international marine industry and the economical aspects in ship design	K1
CO2	Get an understanding of the methodologies for preliminary design of ships	K1
CO3	Have an understanding of the ship design methodologies to develop lines plans of typical merchant ships	K2
CO4	Able to understand the relevant regulations on freeboard, tonnage and stability applicable to ships and to apply these regulations to typical merchant ships	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2										1	
CO2	2	1	1		1							2
CO3	2	2	1		1							2
CO4	2	2	1	1	1							1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Design and Construction	Thomas Lamb (Ed.)	The Society of Naval Architects and Marine Engineers	Edition 1, 2003
2	Elsevier Ocean Engineering Book Series- Volume 1 (Practical Ship Design)	D. G. M. Watson	Elsevier	Edition 1, 1998
3	Ship Design- Methodologies of Preliminary Design	Apostolos Papanikolaou	Springer	Edition 1, 2014
4	Ship Design for Efficiency & Economy	H. Schneekluth & V. Betram	Butterworth-Heinemann	Edition 2, 1998

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Design and Construction	Robert Taggart (Ed.)	The Society of Naval Architects and Marine Engineers	Edition 1, 1980
2	The Maritime Engineering Reference Book	Anthony F. Molland (Ed)	Elsevier	Edition 1, 2008
3	Design Principles of Ships and Marine Structures	S. C. Misra	Taylor & Francis	Edition 1, 2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105004
2	https://nptel.ac.in/courses/114105004
3	https://nptel.ac.in/courses/114105004
4	https://nptel.ac.in/courses/114105004

SEMESTER S7
FUNDAMENTALS OF OFFSHORE STRUCTURES

Course Code	OENST722	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand the basics features of offshore structures
2. Discern the basic principles of fixed offshore platform design
3. Discern the basic principles of Compliant offshore structures design
4. Understand the basics of offshore installation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definition and functions of Offshore structures, Brief Historical Development, Different types of Offshore Structure Configurations: Bottom Supported Fixed Structures, Compliant Structures, Floating Structures and Novel & Small Field Offshore Structures, Offshore field development concepts and design phases	9
2	Wave force on offshore structures: Small diameter and large diameter structures, vertical and arbitrarily oriented cylindrical members, Ocean currents and current forces exerted on offshore structures, Wind forces, Earthquake forces, Ice forces and Tides, Response of offshore structures to ocean environment, Introduction to corrosion and protection of offshore structures	9
3	Design spiral, Various types of loads acting on offshore structures, Introduction to design tools for offshore structure design, Jacket structure: Basic configuration, sizing considerations, design strategy, bracing system, jacket design, pile design, deck design, top sides, appurtenances, Floating Structures: Basic configurations, design strategy, design consideration for components. Compliant Structures: Basic configurations, sizing	9

	considerations, design strategy, design consideration for components. Gravity structures: Basic configuration, sizing considerations, design concepts for foundation, Material considerations for offshore structures	
4	Introduction to construction of various types of offshore structures, Load out methods, Transportation facilities and methods for different types of offshore structures, Installation methods for different offshore structures	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basics features of offshore structures	K2
CO2	Discern the basic principles of fixed offshore platform design	K2
CO3	Discern the basic principles of Compliant offshore structures design	K2
CO4	Understand the basics of offshore installation	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2		1									
CO2	2	1	2									1
CO3	2	1	2									1
CO4	2	1	2		1							2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Offshore Structures Design, Construction and Maintenance	Mohamed A El-Reedy	Elsevier	Edition 1, 2012
2	Introduction to Offshore Structures- Design, Fabrication, Installation	W J Graff	Gulf Publishing Co	Edition 1, 1981
3	Offshore Engineering – An Introduction	Mather A	Wetherby Publishers	Edition 2, 2000
4	Essentials of Offshore Structures - Framed & Gravity Platforms	DV Reddy, A S J Swamidas	CRC Press	Edition 2, 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Dynamics of Offshore Structures	Minoo H Patel	Butterworths	Edition 1, 1989
2	Compliant Offshore Structures	Minoo H Patel Joel Witz	Butterworths	Edition 1, 1991
3	Handbook of Offshore Engineering (Vol I & II), 2005	S.K.Chakrabarti	Elsevier	2005

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114106011
2	https://nptel.ac.in/courses/114106011
3	https://nptel.ac.in/courses/114106015
4	https://nptel.ac.in/courses/114106015

SEMESTER S7

BASICS OF DESIGN OF SUBMERSIBLES

Course Code	OENST723	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand the submersible System.
2. Discern submarine hydrostatics.
3. Understand submarine hydrodynamics.
4. Understand the basic structural aspects and engineering systems of a submarine.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	The submersible system, basic design of manned submersibles, design objectives, characteristics and development of submersibles, environment – physical properties of sea water, dynamical processes	6
2	Submarine hydrostatics, principles of flotation, submarines on the surface, arrangement of the main ballast tanks, submarine submerged, buoyancy elements, weight elements, trim and compensating tanks, special tanks, weight / space relationship, general form and arrangement	9
3	Submarine hydrodynamics, Resistance, appendage resistance, sway and heave resistance, resistance estimation, propulsion, dynamics and control, equation of motion of a submarine, hydrodynamic derivatives, stability and control in the horizontal plane, stability and control in the vertical plane, steering and depth control system.	9

4	Structure and systems, submarine structure, operational requirement for depth, pressure hull design, shape of the pressure vessel, buckling deformation of the shell, failure modes, pressure hull penetration, diving and surfacing system, Hydraulic systems - high pressure air systems - water systems -systems for hydrostatic control - environmental control system - provision for escape, -electrical systems.	12
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the submersible System.	K2
CO2	Discern submarine hydrostatics.	K2
CO3	Understand submarine hydrodynamics.	K2
CO4	Understand the basic structural aspects and engineering systems of a submarine.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											
CO2	2	1	1		1							
CO3	2	2	1		1							
CO4	2	2	1	1	1		1					1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Submarine Design and Development	Norman Friedman	Conway Maritime Press Ltd	Edition , 1984
2	Concepts In Submarine Design - Cambridge Ocean Technology Series 02	Roy Burcher, Louis J. Rydill	Cambridge University Press	Edition 1, 2014
3	Theory of submarine Design	V.N. Kormilitsin, O.A. Kha Lizev	Reivera Maritime Medias	Edition 1, 2001
4	Submersible Vehicle Systems Designs	E Eugene Allmendinger	Society of Naval Architecture and Marine Engineering	1990

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Weapons and Warfare: Submarines - An Illustrated History of Their Impact	Paul E. Fontenoy	ABC-CLIO	Edition 1, 2007
2	Submarines in Color	Bill Gunston	Blandford color series	

SEMESTER S7

BASICS OF SHIP CONSTRUCTION

Course Code	OENST724	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand various types of shipyard layouts, material handling systems.
2. Discern various production management methodologies in modern shipbuilding.
3. Discern the various assembly techniques, welding symbols and standards used in shipbuilding.
4. Explicate the various types of outfitting and launching methodologies adopted in modern shipbuilding.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	General principles for location and layout of a shipyard, evolution of shipyards – first to fourth generation shipyards and their typical layouts, management of shipyard organisation structure – Product Oriented Work Breakdown Structure, Process Oriented Work Breakdown Structure, Integrated Hull construction, Outfitting and Painting (IHOP), grouping and work breakdown structure, zone construction method for hull blocks, outfitting and painting, storage and preparation of material - straightening of plates and rolled sections, sand blasting, painting, lofting, nesting.	9
2	Production management – planning, design , production, quality, material management commercial activities, human resource, quality control, quality assurance, accuracy control, project life cycle, project monitoring – build strategy, progress monitoring, PERT charts, cutting, bending and welding of steel plates and frames, fabrication of component parts, cutting	9

	processes and tools, general description of the various machines, photoelectric and NC control devices, edge preparation, bending of rolled and built up sections - control of the bending process, automation of bending, plate bending, uniaxial bending, biaxial bending devices, cold bending, heat-line bending	
3	Assembly of Ship's Structure, sub-assemblies- built up T bars, web frames, machine foundations etc, welding deformation and straightening, prefabrication of flat sections, pre-assembly of biaxial stiffened panels, assembly of various components, assembly of flat and corrugated sections, flat sections with curvature, double bottom sections, assembly jigs, structural arrangement of the fore and aft end structure, superstructures, machinery foundation, technological processes in the hull installation the installation of main machinery, shafting and propeller, auxiliary machinery, welding in shipbuilding.	9
4	Erection of hull and launching, handling of pre-assembled units in the erection area, Hull assembly – different methods of hull assembly. Block erection methods – floor, ring and pyramid. Single/ multi point erection methods, Technological processes in the installation of main machinery, shafting and propeller, auxiliary machinery, Welding- Weld defects, distortions and their preventive measures. Testing – destructive/ non-destructive, Launching- Mechanical Launching Methods (Slip, Lift), Construction of slipways for end launch and side launch. Launching curves. Arresting arrangements, Tipping, Pivoting/ plunging. Drop off, Other Methods of launching - Launching by Floating Off (Building Dock, Floating Dock), Launching using air bags.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand various types of shipyard layouts, material handling systems.	K2
CO2	Discern various production management methodologies in modern shipbuilding.	K2
CO3	Discern the various assembly techniques, welding symbols and standards used in shipbuilding.	K2
CO4	Explicate the various types of outfitting and launching methodologies adopted in modern shipbuilding	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2				1	1	1					
CO2	2	1	1		1		1				1	
CO3	2	2	1		1	1	1					1
CO4	2	2	2	1	1		1				1	1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Production	Richard Lee	Society of Naval Architects and Marine Engineers	Edition 2, 1995
2	Engineering for Ship Production	Thomas Lamb	Society of Naval Architects and Marine Engineers	Edition 1, 1986
3	Ship Design and Construction Vol I & II	Thomas Lamb (Ed)	Society of Naval Architects and Marine Engineers	Edition 3, 2004
4	Reed's ship construction For marine students	E A Stokoe	Thomas Reed Publications	Edition 5, 1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Construction and Welding	Nisith R. Mandal	Springer Series on Naval Architecture, Marine Engineering, Shipbuilding and Shipping 2	Edition 1, 2017
2	Merchant Ship Construction	H J Pursey	Nautical Press	1978
3	Shipbuilding Technology	Dormidontov VK	Mir Publishers, Moscow	1966

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105004
2	https://nptel.ac.in/courses/114105004
3	https://nptel.ac.in/courses/114105004
4	https://nptel.ac.in/courses/114105004

SEMESTER 8

**NAVAL ARCHITECTURE & SHIPBUILDING
ENGINEERING**

SEMESTER S8
SHIP SURVEY, ESTIMATION AND REPAIR

Course Code	PENST861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST502 PBNST504	Course Type	Theory

Course Objectives:

1. Understand the requirement and various types of marine surveys.
2. Understand the different types of ship repair afloat.
3. Understand ship repairs in dry dock.
4. Discern the principles of work estimates and cost estimates.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Marine Survey: Definition of Elementary Terminologies like Survey Authority, Recognized Organization, Port State control, Flag State control, Various Types of Marine Surveys - Initial Survey, Annual Survey, Periodic Survey, Intermediate Survey, Renewal Survey, Additional Survey. Survey During Transfer of Ships. Marine Surveying Agencies- Roles and Responsibilities of classification societies, Statutory Surveys – Role of MMD, IWAI. Marine Cargo Survey- Dry, Liquid and Container Cargoes.	9
2	Afloat Repairs, Introduction, Causes of Wear and Damage in Ship's Hull. Comparison Between Different Types of Repair Activities (Afloat, Berthed etc.) Testing for Water Tightness and Hull Continuity. Repair of hull and other Parts While Afloat. Underwater Welding – Welding Equipment, Quality Control and Standards; Degree of Automation	9

3	Dry docking and repairing, Dry-docking- Steps to be Taken Before, During and After Dry-docking, Replacement of Hull Plates and Stiffeners, Decks and Bulkheads, Repair of Stem and Stern Frames and Shaft Bracket, Propeller Shaft Sealing Equipment, Rudder– pintle Clearances, Maintenance of Sea Water Suction and Overboard Valves, Safety During Repair – Various Operations Involving Risk, Safety Devices and Plans.	9
4	Work and Cost Estimation, Work Estimation: Dry-docking Works - Docking and Undocking, Hull Preparation, Hull Painting, Propeller Works, Chain Lockers, Steel Works, Planning Charts, Shipbuilding Contracts, Costing and Estimation- Difference between Cost and Price, Top Down and Bottom Top Approaches in Costing, Demarcations and Subdivisions of Costs, Structural Costs, Outfit Costs, Labour Costs, Machinery Costs.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the requirement and various types of marine surveys.	K1
CO2	Understand the different types of ship repair afloat.	K2
CO3	Understand ship repairs in dry dock.	K2
CO4	Discern the principles of work estimates and cost estimates	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	1	1		2	1				1
CO2	2	2	2	1	1	1	2	1			1	1
CO3	2	2	2	1	2	1	2	1			1	1
CO4	2	2	2	1		1	2	1			2	1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Commercial Ship Surveying On/Off-Hire Condition Surveys and Bunker Surveys	Harry Alexander Karanassos	Elsevier	Edition 1, 2016
2	Guide to Ship Repair Estimates	Don Butler	Butterworth-Heinemann	Edition 2, 2012
3	Ship Repair Project Manager's Guide	Mohammed Khamis	IIEC	Edition 2, 2022
4	Ship Repairing - Analyses and Estimates	Arun Kr Dev, Makaraksha Saha, George Bruce	Springer Series on Naval Architecture, Marine Engineering, Shipbuilding and Shipping Vol 12	Edition 1, 2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Rules and Regulations for the Classification of Ships, Part 1, Regulations, Chapters 2 and 3.	Lloyds Register	Lloyds Register	Revised editions
2	Rules and Regulations for the Construction and Classification of Inland Waterways ships - January 1997; IR Class,	Indian Register of Shipping	Indian Register of Shipping	1997
3	Underwater Repair Technology	John H Nixon	Gulf Publishing Company	Edition 1, 2000
4	The Inland Vessels Act, 2021	Ministry of Shipping	Govt of India	2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105004
2	https://nptel.ac.in/courses/114105004
3	https://nptel.ac.in/courses/114105004
4	https://onlinecourses.nptel.ac.in/noc24_mg114/preview

SEMESTER S8
QUALITY ASSURANCE IN SHIPBUILDING

Course Code	PENST862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST502 PBNST504	Course Type	Theory

Course Objectives:

1. Understand the requirements of quality in shipbuilding and repair.
2. Discern quality management principles in shipbuilding.
3. Understand fundamentals of quality management systems.
4. Apply quality assurance processes in various phases of shipbuilding.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Quality & Quality Control: Concept of quality, quality characteristics, quality standards, quality cost, concept of quality control, quality control methodology, quality philosophy and management strategies, Major Quality Problems, Difference between QA and QC.	9
2	Quality Management Principles:- Customer focus, Leadership, Involvement of people, Process approach, System approach to management, Continual improvement, Factual approach to decision making, Mutually beneficial supplier relationships.	9
3	Fundamentals of a Quality Management System: - Definition, Description and Use, Integrated Quality Management System, Advanced QMS Software, Fundamentals of Total Quality Management in Ship Building, Strategic Quality Management and Human Factors in Quality.	9
4	Fundamentals of Quality Assurance in Shipbuilding, Steel process Quality Assurance (Hull and Painting) and Outfit process Quality Assurance, Various Quality Standards and Auditing in shipyards, Overview on ISO standard in Ship Building, Qualification Standards, Training and Certification, Quality Manual and Document Management, Quality in Planning, Design and Operation, Quality Policy of Shipyards.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recall the requirements of quality in shipbuilding and repair.	K1
CO2	Discern quality management principles in shipbuilding.	K2
CO3	Understand fundamentals of quality management systems.	K2
CO4	Apply quality assurance processes in various phases of shipbuilding.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyze, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											1
CO2	2					1	1				2	1
CO3	2										1	1
CO4	2	1	1		1	1					1	1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Production	Richard Lee	Society of Naval Architects and Marine Engineers	Edition 2, 1995
2	Engineering for Ship Production	Thomas Lamb	Society of Naval Architects and Marine Engineers	Edition 1, 1986
3	Shipbuilding Management	George Bruce	Springer	Edition 1, 2021
4	Reed's ship construction For marine students	E A Stokoe	Thomas Reed Publications	Edition 5, 1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Construction and Welding	Nisith R. Mandal	Springer Series on Naval Architecture, Marine Engineering, Shipbuilding and Shipping 2	Edition 1, 2017
2	Merchant Ship Construction	H J Pursey	Nautical Press	1978
3	Shipbuilding Technology	Dormidontov VK	Mir Publishers, Moscow	1966
4	Ship Design and Construction Vol I & II	Thomas Lamb (Ed)	Society of Naval Architects and Marine Engineers	Edition 3, 2004

SEMESTER S8
MARITIME LAW AND SHIPPING MANAGEMENT

Course Code	PENST863	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST205	Course Type	Theory

Course Objectives:

1. Understand the role of international conventions in formulating Maritime Law, the roles of Maritime Labour Convention and international conventions on the Law of the Sea.
2. Understand the roles of international conventions for the Safety of Life at Sea, International Ship and Port Facility Security Code (ISPS Code) and other conventions
3. To enable the student to understand the basic structure of shipping industry and the role of various stakeholders in the industry
4. To provide the student an understanding of shipping economics and the shipping market

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to maritime law; need and source of maritime law; implementation of maritime law through national legislation; role of United Nations (UN), International Maritime Organization (IMO) and International Labour Organization (ILO); role of Flag state and Port state in implementation of law; introduction to SOLAS, MARPOL, STCW and Maritime Labour Convention (MLC); need of survey and certification. Classification Societies, need and role of Classification Societies. United Nations Convention on the Law of the Sea (UNCLOS), various zones and limits. Maritime Labour Convention, minimum requirements for seafarers to work on a ship, conditions of employment, accommodation, recreational facilities, food and catering, Health protection, medical care, welfare and social security.	9

2	SOLAS, Introduction to SOLAS, Subdivision and stability, Fire protection, detection and extinction, Life saving appliances and arrangements, Carriage of grain, Carriage of dangerous goods. Load Line Convention, important definitions and markings, survey and certification. ISPS Code, ship security plan, security levels, declaration of security, ship security alert system, piracy, best management practices to deter piracy. Introduction to other Conventions; International Convention on Salvage, Convention on Limitation of Liability for Maritime Claims; Oil Pollution Compensation Conventions.	9
3	Shipping Management- Types of cargoes, types of ships which carry them, Ship's tonnages (GT, NT), Cargo carrying capacity. Ship registration, ship insurance. Ship owners, operators and managers. Types of shipping services, liner and tramp. Role of intermediaries in shipping business- freight brokers, clearing and forwarding agents, stevedores, shipbrokers, bunker and stores suppliers, shipping agencies.	9
4	Shipping Economics-Sea transport and the global economy, role of shipping in economic development. Shipping market cycles, shipping risk, shipping market supply and demand model. The four shipping markets- freight market, sale and purchase market, newbuilding market and demolition market. Classification of costs, Financing ships and shipping companies, Risk and returns.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basics of international maritime law and describe the roles of international conventions in formulating Maritime Law, the roles of Maritime Labour Convention and international conventions on the Law of the Sea, 1982.	K2
CO2	Describe the roles of international conventions for the Safety of Life at Sea, International Ship and Port Facility Security Code (ISPS Code) and other conventions	K2
CO3	Understand various elements of shipping management and the role of various stakeholders in shipping business	K2
CO4	Understand the basics of shipping economics and the shipping market	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	1	1					1				1
CO2	1	1	1									1
CO3	1	1	1					1				1
CO4	1	1	1					1			1	1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Maritime Economics	Martin Stopford	Taylor & Francis	Edition 3, 2008
2	Economics of Shipping Practice and Management	Alan E. Branch	Chapman and Hall	Edition 2, 1988
3	Elements of Shipping	Alan E. Branch	Routledge	Edition 8, 2007
4	Reeds 21 st Century Ship Management	Capt. J W Dickie	Adlard Coles Nautical	Edition 1, 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Shipping and Logistics Management	Y. H. V Lun, K. H. Lai, & T. C. E. Cheng	Springer	Edition1, 2010
2	The Maritime Engineering Reference Book	Anthony F. Molland (Ed)	Elsevier	Edition1, 2008
3	Ship Design and Construction	Thomas Lamb (Ed.)	The Society of Naval Architects and Marine Engineers	Edition 1, 2003
4	Practical Ship Design - Elsevier Ocean Engineering Book Series- Volume 1	D. G. M. Watson	Elsevier	Edition 1, 1998

SEMESTER S8
TRADITIONAL BOAT BUILDING TECHNIQUES

Course Code	PENST864	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand the history of traditional boatbuilding.
2. Discern the methodologies for design and construction of traditional boats
3. Understand the safety and certification regulations for traditional boatbuilding
4. Understand launching and post launching activities in modern times

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	History/background. Construction techniques; designers' plans; building moulds, jigs and strong-backs. Wooden kit boats, skills required, different types of wood/timber used in boat building. Regional boats of India and other countries – Snake boats, house boats, Urus of Malabar, Kerala; Dhows of Mandvi, Gujarat; Teppas of Ganjam, Odisha; Boats of Zanzibar, Iran, UK and Hong Kong.	9
2	Yard selection, workshop/yard requirements, budgeting and costs, project management, outsourcing parts, buy or lease/machinery and tools, bill of materials/detailed costs. Boats's outfitting; internal layouts; electrical systems; water and sanitation systems; safety equipment and navigation systems; anchoring, mooring and deck fittings; painting of the boat PROPEN?	9
3	CE Certification, Type Approval/Licensing, Legal Requirements, VAT/ Goods Service Tax etc, Warranties/Manuals, Basic Safety at Sea, Skills required & sub-contracting, Insurance, Inland Waterway Regulations.	9

4	Boat relocation, Launching, Ways of lifting & launching a vessel, The actual launch, Stepping the mast, Types of berth/moorings, Sea Trials, Modifications	9
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Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the history of traditional boatbuilding.	K1
CO2	Discern the methodologies for design and construction of traditional boats	K2
CO3	Understand the safety and certification regulations for traditional boatbuilding	K2
CO4	Understand launching and post launching activities in modern times	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1										1
CO2	2	1		1	1	1						
CO3	2	1									1	1
CO4	2	1		1	1	1						1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	How to Design a Boat	John Teale	Adlard Coles Nautical	Edition 2, 1998
2	Traditional Boat Building: An Intersection of Zanzibar's Culture and Environment.	Levenson E	SIT Digital Collections	Edition 1, 2013
3	Boatbuilding Manual	Robert M. Steward	International Marine/McGraw-Hill	Edition 5, 2011
4	How to Build a Wooden Boat	David C. McIntosh	Wooden Boat Magazine	Edition 1, 1987

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A practical course in wooden boat and ship building	Richard M. Van Gaasbeek.	Frederick J Drake & Company	Edition 1, 1918
2	Indian boat building traditions. The ethnological evidence.	Lotika Varadarajan	Topoi. Orient-Occident,	1993
3	Boats of South Asia	Sean McGrail	RoutledgeCurzon	Edition 1, 2003

SEMESTER S8

EXPERIMENTAL TECHNIQUES IN MARINE HYDRODYNAMICS

Course Code	PENST865	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCNST402 PBNST404	Course Type	Theory

Course Objectives:

1. Understand various test facilities in marine hydrodynamics.
2. Discern various resistance tests using models and analyse the results
3. Discern open water tests and self propulsion tests and analyse the results
4. Understand seakeeping and manoeuvring trials and analyse the results.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	History of model testing, towing tank, components of towing tanks including instrumentation, circulating water channel including instrumentation, wind tunnel and instrumentation, cavitation tunnel and instrumentation, wave makers, planar motion mechanism, seakeeping and maneuvering experiments.	9
2	Prediction of Ship Resistance, Resistance Tests, Total Resistance, Resistance Curves, Resistance Coefficients, Ship Models, Laws of Comparisons and Similarity, Extension of Model Results to Ships, Towing Tank, Instrumentation and Method of Measurements. Tests using circulating water channel.	9
3	Open Water Tests: Objectives, Facilities, Test Set Up, Principles, Procedure, Analysis and Conclusions. Self Propulsion Experiments - Objectives, Instruments and Equipment, Test Arrangements, Basic Principles, Experiment, Results. Wake Measurements,	9
4	Cavitation Tests - Cause of Cavitation, Cavitation Number, Classification of Cavitation, Law of Similarities, Cavitation Tests, Prevention of Cavitation. Sea Keeping Tests, PMM tests. Tests in seakeeping and maneuvering basin.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Criteria for evaluation:

1. Problem Definition (K4 - 4 points)

- a. Clearly defines the experimental setup and the problem.
- b. Examine and identifies relevant contextual factors (constraints, resources, objectives).

2. Problem Analysis (K4 - 4 points)

- a. Carry out the experiments on models based on Froude's hypothesis..
- b. Compare and justify the proposed solutions with technical evidence and reasoning.

3. Evaluate (K5 - 4 points)

- a. Thoroughly evaluate the observed model results for extrapolation to ship's results.
- b. Compares trade-offs, advantages, and disadvantages.
- c. Considers feasibility and practical implications.

4. Design Implementation (K5 - 4 points)

- a. Identify and select the most accurate result..
- b. Successfully translates the result into documents.

5. Conclusion (K4- 2 points, K5 – 2 points)

- a. Summarizes findings and insights. State which solution is most appropriate for the problem. **(K4)**
- b. Reflects critical thinking and informed decision-making. **(K5)**

Scoring:

1. **Accomplished (4 points):** Exceptional analysis, clear implementation, and depth of understanding.
2. **Competent (3 points):** Solid performance with minor areas for improvement.
3. **Developing (2 points):** Adequate effort but lacks depth or clarity.
4. **Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. - Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand various test facilities in marine hydrodynamics.	K1
CO2	Discern various resistance tests using models and analyse the results	K2
CO3	Analyse open water tests and self propulsion tests results for accuracy	K4
CO4	Evaluate the results of seakeeping and manoeuvring trials for design implementation.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											
CO2	3	2	1	1	1		1		1			1
CO3	3	2	1	1	1		1	1	1			
CO4	3	2	1		1	1	1		1	1		1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Basic Ship Propulsion	Ghose P. & Gokarn R.P	Knowledge World Publishers Pvt Ltd	Edition 1, 2015
2	Ship Resistance and Propulsion – Practical Estimation of Propulsive Power	Antony F. Moland, Stephen R. Turnock	Cambridge University Press	Edition 1, 2011
3	Marine Propellers and Propulsion	S Carlton	Elsevier Ltd.	Edition 4, 2019
4	Resistance and Propulsion of Ships	Harvald S. A	John Wiley & Sons	Edition 1, 1983
5	Fundamentals of Ship Hydrodynamics - Fluid Mechanics, Ship Resistance and Propulsion	Lothar Birk	JohnWiley & Sons Ltd	Edition 1, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Naval Architecture series – Propulsion	Justin E K, Jacques B H	SNAME, New Jersey	Edition 3, 2010
2	Basic Ship Theory Vol II	K J Rawson, E C Tupper	Butterworth-Heinemann Ltd	Edition 5, 2001
3	Practical Ship Hydrodynamics	Volker Bertram	Butterworth-Heinemann Ltd	Edition 1, 2000
4	Hydrodynamics of High-Speed Vehicles	O M Faltinsen	Cambridge University Press	Edition 1, 2005
5	Hydrodynamics in Ship Design Vol 1-3	H E Saunders	SNAME, New Jersey	Edition 1, 1957

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114106026
2	https://nptel.ac.in/courses/114106026
3	https://nptel.ac.in/courses/114106026
4	https://nptel.ac.in/courses/114106026

SEMESTER S8
JOINING TECHNIQUES IN SHIPBUILDING

Course Code	OENST831	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Explain types of welding, welding process and welding parameters used in shipbuilding.
2. Discern various special welding arrangements in modern shipbuilding.
3. Discern the radiant welding, underwater welding and robotic welding used in shipbuilding.
4. Explicate the various types of weld defects and testing in modern shipbuilding.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Welding. Understand the welding and classification of welding process, development and science in welding, gas welding. Fuel gases, GMAW process, different metal transfers, power source, electrodes, shielding gas, uses of gas in metal arc welding, Different types of oxy-acetylene flames, Fore hand and back hand welding techniques. Function of fillers and fluxes, Arc welding. Various arc welding processes. Carbon arc welding, Shielded metal arc, submerged arc, TIG, MIG, Electro slag, Electro gas, Plasma arc, arc spot welding, stud welding	9
2	Resistance welding - welding arrangement, weld cycle, different types of resistance welding – spot welding, seam welding, projection welding, resistance upset welding, flash butt, percussion welding, tube welding, high frequency welding, solid state welding - principle and welding arrangements, advantages/ disadvantages, different types of solid state welding – cold welding, diffusion welding, ultrasonic welding, explosive, friction and inertia welding, forging, principle and welding arrangements. advantages/ disadvantages, different types	11

	of thermochemical welding – thermit welding, atomic hydrogen welding,	
3	Radiant energy welding - principle and welding arrangements, advantages/ disadvantages, different types of radiant energy welding – Electron beam welding, laser beam welding, Under water welding. Principle and welding arrangements. Advantages/ disadvantages. Specific aspects pertaining to underwater welding. Precautions to be taken. Wet welding, Dry welding, Hyper baric welding, cavity welding, Robotic Welding. Define the Introduction of Robotic welding, Application of Robotic Welding in Ship Production, Robotic Welding System, and Types of Welding Robots, Brazing and soldering. Difference between brazing, soldering and welding, adhesives and bonding methods.	10
4	Weld Defects and measures for their control. Distortion control. Tests- destructive/ non-destructive, Welding standards, welding performance and procedure qualification, Performance Qualification of Welders & Operators, Test Reports, Acceptance Standards, Quality Assurance and Audit, Consumable Classification & Coding. Essential and non-essential variables. Effects of variables, Crystallization of a Pure Metal, Phase Transformation in Iron-Carbon Diagram	7

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain types of welding, welding process and welding parameters used in shipbuilding.	K1
CO2	Discern various special welding arrangements in modern shipbuilding.	K1
CO3	Discern the radiant welding, underwater welding and robotic welding used in shipbuilding.	K2
CO4	Explicate the various types of weld defects and testing in modern shipbuilding.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1									
CO2	2	1	1									
CO3	2	1	1									
CO4	2	2	1		1							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ship Construction and Welding	Nisith R. Mandal	Springer Series on Naval Architecture, Marine Engineering, Shipbuilding and Shipping 2	Edition 1, 2017
2	Welding Engineering and Technology	R.S. Parmar	Khanna Publishers	Edition 3, 2022
3	A Textbook of Welding Technology	O P Khanna	Dhanpat Rai & Sons	Edition 11, 2015
2	Engineering for Ship Production	Thomas Lamb	Society of Naval Architects and Marine Engineers	Edition 1, 1986
3	Ship Design and Construction Vol I & II	Thomas Lamb (Ed)	Society of Naval Architects and Marine Engineers	Edition 3, 2004
4	Reed's ship construction For marine students	E A Stokoe	Thomas Reed Publications	Edition 5, 1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Standard symbols for welding, brazing, and nondestructive examination		American Welding Society	Edition 7, 2012
2	Merchant Ship Construction	H J Pursey	Nautical Press	1978
3	Shipbuilding Technology	Dormidontov VK	Mir Publishers, Moscow	1966

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105004
2	https://nptel.ac.in/courses/114105004
3	https://nptel.ac.in/courses/114105004
4	https://nptel.ac.in/courses/114105004

SEMESTER S8

HULL INSPECTION

Course Code	OENST832	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Understand the requirement and various types of marine surveys.
2. Understand the different types of ship inspection and repair afloat.
3. Understand ship inspection and repairs in dry dock.
4. Discern the principles of work estimates and cost estimates based on inspections

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Marine Survey: Definition of Elementary Terminologies like Survey Authority, Recognized Organisation, Port State control, Flag State control, Various Types of Marine Surveys - Initial Survey, Annual Survey, Periodic Survey, Intermediate Survey, Renewal Survey, Additional Survey. Survey During Transfer of Ships. Marine Surveying Agencies- Roles and Responsibilities of classification societies, Statutory Surveys – Role of MMD, IWAI. Marine Cargo Survey- Dry, Liquid and Container Cargoes.	9
2	Afloat Repairs, Introduction, Causes of Wear and Damage in Ship's Hull. Comparison Between Different Types of Repair Activities (Afloat, Berthed etc.) Testing for Water Tightness and Hull Continuity. Repair of hull and other Parts While Afloat. Underwater Welding – Welding Equipment, Quality Control and Standards; Degree of Automation	9
3	Dry docking and repairing, Dry-docking- Steps to be Taken Before, During and	9

	After Dry-docking, Replacement of Hull Plates and Stiffeners, Decks and Bulkheads, Repair of Stem and Stern Frames and Shaft Bracket, Propeller Shaft Sealing Equipment, Rudder– pintle Clearances, Maintenance of Sea Water Suction and Overboard Valves, Safety During Repair – Various Operations Involving Risk, Safety Devices and Plans.	
4	Work and Cost Estimation, Work Estimation: Dry-docking Works - Docking and Undocking, Hull Preparation, Hull Painting, Propeller Works, Chain Lockers, Steel Works, Planning Charts, Shipbuilding Contracts, Costing and Estimation- Difference between Cost and Price, Top Down and Bottom Top Approaches in Costing, Demarcations and Subdivisions of Costs, Structural Costs, Outfit Costs, Labour Costs, Machinery Costs.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the requirement and various types of marine surveys.	K2
CO2	Understand the different types of ship inspection and repair afloat.	K2
CO3	Understand ship inspection and repairs in dry dock.	K2
CO4	Discern the principles of work estimates and cost estimates based on inspections	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1									
CO2	2	1	1									
CO3	2	1	1									
CO4	2	2	2		1							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Commercial Ship Surveying On/Off-Hire Condition Surveys and Bunker Surveys	Harry Alexander Karanassos	Elsevier	Edition 1, 2016
2	Guide to Ship Repair Estimates	Don Butler	Butterworth-Heinemann	Edition 2, 2012
3	Ship Repair Project Manager's Guide	Mohammed Khamis	IIEC	Edition 2, 2022
4	Ship Repairing - Analyses and Estimates	Arun Kr Dev, Makaraksha Saha, George Bruce	Springer Series on Naval Architecture, Marine Engineering, Shipbuilding and Shipping Vol 12	Edition 1, 2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Rules and Regulations for the Classification of Ships, Part 1, Regulations, Chapters 2 and 3.	Lloyds Register	Lloyds Register	Revised editions
2	Rules and Regulations for the Construction and Classification of Inland Waterways ships - January 1997; IR Class,	Indian Register of Shipping	Indian Register of Shipping	1997
3	Underwater Repair Technology	John H Nixon	Gulf Publishing Company	Edition 1, 2000
4	The Inland Vessels Act, 2021	Ministry of Shipping	Govt of India	2021

SEMESTER S8
BASICS OF FISHING VESSEL DESIGN

Course Code	OENST833	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Acquire basic knowledge on fishing vessels and systems onboard.
2. Understand basic design procedure of fishing vessels
3. Understand resistance and propulsion performance characteristics of fishing vessels.
4. Understand basic structural design concepts of fishing vessels.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: definitions of fishing vessel, special features of fishing vessels, regulations for the safety of fishing vessels, classification of fishing vessels, Characteristics of fish ground; Fishing gear and methods, drift net, long line, drag net, seine net, dredging, with electric light, harpoon/whale catching, trawling (side and stern trawlers, single and pair trawling, pelagic & bottom trawling).	9
2	Design Procedure: Owner's specifications, Economy, fuel efficiency, hull form, investment cost, operating revenues and costs. Design of Main Dimensions and form: parent vessel data analysis, space requirement (capacity) of the whole ship, estimation of main dimensions, estimation of form coefficients, estimation of light ship weight, estimation of dead weight, design of lines. General arrangement: Engine room, fish holds, erections, deck machinery arrangement, crew accommodation, fuel, fresh-water, ballast tanks, bulkhead positions.	9

3	Resistance, powering and propeller selection: friction resistance, wave making resistance, eddy resistance, net resistance, powering calculations, propeller selection, Propulsion Systems and other machinery/equipment, different propulsion systems and selection of main engine, selection of equipment's/instruments for fish finding, navigation, communication, firefighting, lifesaving and net monitoring Seakeeping and manoeuvring considerations	9
4	Material and construction methods: mechanical properties of materials, comparison of hulls of different material, type of construction, details of steel construction, structural design methods, construction methods in FRP/GRP, Aluminum and Ferro-cement.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Acquire basic knowledge on fishing vessels and systems onboard.	K1
CO2	Understand basic design procedure of fishing vessels	K1
CO3	Understand resistance and propulsion performance characteristics of fishing vessels.	K2
CO4	Understand basic structural design concepts of fishing vessels	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1									
CO2	3	1	1									
CO3	3	1	1									
CO4	3	1	1		1							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of small fishing vessels	John F. Fyson	Food and Agriculture Organization	Edition 1, 1985
2	Modern Fishing Gear Technology	M Shahul Hameed		Edition 3, 2022
3	Fishing boat designs 1. Flat bottom boats	A F Haug	Food and Agriculture Organization	Edition 1, 1972
4	Fishing boat designs 2. V-bottom boats of planked and plywood construction (Rev.2)	O Gulbrandsen	Food and Agriculture Organization	Edition 2, 2004
5	Fishing boat designs 3. Small Trawlers	John F. Fyson	Food and Agriculture Organization	Edition 2, 1991
6	Fishing boat designs 3. Small Steel Fishing Boats	D J Eyres	Food and Agriculture Organization	Edition 1, 1984

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fishing boats of the world	Latest edition		
2	Guidelines for the Design, Construction and Equipment of Small Fishing Vessels		FAO/ILO/IMO	2005

SEMESTER S8
BASICS OF SHIP PROPULSION

Course Code	OENST834	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min..
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Discern thrust development mechanism by screw propellers and its geometry.
2. Understand hull-propulsion interaction and resulting efficiencies and cavitation.
3. Understand propulsion system design basics
4. Understand special types of propulsion systems and propeller strength.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Propeller as a thrust producing mechanism, Screw propeller, screw propeller geometry, sections, construction details. Propeller theories-Momentum theory, Blade element theory, Dimensional analysis	9
2	Interaction between Hull and propeller- Wake and wake fraction, Resistance augmentation and thrust deduction, propulsive efficiency in open water and behind conditions, hull efficiency, quasi propulsive coefficient, transmission efficiency; Powering Cavitation-Types, Cavitation Number, Effects of cavitation, Prevention of cavitation	11
3	Design of propellers-Propeller families and series; Open water tests, Presentation of data, Design charts- Kt-Kq, Bp- δ , T-J, P-J charts, Use of charts in propeller design including selection of prime movers and optimisation, propeller drawing familiarisation.	10
4	Special types of propulsion systems - Shrouded propellers, Controllable Pitch propellers, Super cavitating propellers, Waterjet propulsion, Vertical axis propellers, Sail, Paddle wheels, Electromagnetic propulsion etc.	7

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Discern thrust development mechanism by screw propellers and its geometry.	K1
CO2	Understand hull-propulsion interaction and resulting efficiencies and cavitation.	K2
CO3	Be competent in the fundamentals of propulsion system design and propeller drawing as a project	K3
CO4	Understand special types of propulsion systems and propeller strength.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1									
CO2	3	1	1									
CO3	3	2	2		1							
CO4	3	1	1		1							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Basic Ship Propulsion	Ghose P. & Gokarn R.P	Knowledge World Publishers Pvt Ltd	Edition 1, 2015
2	Ship Resistance and Propulsion – Practical Estimation of Propulsive Power	Antony F. Moland, Stephen R. Turnock	Cambridge University Press	Edition 1, 2011
3	Marine Propellers and Propulsion	S Carlton	Elsevier Ltd.	Edition 4, 2019
4	Resistance and Propulsion of Ships	Harvald S. A	John Wiley & Sons	Edition 1, 1983
5	Fundamentals of Ship Hydrodynamics - Fluid Mechanics, Ship Resistance and Propulsion	Lothar Birk	JohnWiley & Sons Ltd	Edition 1, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Naval Architecture series – Propulsion	Justin E K, Jacques B H	SNAME, New Jersey	Edition 3, 2010
2	Basic Ship Theory Vol II	K J Rawson, E C Tupper	Butterworth-Heinemann Ltd	Edition 5, 2001
3	Practical Ship Hydrodynamics	Volker Bertram	Butterworth-Heinemann Ltd	Edition 1, 2000
4	Hydrodynamics of High-Speed Vehicles	O M Faltinsen	Cambridge University Press	Edition 1, 2005
5	Hydrodynamics in Ship Design Vol 1-3	H E Saunders	SNAME, New Jersey	Edition 1, 1957

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/114105004
2	https://nptel.ac.in/courses/114105004
3	https://nptel.ac.in/courses/114105004
4	https://nptel.ac.in/courses/114105004